

FO Modules

Instruments, Market Data, Simulation & Viewer

Target Audience

FO Configuration
P&L Management

- ✓ FO Support team
- ✓ FO Financial Analyst
- ✓ FO Configurators in an implementation team

Prerequisites

MX.3 Core Modules (or basic knowledge of MX.3 Platform)

Very good knowledge of:

- Front-Office desk activity
- Financial products and trade typologies
- Asset classes (e.g. Interest rate swaps, Securities and Options, FX and Commodities)

Duration

3 days

Course Outline

The purpose of this training session is to give trainees a thorough understanding of the Front-Office modules in Murex. This is an essential step for both the implementation and the daily work and support activity surrounding Murex Front-Office modules.

Learning Objectives

At the completion of this training, the attendee will be able to:

- Configure instruments and other static data in Murex
- Design a rate curve and maintain market data in Murex
- Design simulation views for analysis of trades P&L and risk
- Explain portfolio management (position blotters, real time analysis, risk analysis, scenarios, P&L decomposition)

Sample Syllabus

Day 1	Demonstration of an Interest Rate Swap	Follow the flow of trade through the various Front-Office modules from pricing to booking to risk management
	Profit & Loss	P&L components in MX
	Instrument Configuration	FX instruments configuration
Day 2	Instrument Configuration	Interest rate generators
	Market Parameters	
	Rate Curve Overview	
Day 3	Simulation Module	• Simulation module mechanism and features • Design and customization of data to be displayed and monitored
	Simulation Viewer	
	Profit & Loss Variance Module	

MTEK eLearning tutorials included

- Viewer (starting v3.1.23 for Designer): Tutorial 2: First steps
- Viewer (starting v3.1.23 for Designer): Tutorial 3: Format
- Viewer (starting v3.1.23 for Designer): Tutorial 5: Filters
- Viewer (starting v3.1.23 for Designer): Tutorial 7: Formulas



Preparatory Reading List

FO

Title	docid	Content	Key concepts
MX.3 Platform Overview	8564	Overview of the MX.3 platform and how it adapts to clients' needs	One platform - multiple facets, Main features of the MX.3 platform, Global Operating Model, MX Modules, Mx product coverage
Getting Started- (GOM) Global Operating Model Overview	8581	Explanation of what the (GOM) Global Operating Model is and how it functions.	Global Operating Model, GOM building blocks, Users and Groups, GOM organisation overview
Navigation eLearnings	15808	Navigation tutorial I: Starting up in MX.3	Start and stop MX.3 services, Determine mandatory and optional services, Start the MX.3 application, Navigate MX.3 screens
	15812	Navigation Tutorial II: End users in MX.3	Explain the activities of the different types of End-user groups (FO, MO, BO), Name the main menus available in the standard FO, MO and BO group, Explain the difference between an MX and an MLC group
	15809	Navigation Tutorial III: Admin Session	Navigate the ADMIN session, Define a new MX user, Attach an MX user to an MX group, Suspend and lock users, Link and unlink users from user policies, Define a new MLC group, Attach an MLC user to an MLC group
	15810	Navigation Tutorial IV: Supervisor Session part I	Explain the role of a Supervisor user, Explain the different tabs used to define the group rights in the Group definition screen, Explain the Navigation Template and how to access it, Add and rename menus in the End-user session, Change the access right of a menu in an End-user session.
	15814	Navigation Tutorial V: Supervisor Session part II	Define a new Front Office Group, Define a new P&L Control Group, Define a new Processing Center Group, Define a new Configurator Group
	15911	Navigation Tutorial VI: Configurator Session	Navigate the Configurator session, Add a new Counterparty in MX.3, Use the Import/Export Wizard, Explain the GOM screens in the Configurator session
	15815	Navigation Tutorial VII: Housekeeper Session	Explain the main usage of the Housekeeper session



Preparatory Reading List

FO

Title	docid	Content	Key concepts
Navigation Tools	8576	An outline of the basic navigation tools used throughout the application	Login, Accelerators, Navigation, Display, Numerical cells
Trade Hierarchy Overview	8579	Explanation of the key components in the MX.3 Trade hierarchy	Financial Contract Component, Financial Contract and Package, Typologies, Events, Versioning
Trade management eLearnings	8748	Overview of trade management	Define a Financial Contract (FC), Explain the concept of Financial Contract Component (FCC), Distinguish between a simple and a more complex FC, What is a Package?
	8754	Tutorial Events and Versioning	Notion of Events and Versioning, Identify the IDs of Financial Contract, Events Versions, etc.
	15697	Trade Insertion	Trade insertion from e-Tradepad and FDI as well as the Trade Ticket
	15795	Trade Query	How to query trades from the Trade query screen
	14719	Trade Typologies	Overview on the trade typologies and how they are used in the system
Trade & Static Data Life Cycle	8575	A high level, general overview of the key concepts of Trade and Static Data Management in MX.3	Straight Through Processing (STP), One Stop Processing (OSP), Workflow, Settlement Instructions (SI) workflow, Events
Pricing and Structuring Module	8572	Overview of the module Traders use to determine the fair price of a simple or structured trade	eTradepad, Pricing notepads, Structured trade builder (STB), Pricing matrix, Market data, Archiving module, Communication and Distribution hub, Risk analysis
Distribution and Order Management	8569	Explains the key concepts of the Distribution and Order Management module	Structuring, Sales and eCommerce, Deploying eCommerce solutions to clients, Negotiated vs Automated Trading



Preparatory Reading List

FO

Product Description	8566	List of product typologies per asset class with their descriptions	Commodities Credit Derivatives Equities Fixed Income Fund and Fund Derivatives FX Cash FX Derivatives Hybrids Inflation Interest Rate Derivatives Money Market Securities Finance Securitization
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August 2020
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Instructions to the trainer for DEMO

This demo is given at the beginning of a three-day FO modules training. You need to let the trainees know:

- The training begins with a demo that shows the use of the MX.3 application.
- They will explore in details the modules used for the booking and monitoring of a trade.

You need to explain that you will do this demo **twice**. For that we recommend to open two FO sessions, one with the e-Tradepad open, one with the Simulation viewer open (e.g. portfolio PL_TRAINING).

Some parts of the two demos are mandatory, i.e. should be mentioned, and some other parts are optional, i.e. will be covered if there is time.

1st demo: Will be done in regular speed. **This is what the traders do** and see when they book a trade. The benefit of having everything already configured induces that the traders will book their trade smoothly and you as consultants will not get desperate, angry calls.

You will book a trade using e-Tradepad, look at the simulation screen to check the position. **If possible** you will show that the simulation is automatically refreshed after booking the trade. You will also query the trade ticket. If possible query it from the blotter (e-Tradepad). You will run a “Cancel & Reissue” event and query the trade.

Note to trainer: Please do NOT use accelerators.

2nd demo: In slow motion.

Note to trainer: You must pause at each step. At each section, you must pause. You have to explain all the configuration that has been done to achieve this without redoing the configuration yourself (this will be done in the following training presentations).

Before the demo begins, you tell the trainees the following:

This demo will showcase the FO modules and skills necessary for FO users for trading purposes. For the workshop that follows in the next three days, they will learn to use the e-Tradepad, simulation and viewer. Then they will learn how to configure so there is no need to show that in this FO session.

Keep in mind that this session only lasts 1h30. It goes fast.

There are three main areas used by end-users in the system:



Write on the white board:

- 1) e-Tradepad
- 2) Portfolio management (Simulation)
- 3) Trade Query

Say that this training will focus on the two first areas, as they are the ones mainly used by FO end-users.

Explain to trainees that

- e-Tradepad → this is the central pricing and booking screen for FO users. It has the pricers (models that run the computations), accelerators, automatic completion fields etc.
- Portfolio management screen (known as simulation screen) is the position keeping screen of the system, which is widely used as booking for cash products. The FO and MO (Risk) users use simulations to manage their positions. It allows users to book trades from the FDI (lighter version of the e-Tradepad useful for high volume trading), monitor their position and risk thanks to the simulation viewer, and query trades using the daily blotter and queries. The simulation screen can be accessed directly through the e-Tradepad.
- Trade query is most often used by BO end users to search for trades. The trade query screen can also be accessed directly through the e-Tradepad.



Write on the white board:

Screens:	Users
1) e-Tradepad	FO
2) Simulation	FO/MO
3) Trade Query	BO

Instructions for Demo 1-A:

This is a typical demo for an IRD desk, whereby each user will have two open sessions: the e-Tradepad for structuring, pricing and trade booking, and the simulation screen to monitor position and risk. The idea is to highlight the fact that with an appropriate configuration it is very easy and quick to book a trade. No time loss here.

Activity	Content
Insert an IRS trade in e-Tradepad	Login as MUREXFO/MUREXFO Choose an FO group, such as "FO_FXMM_TR" Open the e-Tradepad using the search screen tool at the top right Insert an IRS Vanilla trade. The product is saved as favorites. (in the Default Notepad "Multi", the IRS leg should be already saved and available – you can highlight the fact that pricings are already saved which avoids pricing reconfiguration and inserting legs).
Generator	Mandatory: Select Generator USD LIBOR A 6M Optional: Open the USD LIBOR A 6M generator screen in order to customise the generator rate convention, open the payment flows in the leg in e-Tradepad to check that the change has applied
Rate Curve	Open the rate curve screen Change the estimation curve (USD LIBOR 6M → USD STD)
Mandatory Fields	Portfolio: by default, it is MX_PTF Counterparty: select any counterparty
Simulation	Bottom button: "Risk" Check the impact (risk) of the pricing before the booking. The layout should be configured so that the simulation opens with

	the same layout as the simulation
New Trade Ticket with e-Tradepad template	Bottom Button: "Book" Press "OK" (or "Ctrl+Enter") to book trade Check booking confirmation with contract number
Blotter	The blotter is already displayed by default Identify the trade in the Trade blotter screen
Trade query	Query the trade by its financial contract ID Open the trade ticket Identify the version of the trade Show the P&L of the trade
Perform event	Toolbar (from the blotter): Operations/ Cancel & Reissue Change the nominal amount in the generated trade
Trade Query after event	Query the trade using the FC ID Show the modified nominal and the new P&L of the trade

Instructions for Demo 1-B:

The idea in this section is to demonstrate that trade insertion could also be conducted directly from the portfolio management screen, and that the aforementioned blotter is also directly available in the portfolio management screen. Note that the FDI is quite adapted to desks which have a high volume trading activity. So this second demo would make more sense for clients who have a flow based business such as Secfin, FX, MM and vanilla equity.

Activity	Content
Insert an IRD LN_BR trade from FDI	Login as MUREXFO/MUREXFO group FO Load a simulation by position on portfolio "FXMM_PAR" Make sure that simulation screen opens by default with the loan FDI and blotter activated Book a Loan deal from the FDI (upper part of the screen). Ensure you select the portfolio FXMM_PAR. You should just have to fill in the ctp, the currency (field "Instrument"), the maturity, adjust the rate and the pfolio.
Monitoring views	Check the position refreshing in the different monitoring views. Briefly speak about and show views applicable for the type of trade you have inserted
Blotter	Show that deal displays in the daily blotter
Trade query	Say that we are able to add queries in the bottom part of the screen in order to search for trades inserted in the past
Perform event	Perform a cancel and reissue event on the loan directly from the daily blotter (right click on the transaction> Operations> C&R)

Now you can begin the demo 2 (**SLOW MOTION**):

In the demo 2 we redo the actions in the demo 1 plus we spend some time explaining the configuration that has been done to achieve this without redoing the demo1.

However, in order to explain where the configuration has been done, it is possible to show where the configuration information can be found (navigation).

Instructions for Demo 2-A:

Activity	Content
Insert an IRS trade in e-Tradepad	Login as MUREXFO/MUREXFO Choose an FO group, such as "FO_FXMM_TR" Open the e-Tradepad using the search screen tool at the top right Insert an IRS Vanilla trade. The product is saved as favorites. (in the Default Notepad "Multi", the IRS leg should be already saved and available – you can highlight the fact that pricings are already saved which avoids pricing reconfiguration and inserting legs). → Quickly remind the Group and Desk purpose (rights and configuration) Insert an IRS: ○ Show how to insert additional legs by clicking on the "Insert" icon (explain it has been added by config, explain the buttons are customisable). ○ Choose the product IRS Vanilla from the drop down list (show how to use the search tool). Show how to set products as favorites and how to find them afterwards. ○ Show that the e-Tradepad set of fields are user-definable. Users can add fields to the leg. ○ Explain that notepads and legs can be created and saved, no need to redo everything (which is the case for the IRS in our case). ○ In the instrument field of the leg start typing the generator name: USD LIBOR Click on tab (show how to quickly access the wanted product by typing the first letter then tab). Choose the USD LIBOR A 6M generator by double clicking on it. Explain the list of "preferred products" (Users can have a different "short list" of products). ○ Show everyone how the leg is populated with default values for the fields. Hit space bar on the Template details field to open the USD LIBOR A 6M generator screen.
Generator	○ Open the USD LIBOR A 6M generator. ○ Explain the logic of "generator": A generator is a template that stores the static data used for each kind of trade. The generator accelerates the booking of a trade. ○ Explain the following parts of the generator: 1. Generator naming convention (MXpress): E.g. USD LIBOR A 6M: USD: Currency LIBOR: Market (of the index) A: Frequency of the fixed leg (can be A, S, Q or M) 6M: Tenor of the floating index

	2. Scheduling 3. Rate conventions ○ You must also explain the concept of a default generator that is used for quick booking. ○ Mention that wherever allowed, users are able to modify the generator chosen (the template) at the pricing level for the current deal insertion. The changes will not be kept at the global level for that generator. Add the field “Generator” to the layout to show if the generator has been customised or not. For now, no customisation has been done so this field should indicate “Template”. Customise the generator for the trade. Change the rate convention for example. Now show the field Generator should indicate “Customized”. ○ Show also the customised payment flows (open them by click on the “Flows” field in e-Tradepad). ○ Point out the dates and flow amounts (show how to customise the date, payment and rates).
Rate Curve	Next, show the rate curve (market data). ○ Access the rate curves from the e-Tradepad fields “Discount curve” or “Estimation curve”. Hit space bar to open a rate curve. ○ Explain that these rate curves are assigned to this trade by default using the rules in the rate curve assignment. ○ Explain that you can also open the rate curves from menu in e-Tradepad: Navigation>Market Data> Rates> Revaluation rate curves. ○ Explain that you can also open the rate curves from menu Market data/ Used market data and that you can run scenarios on market data from here. Also show there is an icon for this menu “Used MD”. ○ Insist on the fact that all the related market data and static data can be accessed from the e-Tradepad using the menus. Users do not have to exit this screen. ○ Show how to change the rate curve assigned to this leg by choosing another USD curve. ○ Emphasise that Historical data used at booking are not stored at the deal level. ○ Open the Historical Data screen by accessing it from the Navigation menu. ○ Insert a maturity for the trade. ○ Modify the fixed rate if you want. Explain MX by default calculates the fixed rate of the transaction given NPV=0.
Pricing loaded in simulation	Explain that even if the trade is not booked it is possible to check the impact once booked on the simulation (detailed simulation aka simulation by trade). ○ Click on the “Risk” icon to load the trade in (detailed) simulation. ○ Explain that the layout in the simulation viewer can be defined and changed at the notepad level. ○ Explain that the views in the viewer are entirely customisable based on the data dictionary (if not explained elsewhere). ○ Explain that the view that we see in the simulation screen is an MXpress view. ○ Show the layout, how it consists the different views.

	○ Say that the clients can configure their own views.
Mandatory Fields and trade booking	Next, insert other mandatory fields for trade booking, without which the trade cannot be inserted into the DB (which of course can be overwritten, but not advised). ○ Insert a portfolio (MX_PTF.) ○ Insert a counterparty. ○ Explain that clients can have automated completion rules to fill in the portfolio name or counterparty name, based on some criteria. E.g: LIBOR 6M, maturity >10yrs, then fill in Portfolio A and Counterparty = NewBank. Other fields can also be set as mandatory or auto-completed. ○ Clients can also use completion rules to block trade insertion if certain business rules are not respected (e.g.: maturity >15 years etc). ○ Explain that these rules are defined in pre-trade in the processing template under a configuration session. No need to show where it is configured (next trainings). ○ Say that they can either: 1. Book the trade: By clicking on the green “Book” icon. 2. Generate the trade: By right clicking on the leg and selecting “Book trade”. 3. Capture the trade (no ticket) – not always available, please check your environment first. ○ Explain that the difference is the first method books the trade directly in the DB without generating the trade ticket first. ○ Generate the trade. ○ The new trade ticket screen will appear.
(Optional) New Trade Ticket with e-Tradepad template	Explain the new trade ticket screen: ○ The trade ticket is a representation of the trade that has just been entered by the user. It is broken in three parts: 1. The top and bottom banners: Display all the non-financial information. 2. The middle portion: Resembles the e-Tradepad pricing template and contains all the financial data. The display cannot be modified. ○ Next you will zoom in on the top banner. Point out the contract number and the component number. Note the FC ID and the FCC ID. Explain the difference between both IDs. Note down the FC ID. Say that we will use this ID to query the trade afterwards. ○ Click “Save” or “Ctrl + Enter” to complete the insertion. ○ The trade is now booked. Murex will confirm the contract number.
Blotter	○ Explain how to access the blotter if it is not displayed by default on the e-Tradepad: from the Folders list, go to Tools> Daily ○ The trade now appears in the deal blotter. Point it out. ○ Say that the view in the trade blotter screen is user definable. ○ Click on the “Edit view” design button to open the Data Dictionary screen. ○ Browse the different fields which you can add to the view.

Trade query	○ Explain there are several way to access the trade query from the e-Tradepad screen: 1. Go to the menu: Processing/ Query/ Trade Query 2. From the Folders list, go to Batches> Trade query (it may be already open in the e-Tradepad) ○ Explain briefly that this trade query screen allows the user to select many fields which act as filters to narrow down the search. For our demo, we will perform a trade query on portfolio "MX_PTF". ○ Locate our trade concerned and open the trade ticket. Say aloud "press spacebar". ○ Mention the look of the trade ticket. It is what the trainees have seen when trade was booked. ○ Locate the version number (should be v1) ○ Go to View/ P&L – note the P&L value. ○ Close the trade ticket ○ Explain you can access the trade ticket as well as its P&L directly from the blotter (spacebar to open the ticket ; right click> P&L or "Alt+P" to get the P&L contribution of this deal).
Perform event	○ Mention that there are two ways of performing an event. 1) Toolbar: Operation 2) Right click on trade: Operation Both can be accessed from the trade query or the blotter. ○ Perform a "Cancel and Reissue" on the trade. ○ Explain the events window: - Tab: Target Contract/Package : v1 - Tab: Generated Contract/Packages : v2 Question: Ask the trainees in which tab they should define a new nominal amount. (Answer: the Generated Contract/Package) ○ Change the nominal to \$50 million ○ Save the trade
Trade Query after event	Question: Ask the trainees what should be the trade number (financial contract ID) of our trade now? ○ Get a show of hands for the same ID and one for ID + 1. Explain the answer if necessary. ○ Exit the trade query results screen. Perform the trade query again on portfolio "MX_PTF". ○ Show the results. Exit. ○ Query again: This time query using the FC ID and check the box "All versions". ○ Open the trade ticket with new version. Question: Ask the trainees how to open the trade ticket (Answer: press spacebar) ○ Check the new nominal amount. Open the payment flows. ○ Exit by pressing on ESC button.

Simulation	You have already explained the simulation when loaded from the e-Tradepad, only insist here on the most important aspects and on the different approach: global portfolio, risk monitoring and not trade impact. ○ Go to the menu: Simulation/ Simulation Detailed. ○ Load the portfolio “MX_PTF”. ○ Explain that the screen is populated with “Live” deals and is connected to market data in Real-time mode. Select the view “PL breakdown”. ○ (Optional) Explain that views are user-definable. You click on the edit the view to access the Data Dictionary and add fields such as “Trade number”. ○ (Optional) Access the “Curve risk” tab to show the bottom view “Hedge Basket EUR”. Check if they are pre-configured. ○ Explain that the trade number is relevant in a detailed simulation. It enables the users to track any issues or incoherence and zoom in on specific trades. ○ Detailed simulation is low in performance in the large-flows business (e.g.: FXMM) though. ○ Views can be exported in report batch. In which case, trade number and counterpart name become relevant. (You do not need to show how to export the batch of trades for a report. ○ Explain that market data and static data are also accessible from the simulation screen using the navigation menu or the View/ Popup market data menu. Explain that from the View/ Popup market data menu one can run scenarios on the market data.
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Instructions for Demo 2-B:

Activity	Content
Insert an IRD LN_BR trade from FDI	Login as MUREXFO/MUREXFO group FO Load a simulation by position on portfolio “FXMM_PAR” Make sure that simulation screen opens by default with the loan FDI and blotter activated. From the loan FDI template: ○ Speak about the FDI, basically how to show/hide fields. ○ Briefly speak also about compliance and completion rules (if not yet done) ○ Speak about the limits button ○ Show how to add as many FDI patterns as we want ○ Show how to open the deal ticket before booking, by clicking on the magnifier button Book a Loan deal from the FDI (upper part of the screen). Ensure you select the portfolio FXMM_PAR. You should just have to fill in the ctp, the currency (field “Instrument”), the maturity, adjust the rate and the pfolio. Check the position refreshing in the different monitoring views.
Monitoring views	Briefly introduce the MX viewer (if not yet done): ○ Quick demo of the capabilities of the viewer from an end-user and config perspective would be good to present. ○ Speak about the real-time aspect of the monitoring screens.

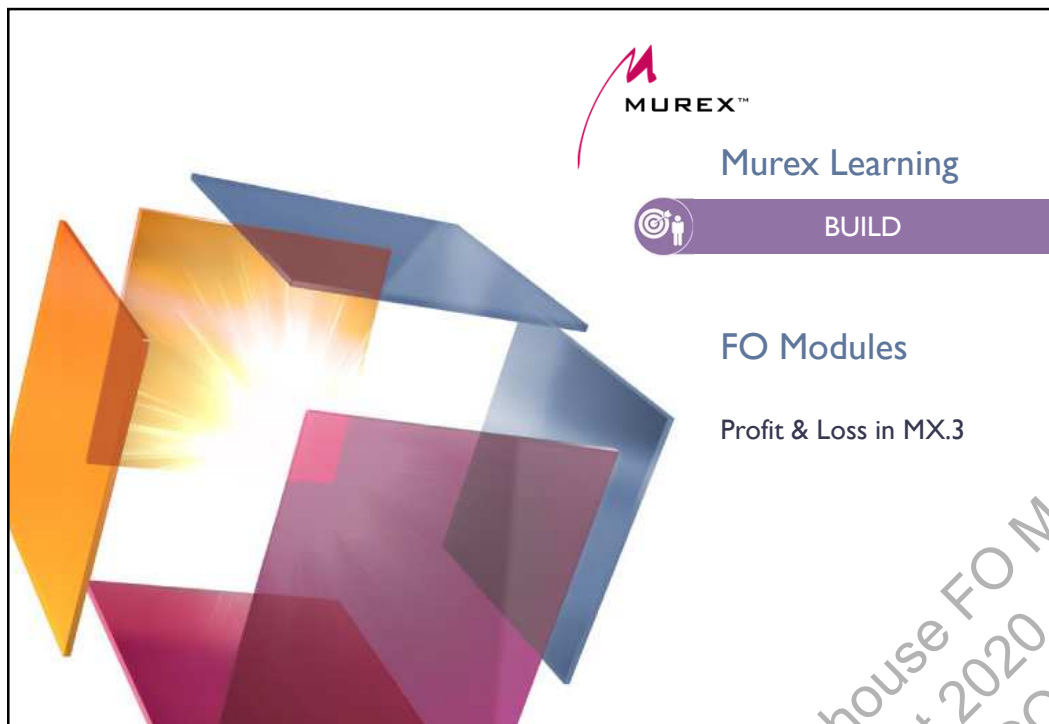
Blotter	Show that the deal is displayed in the daily blotter once booked. Speak about the ability of marking newly inserted deals in a specific color.
Trade query	Show how to add a simple query near the daily tab, allowing to search for previously inserted trades.

After the demo, please recap the steps to book a trade in the e-Tradepad. You DO NOT need to ask for details. The goal is for the trainees to remember the steps to book, they will then “see” the things they will need to learn throughout the different training.

Steps:

Insert an IRS trade in e-Tradepad
○ Generator (static data) – can be customised at the level of the trade
○ Rate curve (market data)
○ Mandatory fields
○ Risk preview in simulation
○ Select Book trade
○ Access trade via blotter or trade query
○ If event to perform, it can be done via trade query or blotter

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 **Learning Objectives**
Profit & Loss in MX.3

At the end of this course, you will be able to:

- ✓ Explain the two P&L calculation methods
- ✓ Differentiate between Accounting and Economic P&L
- ✓ Locate the P&L of a given trade in the application and justify the calculations

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Agenda

P&L Calculation Methods

- Accounting P&L
- Economic P&L

Finding P&L in the System

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3



P&L Calculation Methods

Accounting P&L: Components & breakdown

The accounting P&L can be broken down in two ways:

Accounting P&L	=	Cash	+	Market Value
		OR		
	=	Capital Gain	+	Revenue

The choice of accounting P&L breakdown depends on the asset class and user preference.

MX is able to compute and display both calculation methods for all asset classes.

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 **P&L Calculation Methods**
Accounting P&L: Components & breakdown

$$\text{Accounting P\&L} = \text{Cash} + \text{Market Value}$$

Component	Definition	Example
Market Value	Market Value = Today's trade value	Option's MV = Premium value today Interest rate Swap's MV = All estimated/known future proceeds discounted to currency spot date
Cash	All past payments (from Capital or Revenue source) & future payments <u>not</u> included in the market value (e.g. premium for an option).	Cash of an equity trade = Initial price + All past dividends Dividend of an equity swap when ex-date is in the past and payment still in future.

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 **P&L Calculation Methods**
Accounting P&L: Components & breakdown

$$\text{Accounting P\&L} = \text{Capital Gain} + \text{Revenue}$$

Component	Definition	Example
Capital Gain	Capital Gain = Today's value – Initial value	Option's CG = Premium value today – Premium paid Bonds = Current market price – Initial (purchase/sell) price
Revenue	Revenue = Past returns + Upcoming returns	Stock dividends Bond coupons

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Practice I

Practice I: Which accounting formula is used to calculate the Accounting P&L of the following asset classes?

Accounting P&L = Cash + Market Value
OR
Accounting P&L = Capital Gain + Revenue

FX (spot)	Cash + Market Value
FI (Bond)	Capital Gain + Revenue
Equity Cash	
IR (Interest rates)	
Sec Fin	
CRD (Credit)	
COM (Commodities)	

Solution

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Beyond accounting P&L

Time Value of money:

**\$1 today is not equal to \$1 in one month
or \$1 a month ago!**

The time value of money is the value of money figuring in a given amount of interest earned over a given amount of time.

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Beyond accounting P&L

$\$1 \text{ a month ago} + \$1 \text{ today} + \$1 \text{ in one month} \neq \3

Amounts at various dates should be discounted or capitalized before being totaled and expressed as of current date (Today).

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MUREX™ | **P&L Calculation Methods**
Economic P&L: Components & breakdown

Economic P&L represents the P&L financed and discounted to Today

Economic P&L = **Accounting P&L** + **Financing** + **PV Effect**

Component	Value
Accounting P&L	Cash + Market Value Capital Gain + Revenue
Financing	Cost of carry of past cash Capitalized past cash – Net past cash
PV Effect (Present Value Effect)	Discounted future cash – Net future cash + Discounted MV – Net MV

 Note 1: MV of trade includes sum of all future flows not part of the deal's future cash, discounted to instrument spot date.
Note 2: Discounted MV = MV discounted from the instrument (currency/security) spot date to Today.

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MUREX™ | **P&L Calculation Methods**
Economic P&L: Practice 2

 Calculate the Accounting and Economic P&L of a US Treasury bond that was bought yesterday and settles today.

Buy quantity	3,000,000	* If the bond is sold today, settlement would occur tomorrow (delay of 1 business day is equivalent to the instrument spot day).
Buy price @ buy date	102.0354%	
Accrued coupon (revenue) @ buy date*	0.368%	
Sell price @ spot date	103%	** Discount factor is given as between today and the spot date of USD cash amount.
Accrued coupon (revenue) @ spot date	0.377%	
Discount factor**	0.999990277872299	

Question: Which accounting P&L breakdown would make more sense to use?

Accounting P&L = Cash + Market Value

OR

Accounting P&L = Capital Gain + Revenue

 **Solution**

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MUREX™ | **P&L Calculation Methods**
Economic P&L: Practice 3

 To recap, which formulas calculate the Economic P&L?

A P&L @ T = Accounting P&L + PV Effect + Capital Gain

B P&L @ T = Past Cash + Financing + PV Effect + Future Cash

C P&L @ T = Past Cash + Financing + PV Effect + Future Cash + Market Value

D P&L @ T = Accounting P&L + Economic P&L

E P&L @ T = Financing + Capital Gain + PV Effect + Revenue

 **Solution**

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Agenda

P&L Calculation Methods

Finding P&L in the System

- Overview
- P&L on one deal
- Simulation viewer
- P&L variation

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Finding P&L in the System

Overview

There are different ways to find the P&L in the system. This training will focus on the P&L on one deal and the P&L displayed in the Simulation screen.

P&L in the system	Typical Usage
P&L on one deal	Quick check of one particular deal – mainly for testing purposes.
Simulation	Dynamic view of P&L during trading activity.
Dynamic tables	View of P&L and trade attributes with user definable filter – mainly for testing purposes.
Reports / Extracts	P&L closing positions. P&L reconciliation.
P&L Variance Module	Comprehensive view of P&L variation between 2 consecutive days.
Accounting Module	Book keeping of daily positions.

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MUREX™ | Finding P&L in the System
P&L on one deal

System date: 25 Sep 2012
Horizon (Revaluation date): 25 Sep 2012

Deal identification: Transaction number 3456
Instrument USD LIBOR 5 6M
Buyer IR_SWAPS_US
Seller BARCLAYS CAPITAL NYK
P&L view from Buyer Side

P&L currency: Results in USD

Traditional P&L breakdown:

- Accounting P&L = Past Cash + Future Cash + Market Value
- Economic P&L = Accounting P&L + Financing + PV Effect

Zoom on Past or Future cash flows

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

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MUREX™ | Finding P&L in the System
P&L on one deal

The P&L screen of a trade offers the P&L view for all trade types:

Absolute P&L	Quick check of absolute P&L of any deal.
Is easily accessible	From many parts of the application, via View/ P&L or via a right-click/ P&L.
P&L as of past date	In the same way, you can have a look at a P&L as of a past date by choosing P&L as of past date.

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

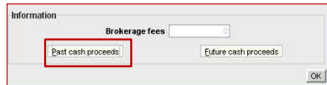
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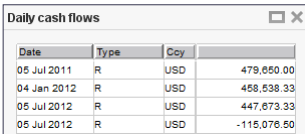
MUREX™ | Finding P&L in the System
P&L on one deal: Past cash proceeds

Past cash proceeds can be easily retrieved in two ways:

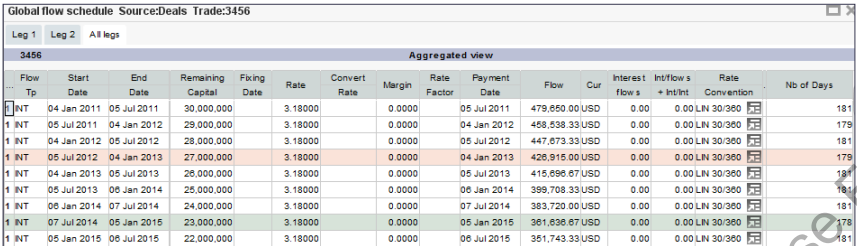
1. By clicking on **Past cash proceeds** button in P&L window



2. By checking deal ticket flows



Date	Type	Ccy	
05 Jul 2011	R	USD	479,650.00
04 Jan 2012	R	USD	458,538.33
05 Jul 2012	R	USD	447,673.33
05 Jul 2012	R	USD	-115,076.50



Flow Tp	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur	Interest flows	Int/flow + Int/Int	Rate Convention	Nb of Days
1 NT	04 Jan 2011	05 Jul 2011	30,000,000		3.18000		0.0000		05 Jul 2011	479,650.00 USD		0.00	0.00 LIN 30/360		181
1 NT	05 Jul 2011	04 Jan 2012	29,000,000		3.18000		0.0000		04 Jan 2012	458,538.33 USD		0.00	0.00 LIN 30/360		179
1 NT	04 Jan 2012	05 Jul 2012	28,000,000		3.18000		0.0000		05 Jul 2012	447,673.33 USD		0.00	0.00 LIN 30/360		181
1 NT	05 Jul 2012	04 Jan 2013	27,000,000		3.18000		0.0000		04 Jan 2013	426,915.00 USD		0.00	0.00 LIN 30/360		179
1 NT	04 Jan 2013	05 Jul 2013	26,000,000		3.18000		0.0000		05 Jul 2013	415,696.67 USD		0.00	0.00 LIN 30/360		181
1 NT	05 Jul 2013	06 Jan 2014	25,000,000		3.18000		0.0000		06 Jan 2014	399,708.33 USD		0.00	0.00 LIN 30/360		181
1 NT	06 Jan 2014	07 Jul 2014	24,000,000		3.18000		0.0000		07 Jul 2014	383,720.00 USD		0.00	0.00 LIN 30/360		181
1 NT	07 Jul 2014	05 Jan 2015	23,000,000		3.18000		0.0000		05 Jan 2015	361,836.67 USD		0.00	0.00 LIN 30/360		178
1 NT	05 Jan 2015	06 Jul 2015	22,000,000		3.18000		0.0000		06 Jul 2015	351,743.33 USD		0.00	0.00 LIN 30/360		181

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

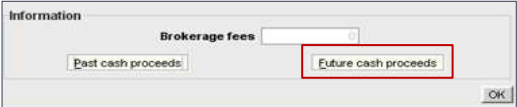
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MUREX™ | Finding P&L in the System
P&L on one deal: Future cash proceeds

Future cash proceeds are future flows received after the system date and not included in the market value:

- ✓ Mainly premiums, additional customized flows (of type premium only) and settlement payments.
- ✓ Some future flows might be implicitly accounted for in the market value. Hence, they are not considered as future cash proceeds.



 For **bond options**: future cash proceeds = premium paid, hence the FC proceeds represent by definition the cash amount invested to enter into the transaction.

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

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MUREX™ | Finding P&L in the System
P&L on one deal: Market value

Market value is either the market quotation or the sum of all futures flows discounted to spot date.

It can be quickly found in a pricing screen or by loading the trade in pricing.

It is also possible to compute manually the flows discounted to spot date using the interpolator to retrieve the discount factors.

Results in USD	
Theoretical	Closing
Past cash proceeds	1,270,785
Future cash proceeds	0
Market value	2,754,130
Accounting P&L	4,024,915
Financing	9,888
Present value effect	-83

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

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MUREX™ | Finding P&L in the System
P&L on one deal: Financing

Financing applies to past cash flows. It is also called “cost of carry”.

It assumes that all past flows have been reinvested every day at the overnight financing rate.

Results in USD	
Theoretical	Closing
Past cash proceeds	1,270,785
Future cash proceeds	0
Market value	2,754,130
Accounting P&L	4,024,915
Financing	9,888
Present value effect	-83
Economic P&L	4,034,719

Past cash flow Falling at t_i

$$CF_t = I + r(O/N)_t * I / 360$$

Today

$$CF_{\text{yesterday}} = I + r(O/N)_{\text{yesterday}} * I / 360$$

To ease the computations, MX provides “Lambda” which is, for a given date, the product of all the capitalization factors between t_i and today:

$$\text{Lambda}_i = CF_t * CF_{t+1} * \dots * CF_{\text{yesterday}}$$

The convention used in the computation is inherited from the short term rate convention of the currency.

End-user session: Configuration/ Forex/ Currencies

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

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MUREX™ | Finding P&L in the System
P&L on one deal: Financing

For each past flow, we can retrieve the lambda coefficient which will give the financed flow.
Hence financing effect is:

$$\text{Financing} = \text{Past flow amount} \times (\text{Lambda} - 1)$$

DATE	RATE	LAMBDA
20 Sep 2012	0.0100000	1.0000000
21 Sep 2012	0.0100000	1.0000000
22 Sep 2012	0.0000000	1.0000003
23 Sep 2012	0.0000000	1.0000003
24 Sep 2012	0.0100000	1.0000003
25 Sep 2012	0.0000000	1.0000000

End-user session: Historical data/ Rates/ O/N financing rates

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MUREX™ | Finding P&L in the System
P&L on one deal: Present value effect

The PV effect applies to future flows (future cash proceeds and MV).

It looks at the difference between these future flows and their discounted values (discounted from value date or spot date to today).

The interpolator can be used to get the discount factor between today and spot on the assigned rate curve.

$$\text{PV Effect} = \text{Future Cash} * (DF_{\text{Value date}} - 1) + \text{MV} * (DF_{\text{Spot}} - 1)$$

Results in USD	
Past cash proceeds	1,270,785
Future cash proceeds	0
Market value	2,754,130
Accounting P&L	4,024,915
Financing	9,886
Present value effect	-93
Economic P&L	4,034,719

End-user session: Processing/ Trades/ Trade query – Select a deal –View/ P&L

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MUREX™ | Finding P&L in the System
P&L on one deal: Practice 4

Practice 4: Calculate the Accounting and Economic P&L of a bond with a coupon payment.

Nominal	1,500,000
Start date (settlement date)	14 Dec 2011
Initial price	102.0354%
Accrual @ settlement date	0.366%
Spot date	19 Jul 2014
Closing price	103.0143%
Accrual @ end date	0.106%
Coupon	9,500*
Discount factor (spot → today)	0.999875009
Lambda (coupon → today)	1.000631585
Lambda (settlement → today)	1.002098327

* Fallen on the 20 May 2014 (past)

Question: Which accounting P&L breakdown would make more sense to use?

Accounting P&L = Cash + Market Value

OR

Accounting P&L = Capital Gain + Revenue

Solution

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MUREX™ | Finding P&L in the System
Simulation viewer

MX Viewer allows users to retrieve their P&L figures in simulation, taking all the benefits from the real time connection.

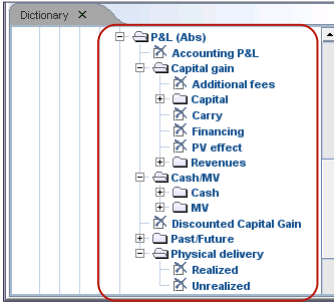
End-user session: Simulation

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MUREX™ | Finding P&L in the System
Simulation viewer: Data dictionary

The Simulation Viewer Dictionary contains fields displaying various P&L components.



Drag and drop different levels from dictionary tree according to granularity of P&L results required.

Economic P&L	Risk Engine.Results.P&L.P&L (Abs)
Market Value	Risk Engine.Results.P&L.P&L (Abs).Cash/MV.MV
Cash	Risk Engine.Results.P&L.P&L (Abs).Cash/MV.Cash
Accounting P&L	Risk Engine.Results.P&L.P&L (Abs).Accounting P&L

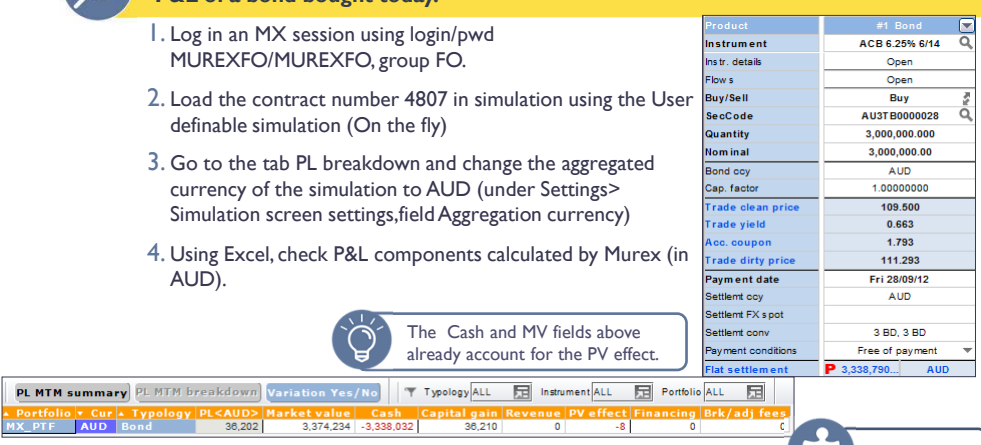
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MUREX™ | Finding P&L in the System
Simulation viewer: Practice 5

Calculate and verify the computation in MX.3 of the Accounting and Economic P&L of a bond bought today.

- Log in an MX session using login/pwd MUREXFO/MUREXFO, group FO.
- Load the contract number 4807 in simulation using the User definable simulation (On the fly)
- Go to the tab PL breakdown and change the aggregated currency of the simulation to AUD (under Settings> Simulation screen settings, field Aggregation currency)
- Using Excel, check P&L components calculated by Murex (in AUD).



The Cash and MV fields above already account for the PV effect.

Solution

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MUREX™ | Finding P&L in the System
Simulation viewer: Practice 6

 Calculate and verify the computation in MX.3 of the Accounting and Economic P&L of a back dated loan/deposit trade.

1. Use the same session as in the previous practice.
2. Load the contract number 4808 in a user definable simulation (like in previous practice)
3. Go to the tab PL breakdown and ensure the aggregation currency is USD.
4. Using Excel, check P&L components calculated by Murex (slight difference due to rounding).

 The Cash and MV fields above already account for the PV effect.

Product	#1 Deposit
Nature	Fixed rate
Instrument	USD
Template details	Open
Payout	Borrow
Nominal	86,000.00 USD
Start period	spot Mon 24/09/12
Maturity period	1Y Tue 24/09/13
Adjusted date	Tue 24/09/13
Coupons payment freq.	Frequency ratio
Rate convention	LIN ACT/360
Rate/Margin	0.5700
Repayment	86,497
Initial capital	Exchange
Intermediate capital	Exchange
Final capital	Exchange
Flows	Open
Amortizing	None

PL MTM summary	PL MTM breakdown	Variation Yes/No	Typology ALL	Instrument ALL	Portfolio ALL						
Portfolio	Cur	Typology	PL<USD>	Market value	Cash	Capital gain	Revenue	PV effect	Financing	Brk/adj fees	
HX_PTF	USD	Depo		-142	-86,143	86,000	-145	0	3	1	0

 **Solution**

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MUREX™ | Finding P&L in the System
P&L variation

The simulation data from the data dictionary provides only information as of the horizon date (updated in real time).
To display values as of different dates, it is possible to display their stored values previously computed at different dates through the Datamart module.
It is also possible to calculate the variation of these displayed values between two dates within the simulation viewer.

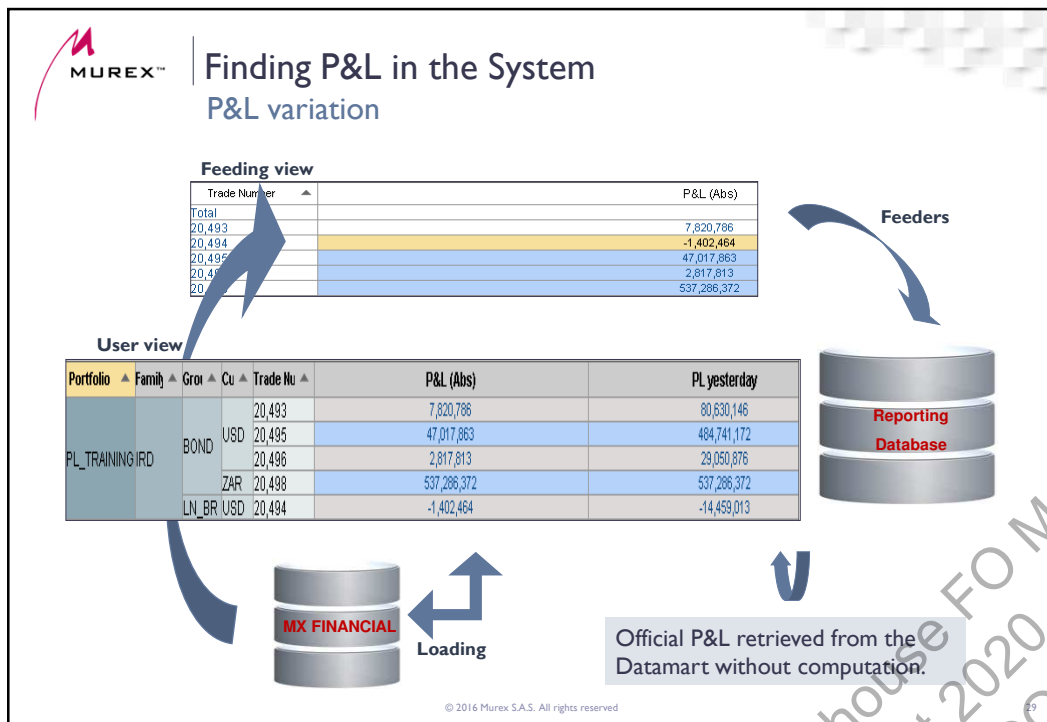
The business is generally interested by the **P&L performance** also known as the **daily P&L**. It is the difference between Today's P&L and Yesterday's P&L.

For example: Let's assume the P&L for the previous business day has been stored (i.e. closing P&L based on official rates).
This stored yesterday P&L can be displayed in the simulation viewer together with today's P&L.
The daily P&L can also be computed directly within the simulation viewer.

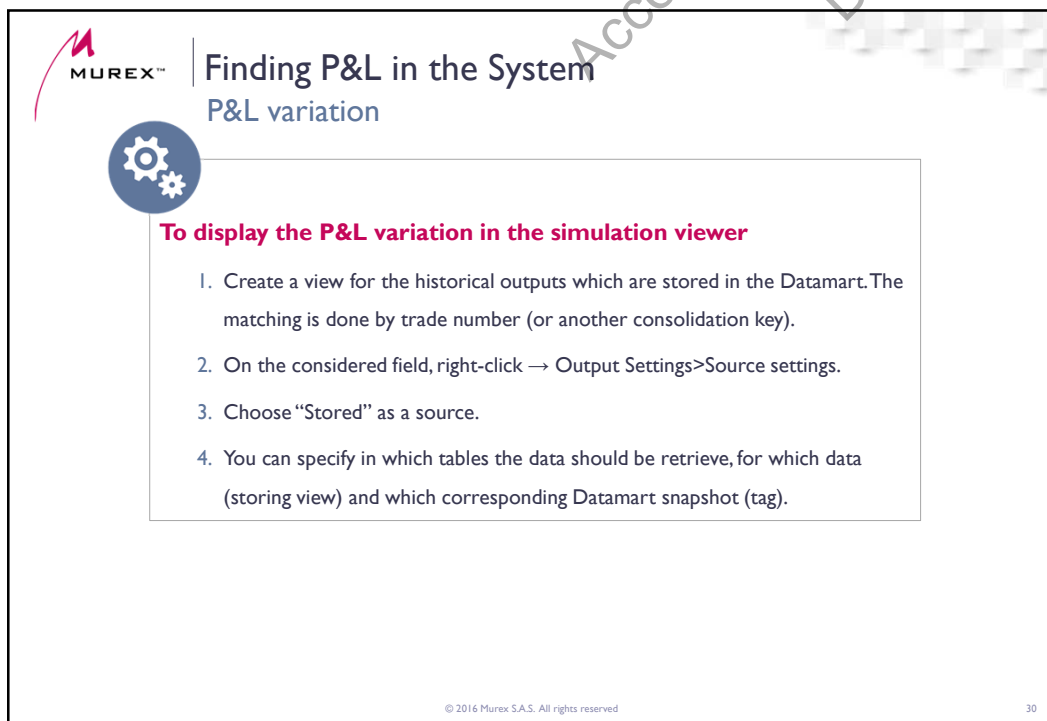
 P&L figures are stored at the close of every business day (within EOD batch).

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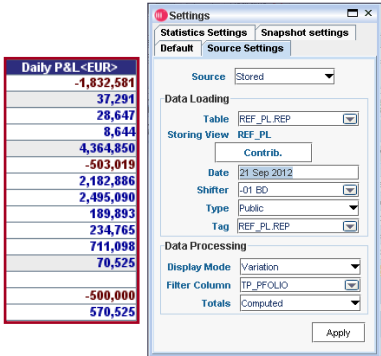


Finding P&L in the System

P&L variation

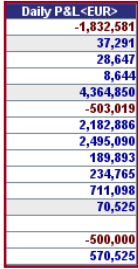


Displaying the P&L Variation in the Simulation viewer:



Output Settings > Source Settings:

Source	Stored
Table	Name of table within which closing P&L results stored.
Date	Populated automatically based on shifter rule.
Shifter	- IBD for daily P&L.
Type	Public and Tag determine more precisely which stored dataset to recover.
Display Mode	Variation for daily P&L.



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EXERCISE SOLUTIONS

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Practice 1

Solution 1: Which accounting formula is used to calculate the Accounting P&L of the following asset classes?

Accounting P&L = Cash + Market Value
OR
Accounting P&L = Capital Gain + Revenue

FX (spot)	Cash + Market Value
FI (Bond)	Capital Gain + Revenue
Equity Cash	Capital Gain + Revenue
IR (Interest rates)	Cash + Market Value
Sec Fin	Cash + Market Value
CRD (Credit)	Cash + Market Value
COM (Commodities)	Cash + Market Value

Exercise

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Practice 2

Solution 2: Calculate the Accounting and Economic P&L of a US Treasury bond that bought yesterday and settles today.

I. Accounting P&L:

Accounting P&L = Capital Gain + Revenue

Component	Equals	Value
Capital Gain	$(-buy\ Price + sell\ Price) * Quantity = (-102.0354\% + 103\%) * 3,000,000$	USD 28,938
Revenue	$(-buy\ Rev + sell\ Rev) * Quantity = (-0.368\% + 0.377\%) * 3,000,000$	USD 270

Hence:
Accounting P&L = 28,938 + 270 = USD 29,208.00

Exercise

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MUREX™ | **P&L Calculation Methods**
Accounting P&L: Practice 2

Solution 2: Calculate the Accounting and Economic P&L of a US Treasury bond that bought yesterday and settles today.

2. Economic P&L:

$$\text{Economic P\&L} = \text{Accounting P\&L} + \text{Financing} + \text{PV Effect}$$

Component	Equals	Value
Financing of Past Cash	0 (since bond settles today)	0
MV of Bond	(Price @ spot date + Accrual @ spot date) * Quantity = (103% + 0.377%) * 3,000,000	USD 3,101,310
PVE	Discounted MV @ spot date – MV = 3,101,310 * DF (spot → today) – 3,101,310 = 3,101,310 * 0.999990277872299 – 3,101,310	USD -30.15

Hence:
Economic P&L = 29,208 + 0 – 30.15 = USD 29,177.85

 **Exercise**

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MUREX™ | **P&L Calculation Methods**
Economic P&L: Practice 3

Solution 3: To recap, which formulas calculate the Economic P&L?

C P&L @ T = Past Cash + Financing + PV Effect + Future Cash + Market Value

E P&L @ T = Financing + Capital Gain + PV Effect + Revenue

 **Exercise**

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MUREX™ | Finding P&L in the System
P&L on one deal: Practice 4

Solution 4: Calculate the Accounting and Economic P&L of a bond with a coupon payment.

1. Accounting P&L:

$$\text{Accounting P\&L} = \text{Capital Gain} + \text{Revenue}$$

Component	Equals	Value
Capital Gain	Quantity * (Closing price – Initial price) = 1,500,000 * (103.0143% – 102.0354%)	USD 14,683.50
Revenue	Quantity * (Accrual @ end – Accrual @ settlement) + Coupon = 1,500,000 * (0.106% – 0.366%) + 9,500	USD 5,600.00

Hence:
Accounting P&L = 14,683.50 + 5,600 = USD 20,283.50

 **Exercise**

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MUREX™ | Finding P&L in the System
P&L on one deal: Practice 4

Solution 4: Calculate the Accounting and Economic P&L of a bond with a coupon payment.

2. Economic P&L:

$$\text{Economic P\&L} = \text{Accounting P\&L} + \text{Financing} + \text{PV Effect}$$

Component	Equals	Value
Financing	Capitalised Past flows – Past flows = where Past flows = Coupon – (Initial price + Accrual) * Quantity → (9,500 * Lambda (coupon → today) – (102.0354% + 0.366%) * Lambda (settlement → today) * 1,500,000) – (9,500 – (102.0354% + 0.366%) * 1,500,000)	USD -3,217.07
MV	Quantity * (Closing price + Accrual @ spot date) = 1,500,000 * (103.0143% + 0.106%)	USD 1,546,804.50
PVE	Discounted MV @ spot date – MV = 1,546,804.50 * DF (spot → today) – 1,546,804.50	USD -193.34

Hence:
Economic P&L = 20,283.50 – 193.34 – 3,217.07 = USD 16,873.09

 **Exercise**

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MUREX™ | Finding P&L in the System
Simulation viewer: Practice 5

Solution 5: Calculate & verify the computation in MX.3 of the Accounting and Economic P&L of a bond bought today.

Accounting P&L: Accounting P&L = Capital Gain + Revenue

Component	Equals	Value
Capital Gain	$(-\text{buy price} + \text{Market price}) * \text{Quantity} = (-109.50\% + 110.7070\%) * 3,000,000$	AUD 36,210
Revenue	No coupon	AUD 0

Economic P&L: Economic P&L = Accounting P&L + Financing + PV Effect

Component	Equals	Value
Financing	0	AUD 0
MV	$\text{Quantity} * (\text{Market price} + \text{Accrual}) = 3,000,000 * (110.7070\% + 1.793\%)$	AUD 3,375,000.00
Future Cash	$\text{Quantity} * (-\text{buy price} - \text{Accrual}) = 3,000,000 * (-109.50\% - 1.793\%)$	AUD -3,338,790.00
PVE	$(\text{Discounted Future Cash @ settlement date} - \text{Future Cash}) + (\text{Discounted MV @ spot date} - \text{MV}) = (-3,338,790.00 * \text{DF}(\text{settlement} \rightarrow \text{today}) - (-3,338,790.00)) + (3,375,000.00 * \text{DF}(\text{spot} \rightarrow \text{today}) - 3,375,000.00)$	AUD -8.22

Hence: Economic P&L = 36,210.00 – 8.22 = AUD 36,201.78

Exercise

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MUREX™ | Finding P&L in the System
Simulation viewer: Practice 6

Solution 6: Calculate and verify the computation in MX.3 of the Accounting and Economic P&L of a back dated loan/deposit trade.

Accounting P&L: Accounting P&L = Cash + Market Value

Remember:
Cash = All past payments (from Capital or Revenue source) & future payments not included in the market value.
MV = All estimated/known future proceeds discounted to currency spot date.

	Equals	Value
Cash	Borrowed amount received	USD 86,000
MV	$\text{Borrowed amount paid back} + \text{Interest} = (-86,000 - 497.01) * \text{DF}(\text{Mat} \rightarrow \text{Spot date}) = (-86,000 - 497.01) * 0.995935067377494$	USD -86,145.40

Hence:
Accounting P&L = Cash + MV = 86,000 – 86,145.40 = USD -145.40

Exercise

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MUREX™ | Finding P&L in the System
Simulation viewer: Practice 6

Solution 6: Calculate and verify the computation in MX.3 of the Accounting and Economic P&L of a back dated loan/deposit trade.

Economic P&L:

$$\text{Economic P\&L} = \text{Accounting P\&L} + \text{Past cash Financing} + \text{PV Effect}$$

	Equals	Value
Financing of Past Cash	Capitalised Past Cash – Past Cash = $86,000 * \text{Lambda (start} \rightarrow \text{today)} - 86,000 =$ $86,000 * 1.0000075 - 86,000$	USD 0.64
PV Effect	Discounted MV @ spot date – MV = $-86,145.40 * \text{DF (spot} \rightarrow \text{today)} - (-86,145.40) =$ $-86,145.40 * 0.999970020905769 + 86,145.40$	USD 2.58

Hence:
 Economic P&L = $-145.40 + 0.64 + 2.58 = \text{USD } -142.18$

 **Exercise**

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Q&A

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Thank You!

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Accenture: In-house L2 Modules
 August 2020
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Profit & Loss in MX.3

Practice 2: P&L of a bond	
Buy quantity	3,000,000.00
Buy price @ buy date	102.0354%
Accrued coupon (revenue) @ buy date	0.368%
Sell price @ spot date	103%
Accrued coupon (revenue) @ spot date	0.377%
Discount factor	0.999990278

Accounting P&L	
Capital Gain	
Revenue	
Accounting P&L	

Economic P&L	
Accounting P&L	
Financing	
Future Cash	
Discounted Future Cash	
MV	
Discounted MV	
PV Effect	
Economic P&L	

Practice 4: P&L of a bond with coupon	
Nominal	1,500,000.00
Start date (settlement date)	12/14/2011
Initial price	102.0354%
Accrual @ settlement date	0.366%
Spot date	7/19/2014
Closing price	103.0143%
Accrual @ end date	0.106%
Coupon on 20/05/2014	9,500.00
DF (spot -> today)	0.999875009
Lambda (coupon -> today)	1.000631585
Lambda (settlement -> today)	1.002098327

Accounting P&L	
Capital Gain	
Revenue	
Accounting P&L	

Economic P&L	
Accounting P&L	
Coupon financed	
Past payment financed	
Financing	
Future Cash	
Discounted Future Cash	
MV	
Discounted MV	
PV Effect	
Economic P&L	

Practice 5: P&L of a bond bought today (in MX.3)	
Buy quantity	3,000,000.00
Buy price	109.5000%
Accrued coupon (revenue)	1.793%
Current market price	110.7070%
Discount factor (settlement -> today)	0.999773053
Today	9/25/2012
Settlement date = Spot date	9/28/2012
FX spot AUD/USD	1.0453

Accounting P&L in AUD	
Capital Gain	
Revenue	
Accounting P&L	

Economic P&L in AUD	
Accounting P&L	
Financing	
Future Cash	
Discounted Future Cash	
MV	
Discounted MV	
PV Effect	
Economic P&L	

Practice 6: P&L of a back dated loan/deposit in MX.3	
Nominal	
Start date	
Current date (Today)	
Maturity date	
Interest rate	
Rate convention	
Interest amount to pay at maturity	
USD currency spot date	
DF (maturity -> spot)	
DF (spot -> today)	
Lambda (start -> today)	

USD LIBOR 3M curve

Accounting P&L	
Past Cash	
Future Cash	
Cash	
Market Value	
Accounting P&L	

Economic P&L	
Accounting P&L	
Past Cash financed	
Financing	
Discounted Future Cash	
Discounted MV	
PV Effect	
Economic P&L	

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Profit & Loss in MX.3

Practice 2: P&L of a bond	
Buy quantity	3,000,000.00
Buy price @ buy date	102.0354%
Accrued coupon (revenue) @ buy date	0.368%
Sell price @ spot date	103%
Accrued coupon (revenue) @ spot date	0.377%
Discount factor	0.999990278

Accounting P&L	
Capital Gain	28,938.00
Revenue	270.00
Accounting P&L	29,208.00

Economic P&L	
Accounting P&L	29,208.00
Financing	-
Future Cash	-
Discounted Future Cash	-
MV	3,101,310.00
Discounted MV	3,101,279.85
PV Effect	- 30.15
Economic P&L	29,177.85

Practice 4: P&L of a bond with coupon	
Nominal	1,500,000.00
Start date (settlement date)	12/14/2011
Initial price	102.0354%
Accrual @ settlement date	0.366%
Spot date	7/19/2014
Closing price	103.0143%
Accrual @ end date	0.106%
Coupon on 20/05/2014	9,500.00
DF (spot -> today)	0.999875009
Lambda (coupon -> today)	1.000631585
Lambda (settlement -> today)	1.002098327

Accounting P&L	
Capital Gain	14,683.50
Revenue	5,600.00
Accounting P&L	20,283.50

Economic P&L	
Accounting P&L	20,283.50
Coupon financed	9,506.00
Past payment financed	- 1,539,244.07
Financing	- 3,217.07
Future Cash	-
Discounted Future Cash	-
MV	1,546,804.50
Discounted MV	1,546,611.16
PV Effect	- 193.34
Economic P&L	16,873.09

Practice 5: P&L of a bond bought today (in MX.3)	
Buy quantity	3,000,000.00
Buy price	109.5000%
Accrued coupon (revenue)	1.793%
Current market price	110.7070%
Discount factor (settlement -> today)	0.99973053
Today	9/25/2012
Settlement date = Spot date	9/28/2012
FX spot AUD/USD	1.0453

Accounting P&L	
in AUD	
Capital Gain	36,210.00
Revenue	-
Accounting P&L	36,210.00

Economic P&L	
in AUD	
Accounting P&L	36,210.00
Financing	-
Future Cash	- 3,338,790.00
Discounted Future Cash	- 3,338,032.27
MV	3,375,000.00
Discounted MV	3,374,234.05
PV Effect	- 8.22
Economic P&L	36,201.78

Practice 6: P&L of a back dated loan/deposit in MX.3	
Nominal	86,000.00
Start date	9/24/2012
Current date (Today)	9/25/2012
Maturity date	9/24/2013
Interest rate	0.570%
Rate convention	LIN ACT/360
Interest amount to pay at maturity	497.01
USD currency spot date	9/27/2012
DF (maturity -> spot)	0.995935067
DF (spot -> today)	0.999970021
Lambda (start -> today)	1.0000075

USD LIBOR 3M curve

Accounting P&L	
Past Cash	86,000.00
Future Cash	-
Cash	86,000.00
Market Value	- 86,145.40
Accounting P&L	- 145.40

Economic P&L	
Accounting P&L	- 145.40
Past Cash financed	86,000.65
Financing	0.64
Discounted Future Cash	-
Discounted MV	- 86,142.82
PV Effect	2.58
Economic P&L	- 142.18

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1

A presentation slide titled 'Learning Objectives' for 'FX Configuration'. The slide has a clean, white background with the Murex logo in the top left corner. The title 'Learning Objectives' is in a large, bold, dark blue font, with 'FX Configuration' in a smaller, dark blue font below it. The main content is a list of six learning objectives, each preceded by a blue checkmark. At the bottom of the slide, there is a small copyright notice and a page number '2'. A large, diagonal watermark across the right side of the slide reads 'Accenture: In-house FO Modules August 2020 DO NOT COPY'.

2

MUREX™ | Agenda

Currency Configuration	• Underlying definition • Default conventions • Rate conventions • Spot schedule • Underlying description
Currency Pair Definition	
FX Listed Contracts	
Settings	

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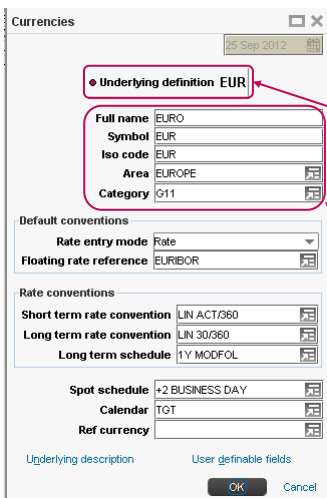
3

MUREX™ | Currency Configuration
Underlying definition

Currency = medium of exchange

Label or name of the currency. Three-letter name consistent with market practice.

Naming conventions (optional fields):
Full name
Symbol
ISO code
Area
Category

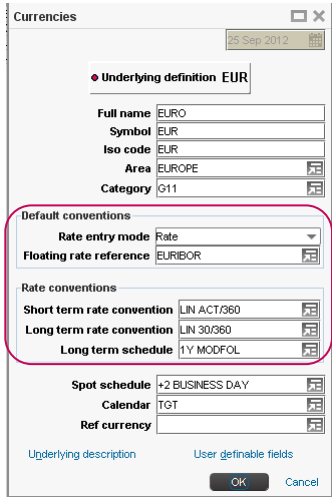


End-user session: Configuration/ Forex/ Currencies

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MUREX™ | **Currency Configuration**
Default conventions



Default conventions:

Rate entry mode	Constant swap points vs. constant interest rates, when US rates move
Floating rate reference	Default archiving group for FRA transactions
Short & Long term rate conventions	Default interest rate calculation convention for short/long term loans
Long term schedule	Default frequency of coupon flows

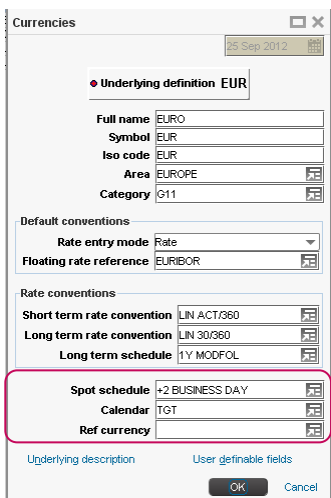
End-user session: Configuration/ Forex/ Currencies

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MUREX™ | **Currency Configuration**
Rate conventions

The currency spot schedule and calendar are used only when the product is mono-currency. These settings are default ones but can be overwritten in a rate instrument generator.

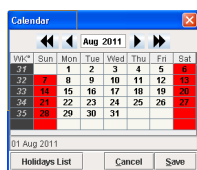


Spot schedule Defines the way to compute spot value date from a given date

EUR: Horizon → Spot (+2 o.d.)

CAD: Horizon → Spot (+1 o.d.)

Calendar Links the currency to its corresponding calendar



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MUREX™ | **Currency Configuration**
Underlying description

Precision on amount	Determines the numbers of decimals of amounts expressed in this currency
Flow rounding rule	Determines if the amount is rounded (nearest, by excess, by default)
Pricing conventions	Rate display

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MUREX™ | **Currency Configuration**
User definable fields

CLS eligible	Continuous linked settlement eligible
CLS start date	Start date

End-user session: Configuration/ Forex/ Currencies

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MUREX™ | **Currency Configuration**
Practice I

 Validate your understanding of the currency definition screen options.

1. Log on using MUREXFO/MUREXFO
2. Select the group FO.
3. Answer the questions below:

What is the rate entry mode for CHF?	
What is the calendar used for the GBP currency? What does it do?	
What is the spot schedule used for the JPY currency? What does it imply?	
How many decimals will appear for the sum 135267.6437 CAD in MX.3?	

 **Solution**

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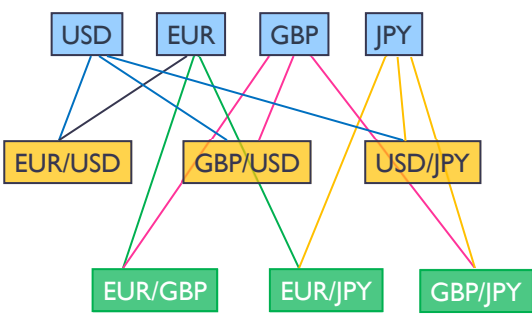
9

MUREX™ | **Currency Configuration**
From currency to currency pair

Step 1: Currencies

Step 2: Leading contracts
(currency pairs including USD)

Step 3: Cross contracts
(not including USD)



 **Note:** Leading and cross contracts are defined according to the spot scenario. Usually, all currencies are quoted against USD, called the root currency, as assumed in the example above. As cross spots are not considered as relevant parameters, it makes more sense to insert leading contracts before the cross contracts.

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Agenda

Currency Configuration

Currency Pair Definition

- Quotation
- Date conventions
- OTC contract description

FX Listed Contracts

Settings

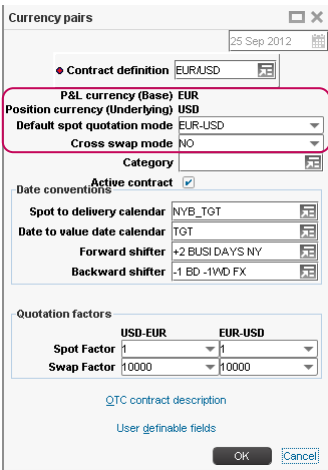
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Currency Pair Definition

Quotation





Label of the contract should be consistent with market practice.

P&L currency (Base)

Position currency (Underlying)

Currency in which the P&L for this currency pair will be expressed.

Position currency for this currency pair. Also known as underlying, risky, or foreign currency.



This choice is specific to each bank, the base currency is considered as less risky from the bank point of view.



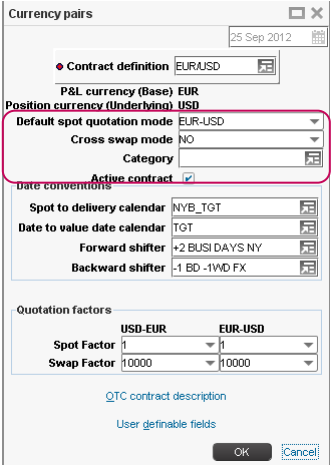
P&L and position currencies cannot be changed or reversed once the contract has been defined. They have a great impact on the P&L and the Greeks (explained in the second part of this course). It is very important to set the correct values at inception!

End-user session: Configuration/ Forex/ Currency pairs

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MUREX™ | **Currency Pair Definition**
Quotation



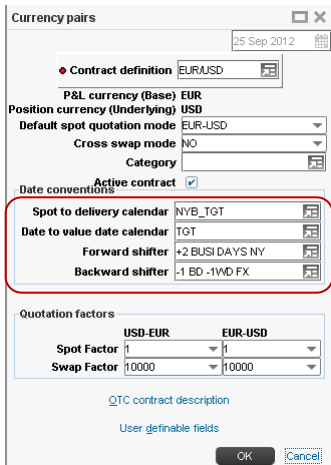
Default spot quotation mode	Spot quotation mode EUR-USD ⇔ 1 EUR = x USD ⇔ x USD per EUR
Cross swap mode	Ability to see cross swap points in the simulation zoom screen
Category	Indicative purposes
Active contract	Contract available/unavailable for selection (in pricing, trade entry...)

End-user session: Configuration/ Forex/ Currency pairs

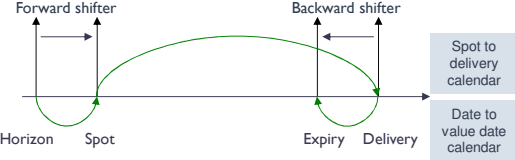
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MUREX™ | **Currency Pair Definition**
Date conventions



Spot to delivery calendar	Computes the Delivery date from the Spot date
Date to value date calendar	Computes the Spot date from the Horizon date or the Delivery date from the Expiry date (or the other way around)
Forward shifter	Computes the Spot date from the Horizon date or the Delivery date from the Expiry date. More generally to compute the Value date. It relies on the Date to value date calendar
Backward shifter	For FX Derivatives only. Computes the Expiry date from the Delivery date and to compute the Fixing date from the Delivery date (NDF). It relies on the Date to value date calendar



End-user session: Configuration/ Forex/ Currency pairs

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MUREX™ | **Currency Pair Definition**
Date conventions: Spot date computation

General market rules to compute the spot date:

Leading contract (including USD) using the forward shifter +2 BUSINESS DAYS NY	+2 days calculation. If day 1 is a holiday for the USD it is counted as a business day	Determination of the spot date takes both currencies calendars into account. The Spot date should be an open day for both currencies
Cross contract (XXX/YYY)	The first shifter (+2 BUSI DAYS FX) finds the spot date of the cross currency pair. If the date falls on a USD holiday, it is moved to next USD open day	The USD calendar is taken into account as well as the XXX and YYY currencies calendars

End-user session: Configuration/ Forex/ Currency pairs

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MUREX™ | **Currency Pair Definition**
Date conventions: Shifter definition

Simple shifter / Relative shifter	A relative shifter first calculates the date according to the underlying shifter then applies the shift taking into account the rules defined in the current shifter definition
Calendar checking	Check whether the generated date is Holiday
Adjustment before / after shift	A date adjustment rule may apply on the original date / on the shifted date. Not generally used in the shifter definition.
Shift definition	Sets the step (Number x Unit) to be added or subtracted (Forward/Backward) to the initial date.
Unit	Business day, Day, Week, Month, Quarter, Semester or Year
Number	The shift is in Number of Units.
If non business day	If the result date on the calendars in Calendar checking is not a business day, this setting defines the rule to be applied
Algorithm	Determines how to shift the dates: Union or parallel

End-user session: Configuration/ Market conventions/ Date shifters

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MUREX™ | **Currency Pair Definition**
Date conventions: Delivery date computation

General market rules to compute the delivery date:

- ✓ Should be open for both countries
- ✓ Should also be a working day in the United States

Currency pairs (25 Sep 2012)

Contract definition: EURUSD

P&L currency (Base): EUR
Position currency (Underlying): USD
Default spot quotation mode: EUR-USD
Cross swap mode: NO
Category:

Active contract ☒

Date conventions

Spot to delivery calendar: NYB_TGT
Date to value date calendar: TGT
Forward shifter: +2 BUSI DAYS NY
Backward shifter: -1 BD -1WD FX

Quotation factors

	USD-EUR	EUR-USD
Spot Factor	1	1
Swap Factor	10000	10000

[QTC contract description](#)
[User definable fields](#)

OK Cancel

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Leading

Date conventions

Spot to delivery calendar: NYB_TGT
Date to value date calendar: TGT
Forward shifter: +2 BUSI DAYS NY
Backward shifter: -1 BD -1WD FX

Cross

Date conventions

Spot to delivery calendar: UNENYBTGT
Date to value date calendar: TGT
Forward shifter: +2 BUSI DAYS NY
Backward shifter: -1 BD -1WD FX

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MUREX™ | **Currency Pair Definition**
Quotation factors

Currency pairs (25 Sep 2012)

Contract definition: EURUSD

P&L currency (Base): EUR
Position currency (Underlying): USD
Default spot quotation mode: EUR-USD
Cross swap mode: NO
Category:

Active contract ☒

Date conventions

Spot to delivery calendar: NYB_TGT
Date to value date calendar: TGT
Forward shifter: +2 BUSI DAYS NY
Backward shifter: -1 BD -1WD FX

Quotation factors

	USD-EUR	EUR-USD
Spot Factor	1	1
Swap Factor	10000	10000

[QTC contract description](#)
[User definable fields](#)

OK Cancel

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Spot factor

If the spot factor is Y, the spot is X, and the quotation mode CUR1-CUR2, this means that there are X units of currency 2 for Y units of currency 1

Spot factor

- Spot factor = 1
1.4 EUR-USD → 1 EUR = 1.4 USD
- Spot factor = 1/1000
0.979 USD-KRW → 1/1000 USD = 0.979 KRW

Swap factor

The swap factor is a multiplying factor for display. If quotation mode is CUR1-CUR2, swap points are expressed in CUR2 per Y CUR1

Swap factor

- Swap factor = 10,000
EUR-USD: $F_{1M} = 1.4237$; $S = 1.4223$
- Swap points = $(F_{1M} - S) * 10,000 = 0.0014 * 10,000 = 14$

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MUREX™ | Currency Pair Definition
OTC contract description

OTC: Over the Counter

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MUREX™ | Currency Pair Definition
OTC contract description

OTC contract description

Default premium quotation	Premium quotation mode displayed by default in the Pricing screen, the Deal entry ticket, and in Simulation
CUR traded quotation	Default quotation mode displayed when CUR is traded as nominal amount
Premium decimals	Number of decimals displayed for the premium by display quotation mode
Premium form factor	For quotation mode CUR1-CUR2, a Y form factor means the premium is expressed in CUR2 per Y CUR1
Format Spot / Strike / Forward / Swap / Volatility	Display of spot, strike, forward and swap points is defined for both quotation modes.
Delta / Gamma / Vega / Rho / Theta	Rounding conventions for greeks. Options are Decimal, 1/2 and 1/4.

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MUREX™ | **Currency Pair Definition**
OTC contract description

OTC contract description

Pricing conventions Back office conventions Simulation conventions Volatility conventions

Default rollover quotation **EUR-USD**

Settlement **Delivery**

Asian Past Average Rounding **No**

Default rollover quotation	Default quotation mode used within the rollover procedure
Settlement	Available options are Delivery and Cash. This field indicates the settlement style by default of the spot forward on this contract
Asian Past Average Rounding	impacts the behavior of the application when an FX Asian option is exercised

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MUREX™ | **Currency Pair Definition**
OTC contract description

Impacts the Simulation and the Notepad Greeks display.
Commonly, the P&L and the Vega currencies are set to the P&L currency of the contract, whereas the Delta and Gamma currencies are set to the first currency of the default quotation.

OTC contract description

Pricing conventions Back office conventions Simulation conventions Volatility conventions

Delta **EUR** P&L **USD**

Gamma **EUR** Vega **USD**

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MUREX™ | **Currency Pair Definition**
OTC contract description

The Volatility conventions tab concerns the smile curve definition.

OTC contract description

Pricing conventions | Back office conventions | Simulation conventions | **Volatility conventions**

Smile quotation conventions

Smile for a call on: EUR
Premium in delta: No
Central point for short term: ATM straddle
Central point for long term: ATM forward (F+K)
Central point long term (expiry >): 10Y
Spot delta for short term: Yes
Spot delta for long term: No
Spot delta long term (expiry >): 1Y
Broker / Smile mode: Smile
Positive RR in favor of: Call

Atm volatility conventions

Correlation and volatility link: Off
Volatility convention: Volatility on calendar days
Volatility Shift Mode: Inherited

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MUREX™ | **Currency Pair Definition**
Practice 2

 Determine the Spot date of each currency pair.
The table below shows the holidays for EUR and USD, where current date = Day0. The date to value date calendar is TGT (EUR calendar). In the below tables, the "?" symbol means that a holiday on this day would not change the spot date

1. **Case 1**

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)					
NYB (USD)	?				

Spot date: _____

2. **Case 2**

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)		Holiday			
NYB (USD)	?	?			

Spot date: _____

3. **Case 3**

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)			Holiday		
NYB (USD)	?	Holiday	?		

Spot date: _____

 **Solution**

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Currency Pair Definition

Practice 3

 Determine the Spot date of each currency pair.
 The table below shows the holidays for EUR and JPY, where System date = Day0. The date to value date calendar is TGT_TKB which is the union of TGT (EUR calendar) and TKB (JPY calendar).

1. Case 1

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)	Holiday				
TKB (JPY)					
NYB (USD)	?	?			

Spot date: _____

2. Case 2

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)	Holiday				
TKB (JPY)		Holiday			
NYB (USD)	?	?			

Spot date: _____

3. Case 3

Calendar	Day1	Day2	Day3	Day4	Day5
TGT (EUR)	Holiday			Holiday	
TKB (JPY)					
NYB (USD)	?	?	Holiday		

Spot date: _____

 **Solution**

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Agenda

Currency Configuration

Currency Pair Definition

FX Listed Contracts

- Market definition
- FX futures
- Listed options

Settings

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MUREX™ | **FX Listed Contracts**
Market definition

FX Listed contracts (Futures and Options) are standardized and exchanged on the futures market.

Examples

CME (Chicago Mercantile Exchange) Chicago, USA	Futures and Options Market	Reference Currency Calendar	 
BMF (Bolsa de Mercadorias & Futuros) São Paulo, Brazil	Futures Market	Reference Currency Calendar	 

End-user session: Configuration/ Market conventions/ FX trading conventions

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MUREX™ | **FX Listed Contracts**
FX futures

Examples of Options & Futures Exchange markets:
Common FX Futures contracts:

- ✓ CME EURUSD, CME USDMXN
- ✓ BMF DOL

CBOE	Chicago Board Options Exchange
CME	Chicago Mercantile Exchange
Euronext.liffe	Takeover from the London International Financial Futures and Options exchange (LIFFE)
BMF	Bolsa de Mercadorias & Futuros (São Paulo)

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FX Listed Contracts

FX futures

Listed FX futures configuration:

OTC Contract	Underlying OTC currency pair.
Face amount	Contract size. Can be expressed in the underlying currency or base currency according to market conventions.
Market	Organized Market on which the contract is traded.
Settlement	Cash or Delivery.
Value date	Shifter used to compute Spot date from Horizon date and Expiry date from Delivery date.
Default quotation	Quotation mode of the futures
Underlying quotation	Quotation mode of the underlying currency pair.
Price form factor	Factor to apply to the spot notation
Swap form factor	Factor to apply to the swap points notation
Strike increment	For indicative purposes only
Payment currency	Currency in which cash settlements are made.
Calendar	Calendar used to compute spot date from horizon date, delivery date from spot date and expiry date from delivery date.
Settlement calendar	Set of settlement dates for the contract
Contract active	If not ticked, the contract cannot be booked

End-user session: Configuration/ Futures/ FX futures

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FX Listed Contracts

FX futures

Listed volatilities	This flag determines which volatilities will be used. The values are: - Interpolated from OTC - Listed curve + OTC shift
Future prices	This flag determines which prices will be used and impact the futures contract prices The values are: - Interpolated from OTC - Listed curve + OTC shift



In *Configuration/ Settings/ Simulation settings/ Fx configuration*, the flag *Future* should be set to *Inherited from contract settings*.

In *Configuration/ Settings/ FX Pricing settings*, the flag *Future* on line with forward should be unticked in order to inherit the configuration done at contract level

End-user session: Configuration/ Futures/ FX futures

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MUREX™ | **FX Listed Contracts**
Listed options

Configuration of the listed options (options on futures):

Fx future definition

25 Sep 2012

• Future CME EURUSD

Future Option

Option type: American

Settlement calendar: CME OF

Premium quotation: USD/EUR Decimal

Premium form factor: 1 EUR-USD

Option Cutoff: 1

Margining: No

Listed volatilities: Interpolated from OTC Future prices: Interpolated from OTC

User definable fields

Option type	American or European
Settlement calendar	Set of settlement dates for the contract
Premium quotation	Default premium quotation mode
Premium form factor	Factor to apply to the premium notation
Option cutOff	Cut off time of the option (if any)
Margining	Option to configure listed option with margining. Margining allows to pay the option premium in MtM

End-user session: Configuration/ Futures/ FX futures

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MUREX™ | **FX Listed Contracts**
Practice 4

 Insert FX future trade.

1. Insert the following trade: Buy 10 DEC12 CME EURUSD contracts from CLEARER1.
2. Explain how the fields are populated in the trade ticket based on the configuration of the contract
 - ✓ How is the maturity date defined? And the delivery date?
 - ✓ How is the nominal determined?
 - ✓ What will be delivered on delivery date?
 - ✓ What is the price of the FX future? What does it correspond to?

 **Solution**

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Agenda

- Currency Configuration
- Currency Pair Definition
- FX Listed Contracts
- Settings**
 - Global vs. local
 - Settings types
 - Example

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Settings

Global vs. Local

There are three scopes for the different types of settings:

Global	Changes impact all users. End-user session: Configuration/ Settings/ General settings
User specific	Impacts current user only. Configurator session: Infrastructure/ FO desk/ Settings
Current session	Current session settings are temporary and apply to the current session only. End-user session: Configuration/ Settings/ Current session settings

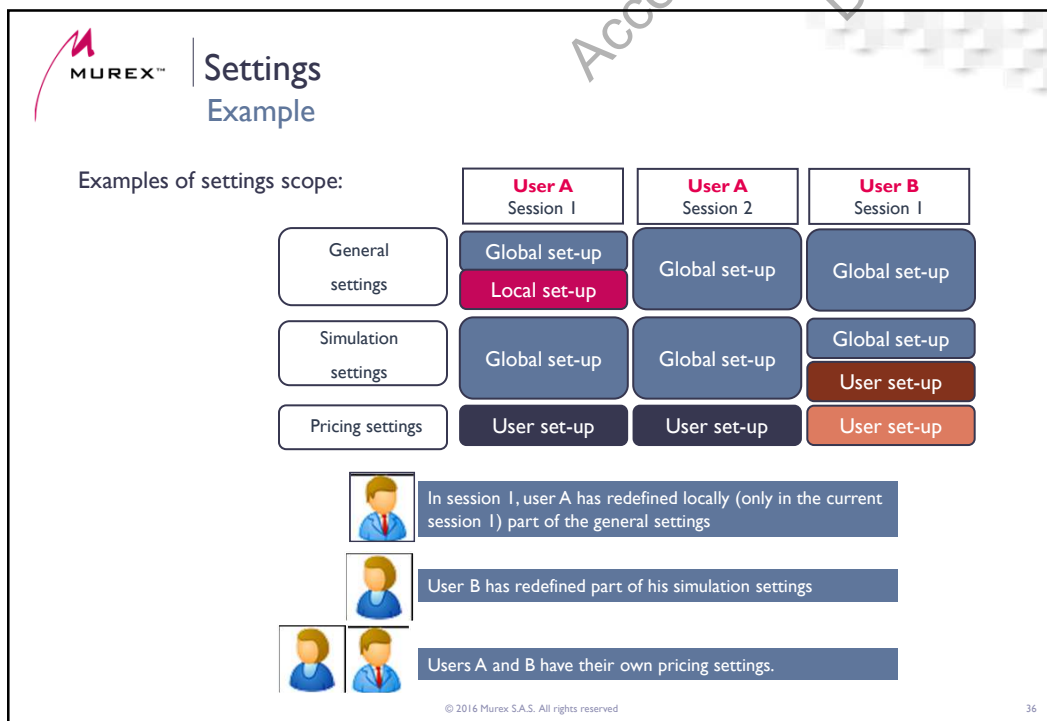
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 Settings Settings types	
General settings	General settings are configured globally at the FO desk level. They can be customized locally by a given end user if he has the appropriate rights, given in the configurator session.
Current session settings	Current session settings apply to the current session only. If the user opens a new session, he will inherit the settings from the general settings.
FX pricing settings & FX user pricing settings	Configuration flags mainly impact trade evaluation rules, cover models and specific market parameters assumptions. Can be modified locally for one user in the menu Configuration/Settings/FX user pricing settings. End-user session: Configuration/ Settings/ FX User Pricing Settings End-user session: Configuration/ Settings/ FX Pricing Settings
Simulation settings	Simulation settings apply to the simulation screens. They are organized in a preliminary common configuration part, shared by all transaction types and additional setting screens specific to each activity (FX, Rates, FXMM, credit...) End-user session: Configuration/ Settings/ Simulation Settings End-user session: Simulation/ Settings/ Simulation screen Settings (per user)
Reporting setup	Provides default calculation assumptions for reporting: printing reports, report batches, risk matrices, running a notepad End-user session: Middle Office/ Reporting/ Reporting Setup

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Thank You!

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EXERCISE SOLUTIONS

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MUREX™ | **Currency Configuration**
Practice 1

 Validate your understanding of the currency definition screen options.

What is the rate entry mode for CHF?	rate
What is the calendar used for the GBP currency? What does it do?	LNB. The calendar is used to compute dates for interest rate curve. It is usually a single calendar defined with the currency holidays
What is the spot schedule used for the JPY currency? What does it imply?	+2 BUSINESS DAYS. This shifter is used to calculate the spot date of the currency when it is used individually.
How many decimals will appear for the sum 135267.6437 CAD in MX.3?	2 - the pricing convention is 7:2. So the amount will appear as 135267.64

 **Exercise**

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MUREX™ | **Currency Pair Definition**
Practice 2

 **Practice 2:** Determine the Spot date of each currency pair.
The table below shows the holidays for EUR and USD, where current date = Day0. The date to value date calendar is TGT (EUR calendar). In the below tables, the "?" symbol means that a holiday on this day would not change the spot date.

1. Case 1: spot date = Day 2
Algorithm:
+2 BUSI DAYS FX is applied on TGT: Count 2 business days on TGT which leads to Day2
+2 BUSI DAYS NY is applied on the date found above: As TGT and NYB are both open, spot date is Day2.
Holiday on Day1 in NYB has no effect on spot date calculation in this case

2. Case 2: spot date = Day 3
Algorithm:
+2 BUSI DAYS FX is applied on TGT: Count 2 business days on TGT which leads to Day3
+2 BUSI DAYS NY is applied on the date found above: As TGT and NYB are both open on Day3, spot date is Day3
Holidays on Day1 and Day2 in NYB have no effect on spot date calculation in this case

3. Case 3 : spot date = Day 4
Algorithm:
+2 BUSI DAYS FX is applied on TGT: Count 2 business days on TGT which leads to Day2
+2 BUSI DAYS NY is applied on the date found above: As Day2 is a holiday in NYB, the system moves forward until the first day where TGT and NYB are both open (Algorithm set to Union in the forward shifter definition. This leads to spot date being Day4

 **Exercise**

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Currency Pair Definition

Practice 3



Practice 3: Determine the Spot date of each currency pair
The table below shows the holidays for EUR and JPY, where System date = Day0. The date to value date calendar is TGT_TKB which is the union of TGT (EUR calendar) and TKB (JPY calendar).

1. Case 1: spot date = Day 3
Algorithm:

- +2 BUSI DAYS FX is applied on TGT_TKB: Count 2 business days on TGT and count 2 business days on TKB, then take the latest day (Algorithm set to Parallel). This leads to Day3
- +2 BUSI DAYS NY is applied on the date found above: As TGT, TKB and NYB are all open, spot date remains Day3

Holidays on Day1 and Day2 in NYB have no effect on spot date calculation in this case

2. Case 2: spot date = Day 3
Algorithm:

- +2 BUSI DAYS FX is applied on TGT_TKB: Count 2 business days on TGT and count 2 business days on TKB, then take the latest day (Algorithm set to Parallel). This leads to Day3
- +2 BUSI DAYS NY is applied on the date found above: As TGT, TKB and NYB are all open, spot date remains Day3

Holidays on Day1 and Day2 in NYB have no effect on spot date calculation in this case

3. Case 3: spot date = Day 5
Algorithm:

- +2 BUSI DAYS FX is applied on TGT_TKB: Count 2 business days on TGT and count 2 business days on TKB, then take the latest day (Algorithm set to Parallel). This leads to Day3
- +2 BUSI DAYS NY is applied on the date found above: As TGT, TKB and NYB are not all open on Day3, the system moves forward up to Day5 to have TGT, TKB, NYB all open at the same time (Algorithm set to Union). This leads to spot date being Day5


Exercise

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FX Listed Contracts

Practice 4



Insert an FX future trade.

- Open a session MUREXFO/MUREXFO
- Go to Processing/ Trades/ Insert trades and select FX future (in FX derivatives) and note the contract number
- Type the instrument CME EURUSD
- Select Buy and the maturity DEC 12
- Type the quantity (e.g 10 contracts)
- Select the clearer « CME » and the portfolio « CLIENT1_PAR » and book the trade by doing Ctrl Enter
- Go to Processing/ Trades/ Trade query and perform a search on the contract number
- Open the deal ticket and check that it is the deal you just inserted


Exercise

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FX Listed Contracts

Practice 4


Insert an FX future trade.

Product	#1 FX Futures		
Instrument Market	CME EURUSD	US CMEC	
Buy/Sell	Buy		
Maturity	DEC 12	Mon 17/12/12	
Delivery	Wed 19/12/12		
Price	1.29743273366		
Quantity	10		
Nominal	Buy	1,250,000	EUR
Other nominal	Sell	1,621,791	USD
First margin call	Tue 25/09/12		
Market price	1.2974		
Market spot	1.2963		
Clearer	CME		
Portfolio	CLIBIT1 PAR		


Exercise

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FX Listed Contracts

Practice 4


Practice 4: Insert FX future trade

How is the maturity date defined? And the delivery date?
 The maturity date & delivery dates are deduced from the maturity set. In the config of the FX future, open the settlement calendar.

How is the nominal determined?
 The nominal in EUR appears. It is equal to $10 * 125,000$ which is the face amount defined in the contract.

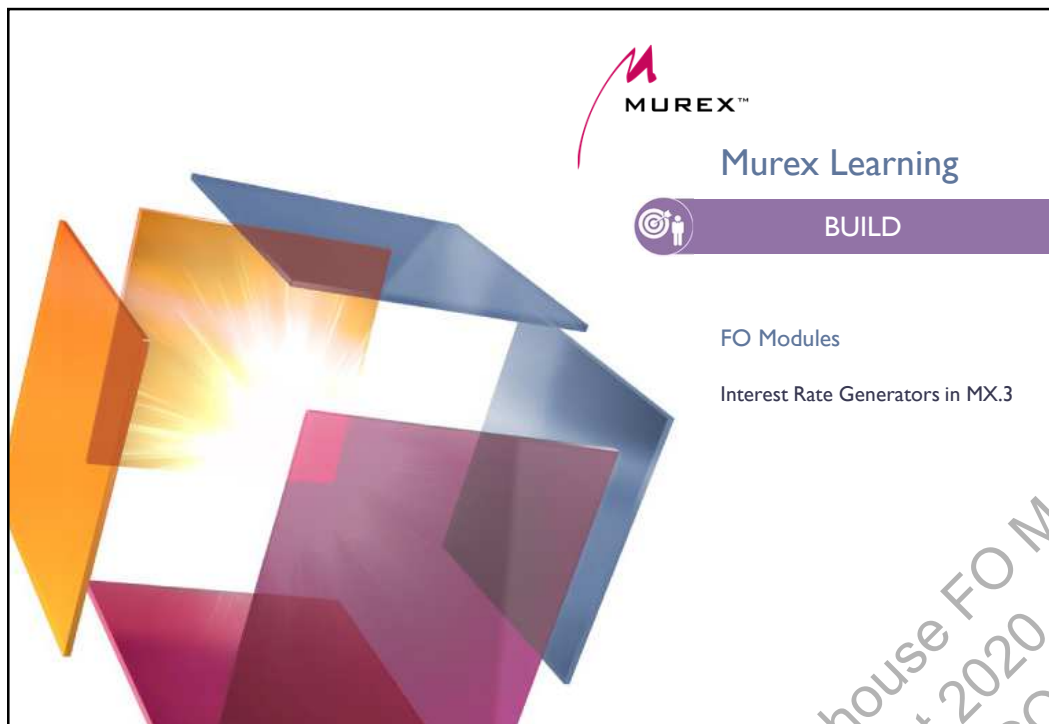
What will be delivered on delivery date?
 On delivery date, the EUR and USD amounts will be delivered, because the FX future is configured to have a delivery settlement.

What is the price of the FX future? What does it correspond to?
 The market price we see in the deal ticket is the same as the market price of a EUR/USD forward trade maturing on 19/12/12, because the futures prices are configured as « interpolated from OTC ».


Exercise

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MUREX™ | Learning Objectives
IR Generators

At the end of this course, you will be able to:

- ✓ Explain a generator and the roles it plays in MX.3
- ✓ Access and modify a pre-configured generator
- ✓ Configure a generator from scratch

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Agenda

Introduction	• Definition • Example: Booking a swap • Global vs. Local change • Link between generator and contract
Swap Generator	
Currency Swap Generator	
Bond Generator	
Cap/Floor, FRA, Loan Generators	
Practice	

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Introduction

Definition

A **Generator** is a template/model which defines a swap or a bond.
It can be used "as is" in a trade or be locally modified.

Enables	Fast insertion of trades	
Contains all the data needed to handle the deal flows	• Calendar, Payment fixing and Calculation schedule • Stub period • Rate convention • In arrears or up front • An index • All... except figures	
Can be changed "globally" or "locally"	Global change:	Means modifying the generator template, which affects all new trades, including existing booked trades with the same generator.
	Local Change:	Means modifying the generator at trade level. A local change only affects that particular trade.



Rights have to be granted in order to change a generator.

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MUREX™ | Introduction
Example: Booking a swap

Product	#1 IRS Vanilla	
Instrument	EURIBOR A 6M	
Maturity period	3Y8M	Fri 27/05/16
Adjusted maturity	Fri 27/05/16	
Nominal	100,000,000	EUR
Flows	Open	
Leg	Pay	Receive
Leg type	Fixed	Floating
Currency	EUR	EUR
Index	EUR EURIBOR 6M	
First fixing	0.0000	
Calculation frequency	1y	6m
Rate/Spread	4.7003	0.0000
Discount curve	EUR EONIA	EUR EONIA
Estimation curve	EUR EURIBOR 6M	
Leg npv	-13,141,656	13,141,656
Legs details	No	
Amortizing	None	
Additional flows	Open	
Customized flows	No	
Mutual break	No	
Npv	0	EUR
Bpv	-27,959	
Typology / Usage	IRS	Libor
Fees	Open	
CSA Type		
Int. Counterparty	No	ABBEY NATIONAL PI
Portfolio	MX_PTF	

- 1) Select the generator
- 2) Choose Start and Maturity Dates
- 3) Input the nominal
- 4) The result of the pricing is
 - ✓ The fixed rate
 - ✓ Or the NPV on settlement date
 - ✓ Or the Margin on the floating leg

Right-click/Set as Implied Parameter on the NPV or the Rate margin in order to fix one and calculate the other.

End-user session: Pricing/ e-Tradepad

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MUREX™ | Introduction
Example: Booking a swap

The generator is called from the trade insertion and pricing screens. It conveys data to the trade (in order, for instance, to compute flows schedules).

If the generator is customized in the trade, the label of the generator is displayed in *italics* and the customized version of it is accessible by clicking on the generator details.

Interest rate swap

Financial definition

Generator: **EURIBOR A 3M** Generator details

Start date: Spot 27 Sep 2012

Maturity: 4Y 27 Sep 2016 27 Sep 2016

Nominal: 100,000,000 EUR

Plain vanilla | Exotic | Market data | Leg characteristics

End-user session: Pricing/ e-Tradepad~Instrument details

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MUREX™ | Introduction
Example: Booking a swap

Interest rate swap

Financial definition

Generator: EURBOR A 3M

Start date: 27 Sep 2012

Maturity: 27 Sep 2016

Nominal: 100,000,000

Characteristics

Amortized ☒

Exotic ☒

Flex ☐

Additional flows ☐

Compute ☒

Disable Cnw adj ☐

Unlink pay/receive ☐

Plan vanilla | Exotic | Market data | Leg characteristics

Plain vanilla | Exotic | Market data | Leg characteristics

Leg 1/2

Phase 1/1

Settlement: 27 Sep 2012

Global flows

NPV: -1,484,710

BPV: -38,069

End-user session: Pricing/ e-Tradepad-Instrument details

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MUREX™ | Introduction
Example: Global vs. Local change

Global change	Affects all existings trades (non customised) and future ones
Local change	Affects only the specific trade

IRS Generator

Floating Index: EUR EURBOR 6M

Floating Index: EUR EURBOR 3M

Deal 1

Floating Index: EUR EURBOR 6M

Floating Index: EUR EURBOR 3M

Deal 2

Floating Index: EUR EURBOR 6M

Floating Index: EUR EURBOR 3M

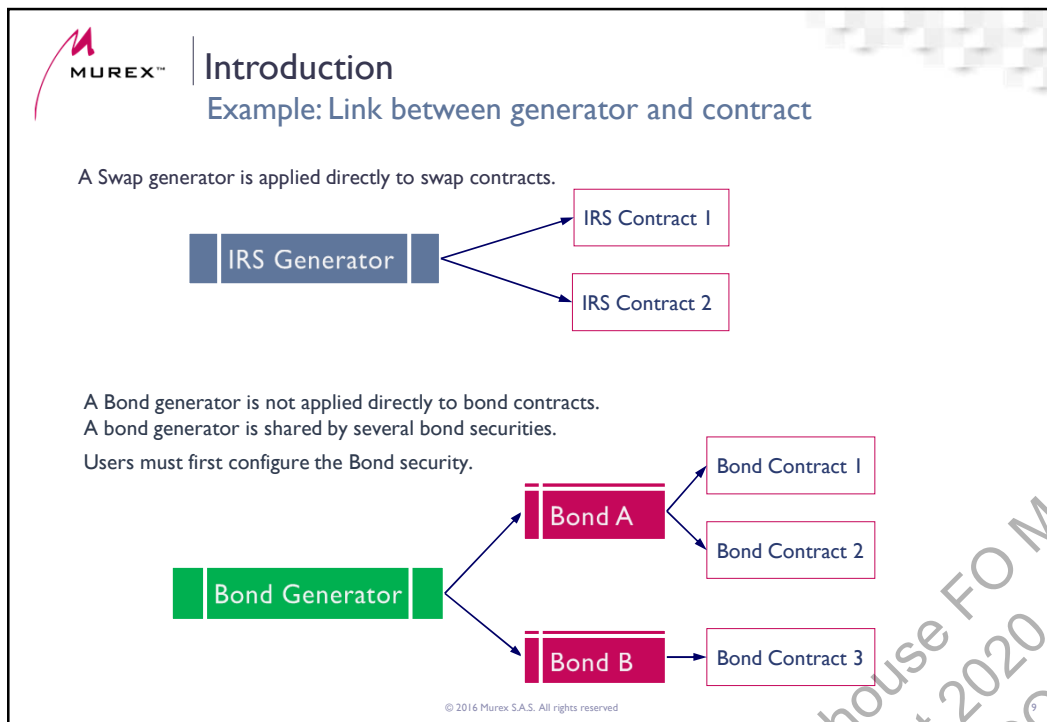
Deal 3

Floating Index: EUR EURBOR 1M (Customisation)

Floating Index: EUR EURBOR 1M (Customisation)

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MUREX™ | Agenda

Introduction	
Swap Generator	• Anatomy of a swap generator • Fixing and payment conventions • Payment date customisation • Compounding • Stub periods • Margin mode • Rate formula • Amortization • Other useful components
Currency Swap Generator	
Bond Generator	
Cap/Floor, FRA, Loan Generators	
Practice	

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MUREX™ | **Swap Generator**
Anatomy of a swap generator

General

Leg 1

Leg 2

End-user session: Configuration/ Generators/ Swap generators

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MUREX™ | **Swap Generator**
Anatomy of a swap generator: General

Schedules
drive legs schedules and capital dependency.

Amortizing
is used only when schedules are "independent sets across legs". Nominals on both legs can be independent: amortizing details are configured in each leg.

Phases / Legs
are MX concepts. On the market, a swap has 2 legs. Phases are used to represent swaps whose the structure changes over time. Legs are used as a workaround for some structured products, e.g. 2 legs to accumulate several types of coupons.

[Estimation forced]
this flag conducts the estimation of the floating rate and thus impacts the revaluation of the deal.

Stub periods
are common to both legs.

Future Cash proceed cut-off
indicates the period for which flows generated by the deal will be accounted for Future proceeds instead of Market value.

End-user session: Configuration/ Generators/ Swap generators

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MUREX™ Swap Generator
Anatomy of a swap generator: Fixed leg

Fixed leg:

- Payment frequency: Can be either monthly, quarterly, semi-annually or annually.
- Stub period: Broken period remaining

End-user session: Configuration/ Generators/ Swap generators

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MUREX™ Swap Generator
Anatomy of a swap generator: Floating leg

Floating leg:
Floating leg rates need to be fixed, using the:

- ✓ Fixing calendar
- ✓ Fixing: *In advance/ in arrears*
- ✓ Fixing schedule

A margin can be added
An “on-the-fly” formula on the floating index can be configured

Stub period characteristics

Coupon	Short coupon
Stub period rate	Interpolated
Stub isda convention	Current index
	Closest index
	Next index
	Previous index
Currency	Interpolated
	Max(Next,Previous)
	SS DAY

End-user session: Configuration/ Generators/ Swap generators

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MUREX™ Swap Generator
Fixing and payment conventions

The Schedules definition screen defines:

- ✓ Schedules for calculation start, calculation end, payment and fixing: Driving/ Equal to/Deduced from/Copy of
- ✓ Frequency ratio

There are three sets of dates for a floating flow:

Schedules definition

Calculation start schedule: Driving schedule
Schedule generator: 3M MODFOL
Calculation end schedule: Equal to
Payment schedule: Deduced from
Deduction formula: +0 BUSINESS DAY
Frequency ratio: 1
Fixing schedule: Deduced from
Deduction formula: -2 BUSINESS DAY
Frequency ratio: 1
Calculations follow fixings: No (time series)
Capital schedules: No
Single period: No

Start/End date: Calculation period
Fixing date: When the floating rate is observed
Payment date: Value date of funds transfer

Flow Tp	Calculation Start date	Calculation End date	Remaining Capital	Fixing Date	Rate	Margin	Rate Factor	Payment Date	Flows
INT	27 Sep 2012	27 Dec 2012	100,000,000.00	25 Sep 2012	1.53000	0.0000	1.0000	27 Dec 2012	386,750.00
INT	27 Dec 2012	27 Mar 2013	100,000,000.00	21 Dec 2012	1.68146	0.0000	1.0000	27 Mar 2013	420,364.97
INT	27 Mar 2013	27 Jun 2013	100,000,000.00	25 Mar 2013	1.80335	0.0000	1.0000	27 Jun 2013	460,856.51
INT	27 Jun 2013	27 Sep 2013	100,000,000.00	25 Jun 2013	1.93643	0.0000	1.0000	27 Sep 2013	494,866.28
INT	27 Sep 2013	27 Dec 2013	100,000,000.00	25 Sep 2013	2.06878	0.0000	1.0000	27 Dec 2013	522,940.87
INT	27 Dec 2013	27 Mar 2014	100,000,000.00	23 Dec 2013	2.19970	0.0000	1.0000	27 Mar 2014	549,925.54
INT	27 Mar 2014	27 Jun 2014	100,000,000.00	25 Mar 2014	2.33160	0.0000	1.0000	27 Jun 2014	595,854.36
INT	27 Jun 2014	29 Sep 2014	100,000,000.00	25 Jun 2014	2.46648	0.0000	1.0000	29 Sep 2014	644,025.84

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MUREX™ Swap Generator
Payment dates customisation

Payment dates do not always follow the calculation periods. There are 3 possible payment schedules:

Payment schedule: Equal to
Calculation start schedule: Frequency ratio
Frequency ratio: 3
Backward (broken periods included): ☐
Fixing schedule: Deduced from
Deduction formula: -2 BUSINESS DAY
Frequency ratio: Zero coupon
Formula adjusted: ☐

Zero-coupon	Interest flows are not paid but capitalised at the leg rate at the original payment frequency.
Frequency ratio with ratio n	Flows are only paid every n periods. When not paid, they are capitalised at the leg rate at the original payment frequency.
Formula adjusted	Not relevant for swaps (used for CDS).

Interest flow Zero-coupon

Start date Interest flows Maturity
Frequency ratio = 3

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MUREX™ Swap Generator Compounding

Frequency ratio is an integer that defines the frequency at which the flows are being paid.
Zero-coupon is a mode in which no intermediate flow is paid until the trade maturity.

(I+M) at (I+M)	Interests of period 1 are not paid. They are added to the capital on period 2
No compounding	Interests of period 1 are not paid. They are not used for following periods

Compounding (I+M) at (I+M) ▼

(I+M) at (I+M)

(I+M) at I

(I at I) + M

I at (I+M)

I at (I+M) + M

No compounding

(I-M1) at (I-M2)

(I-M) at C

Rate convention ☐ **Rate conversion**

Floating interest payments EUR												
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur	Interest flows
INT	27 Sep 2012	27 Dec 2012	100,000,000	25 Sep 2012	1.53000	1.53000	0.0000	1.0000	27 Dec 2012	386,750.00 EUR		386,750.00
INC	27 Dec 2012	27 Mar 2013	100,000,000	21 Dec 2012	1.68146	1.68146	0.0000	1.0000		420,364.97 EUR		420,364.97
INC	27 Mar 2013	27 Jun 2013	100,000,000	25 Mar 2013	1.80335	1.80335	0.0000	1.0000		883,158.76 EUR		460,856.51
INT	27 Jun 2013	27 Sep 2013	100,000,000	25 Jun 2013	1.93643	1.93643	0.0000	1.0000	27 Sep 2013	1,382,395.50 EUR		494,866.28
INC	27 Sep 2013	27 Dec 2013	100,000,000	25 Sep 2013	2.06878	2.06878	0.0000	1.0000		522,940.87 EUR		522,940.87
INC	27 Dec 2013	27 Mar 2014	100,000,000	23 Dec 2013	2.19970	2.19970	0.0000	1.0000		1,075,742.20 EUR		549,925.54
INT	27 Mar 2014	27 Jun 2014	100,000,000	25 Mar 2014	2.33160	2.33160	0.0000	1.0000	27 Jun 2014	1,678,006.42 EUR		595,854.38

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MUREX™ Swap Generator Practice I

Identify the payment schedules: which is zero-coupon and which is using a frequency ratio?

A

Floating interest payments EUR												
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur	Interest flows
INC	08 Apr 2010	08 Jun 2010	100,000,000	06 Apr 2010	1.10000	1.10000	0.0000	1.0000		186,388.89		
INT	08 Jun 2010	08 Dec 2010	100,000,000	04 Jun 2010	1.64250	1.64250	0.0000	1.0000	08 Dec 2010	1,022,882.62		
INC	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.0000		505,555.56		
INC	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.0000		2,183,121.34		
INT	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.0000	08 Jun 2012	3,593,556.31		
INC	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.0000		1,578,952.21		
INC	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.0000		3,298,192.25		
INT	10 Jun 2013	08 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.0000	09 Dec 2013	5,111,822.68		

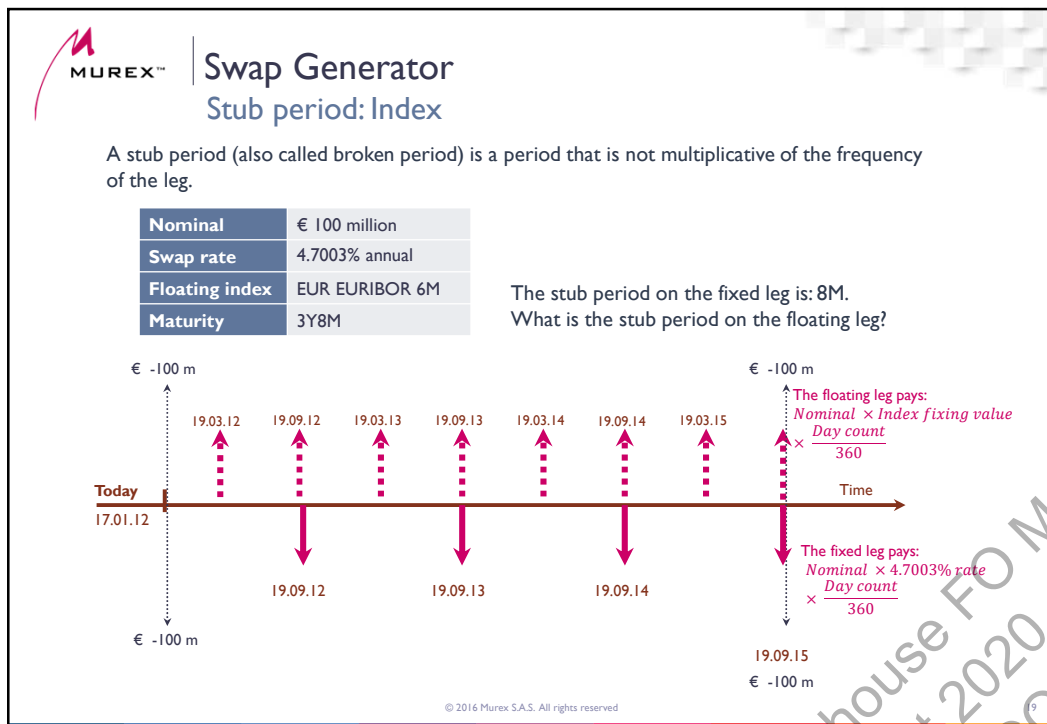
B

Floating interest payments EUR												
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur	Interest flows
INC	08 Apr 2010	08 Jun 2010	100,000,000	06 Apr 2010	1.10000	1.10000	0.0000	1.0000		186,388.89		
INC	08 Jun 2010	08 Dec 2010	100,000,000	04 Jun 2010	1.64250	1.64250	0.0000	1.0000		1,022,882.62		
INC	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.0000		1,533,609.42		
INC	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.0000		3,228,394.73		
INC	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.0000		4,653,196.80		
INC	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.0000		6,345,395.21		
INC	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.0000		8,198,192.25		
INT	10 Jun 2013	08 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.0000	09 Dec 2013	10,000,000.00		

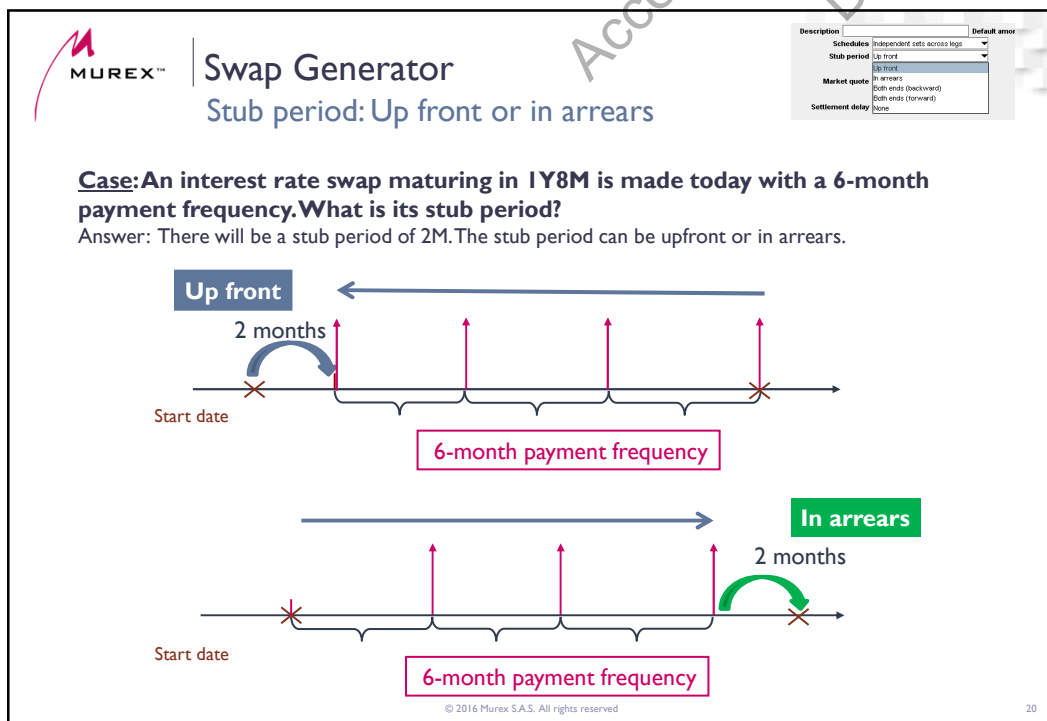
Solution

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MUREX™ Swap Generator Practice 2

Match the right flow schedules to the correct corresponding stub periods
Which is the right flow schedule for "in arrears" stub period?

A

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Mar 2006	100,000,000.00	30 Dec 2005	3.52524	0.0000	1.000000	01 Mar 2006	558,163.00	EUR
INT	01 Mar 2006	01 Sep 2006	100,000,000.00	27 Feb 2006	3.81172	0.0000	1.000000	01 Sep 2006	1,948,214.42	EUR
INT	01 Sep 2006	01 Mar 2007	100,000,000.00	30 Aug 2006	4.64828	0.0000	1.000000	01 Mar 2007	2,337,050.20	EUR
INT	01 Mar 2007	03 May 2007	100,000,000.00	27 Feb 2007	5.31309	0.0000	1.000000	03 May 2007	929,790.57	EUR

B

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	03 May 2006	100,000,000.00	30 Dec 2005	3.61324	0.0000	1.000000	03 May 2006	1,204,413.33	EUR
INT	03 May 2006	03 Nov 2006	100,000,000.00	28 Apr 2006	4.11184	0.0000	1.000000	03 Nov 2006	2,101,605.50	EUR
INT	03 Nov 2006	03 May 2007	100,000,000.00	01 Nov 2006	4.88141	0.0000	1.000000	03 May 2007	2,454,264.10	EUR

C

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	03 Jul 2006	100,000,000.00	30 Dec 2005	3.65856	0.0000	1.000000	03 Jul 2006	1,839,441.44	EUR
INT	03 Jul 2006	03 Jan 2007	100,000,000.00	29 Jun 2006	4.40626	0.0000	1.000000	03 Jan 2007	2,252,088.23	EUR
INT	03 Jan 2007	03 May 2007	100,000,000.00	29 Dec 2006	4.97030	0.0000	1.000000	03 May 2007	1,656,766.67	EUR

Solution

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MUREX™ Swap Generator Practice 3

Match the right flow schedules to the correct corresponding stub periods frequency ratio?
Which is the right flow schedule for "up front" stub period?

A

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Mar 2006	100,000,000.00	30 Dec 2005	3.52524	0.0000	1.000000	01 Mar 2006	558,163.00	EUR
INT	01 Mar 2006	01 Sep 2006	100,000,000.00	27 Feb 2006	3.81172	0.0000	1.000000	01 Sep 2006	1,948,214.42	EUR
INT	01 Sep 2006	01 Mar 2007	100,000,000.00	30 Aug 2006	4.64828	0.0000	1.000000	01 Mar 2007	2,337,050.20	EUR
INT	01 Mar 2007	03 May 2007	100,000,000.00	27 Feb 2007	5.31309	0.0000	1.000000	03 May 2007	929,790.57	EUR

B

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	03 May 2006	100,000,000.00	30 Dec 2005	3.61324	0.0000	1.000000	03 May 2006	1,204,413.33	EUR
INT	03 May 2006	03 Nov 2006	100,000,000.00	28 Apr 2006	4.11184	0.0000	1.000000	03 Nov 2006	2,101,605.50	EUR
INT	03 Nov 2006	03 May 2007	100,000,000.00	01 Nov 2006	4.88141	0.0000	1.000000	03 May 2007	2,454,264.10	EUR

C

Flow schedule Source:Notepad Leg:2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tp	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Feb 2006	100,000,000.00	30 Dec 2005	3.47346	0.0000	1.000000	01 Feb 2006	279,806.47	EUR
INT	01 Feb 2006	01 Aug 2006	100,000,000.00	30 Jan 2006	3.75339	0.0000	1.000000	01 Aug 2006	1,887,120.25	EUR
INT	01 Aug 2006	01 Feb 2007	100,000,000.00	28 Jul 2006	4.52827	0.0000	1.000000	01 Feb 2007	2,312,914.16	EUR
INT	01 Feb 2007	03 May 2007	100,000,000.00	30 Jan 2007	5.21492	0.0000	1.000000	03 May 2007	1,318,215.67	EUR

Solution

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MUREX™ Swap Generator
Stub period: Both ends backward or forward

Case: An interest rate swap maturing in 1Y8M is made today with a 6-month payment frequency. What is its stub period?

Answer: There will be a stub period of 2M. The stub period can be:

Both ends backward

Both ends forward

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MUREX™ Swap Generator
Practice 4

Practice 4: Match the right flow schedules to the correct corresponding stub periods.

Both ends (backward) (IRS 1Y4M)

Generator: EURIBOR A 6M
Start date: Spot (03 Jan 2006)
Maturity: 1y4m (03 May 2007)
Nominal: 100,000,000
Currency: EUR

Roll date: 01 Mar 2007
Payment calendar: TGT

Both ends (forward) (IRS 1Y4M)

Generator: EURIBOR A 6M
Start date: Spot (03 Jan 2006)
Maturity: 1y4m (03 May 2007)
Nominal: 100,000,000
Currency: EUR

Roll date: 01 Feb 2006
Payment calendar: TGT

A

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tr	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Feb 2006	100,000,000.0030	Dec 2005	3.47346	0.0000	1.0000001	Feb 2006	279,806.47	EUR
INT	01 Feb 2006	01 Aug 2006	100,000,000.0030	Jan 2006	3.75339	0.0000	1.0000001	Aug 2006	1,887,120.25	EUR
INT	01 Aug 2006	01 Feb 2007	100,000,000.0028	Jul 2006	4.52527	0.0000	1.0000001	Feb 2007	2,312,914.16	EUR
INT	01 Feb 2007	03 May 2007	100,000,000.0030	Jan 2007	5.21492	0.0000	1.0000003	May 2007	1,318,215.67	EUR

B

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tr	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Mar 2006	100,000,000.0030	Dec 2005	3.52524	0.0000	1.0000001	Mar 2006	559,806.47	EUR
INT	01 Mar 2006	01 Sep 2006	100,000,000.0027	Feb 2006	3.81172	0.0000	1.0000001	Sep 2006	1,948,120.25	EUR
INT	01 Sep 2006	01 Mar 2007	100,000,000.0030	Aug 2006	4.64826	0.0000	1.0000001	Mar 2007	2,337,050.20	EUR
INT	01 Mar 2007	03 May 2007	100,000,000.0027	Feb 2007	5.31309	0.0000	1.0000003	May 2007	929,790.57	EUR

Solution

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MUREX™ Swap Generator
Stub period: Coupon

Case: An interest rate swap maturing in 8M is made today with a 6-month payment frequency.

Answer: There will be a stub period of 2M. The last coupon can be:

Stub period characteristics

Coupon	Short coupon
Stub period rate	Interpolated
Stub index convention	Current index
Evaluation	Inherited
	Interpolated (Max(Next, Previous))

Short coupon: 2 months stub, 6-month payment frequency

Long coupon: 8 months stub, 6-month payment frequency

Full coupon: 6 months stub, 6-month payment frequency

Conditional is used for CDS:
If the stub is shorter than X, then the user can specify the type of coupon.
Otherwise, the user should specify another coupon.

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MUREX™ Swap Generator
Practice 5

Practice 5: Match the right flow schedules to the correct stub periods coupon. Which is the right flow schedule for “short coupon” and “long coupon”?

A

Global flow schedule Source: Notepad

Flow Tp	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur
JNT	08 Apr 2010	08 Dec 2010	100,000,000	06 Apr 2010	1.50900	1.50900	0.0000	1.00000	08 Dec 2010	1,022,766.67	EUR
JNT	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.00000	08 Jun 2011	505,555.56	EUR
JNT	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.00000	08 Dec 2011	1,669,127.42	EUR
JNT	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.00000	08 Jun 2012	1,380,301.32	EUR
JNT	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.00000	10 Dec 2012	1,578,952.21	EUR
JNT	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.00000	10 Jun 2013	1,692,516.02	EUR
JNT	10 Jun 2013	08 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.00000	08 Dec 2013	1,755,723.28	EUR

B

Global flow schedule Source: Notepad

Flow Tp	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur
JNT	08 Apr 2010	08 Jun 2010	100,000,000	06 Apr 2010	1.10000	1.10000	0.0000	1.00000	08 Jun 2010	186,388.89	EUR
JNT	08 Jun 2010	08 Dec 2010	100,000,000	04 Jun 2010	1.64250	1.64250	0.0000	1.00000	08 Dec 2010	834,937.50	EUR
JNT	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.00000	08 Jun 2011	505,555.56	EUR
JNT	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.00000	08 Dec 2011	1,669,127.42	EUR
JNT	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.00000	08 Jun 2012	1,380,301.32	EUR
JNT	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.00000	10 Dec 2012	1,578,952.21	EUR
JNT	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.00000	10 Jun 2013	1,692,516.02	EUR
JNT	10 Jun 2013	08 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.00000	08 Dec 2013	1,755,723.28	EUR

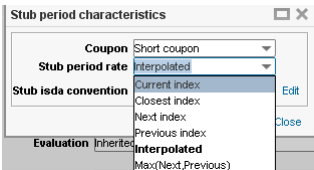
Solution

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MUREX™ Swap Generator
Stub period: Index

Stub period rate (archiving group) – which index to use?



Current index	The index defined in the floating leg.
Closest index (resp Next/Previous)	The index tenor in the SAME archiving group of the index of the floating leg, whose tenor length is the closest (resp. immediately longer/shorter) of the length of the stub period.
Interpolated	Two index tenors in the same archiving group of the index of the floating leg, whose tenors are immediately shorter and longer than the length of the stub period. Then interpolation pro-rata temporis is performed.

In our example for the stub period of 2M:

- ✓ Current index: EURIBOR 6M
- ✓ Closest index: EURIBOR 2M
- ✓ Interpolated: Interpolated between EURIBOR 2M & EURIBOR 3M

End-user session: Configuration/ Generators/ Swap generators/Stub period:Details/ Stub period rate

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MUREX™ Swap Generator
Stub period: Index

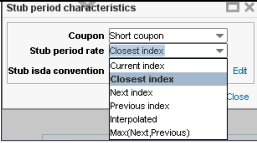
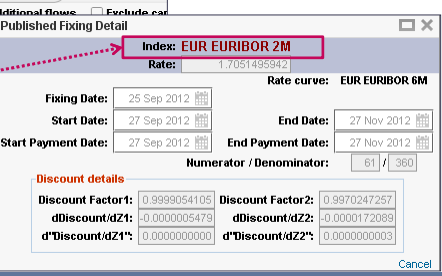
The **Closest index** is the closest index stored in the system in terms of maturity.

Leg 2 | All legs

Floating interest payments EUR								
Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date
27 Sep 2012	27 Nov 2012	100,000,000	25 Sep 2012	1.10896	1.10896	0.0000	1.0000	27 Nov 2012
27 Nov 2012	27 May 2013	100,000,000	23 Nov 2012	1.64250	1.64250	0.0000	1.0000	27 May 2013
27 May 2013	27 Nov 2013	100,000,000	23 May 2013	1.75546	1.75546	0.0000	1.0000	27 Nov 2013
27 Nov 2013	27 May 2014	100,000,000	25 Nov 2013	2.32638	2.32638	0.0000	1.0000	27 May 2014
27 May 2014	27 Nov 2014	100,000,000	23 May 2014	2.71726	2.71726	0.0000	1.0000	27 Nov 2014

Evaluation flows

EUR EURIBOR 2M

End-user session: Configuration/ Generators/ Swap generators/Stub period:Details/ Stub period rate

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MUREX™ Swap Generator
Stub period: Index

Stub period characteristics

Coupon	Short coupon
Stub period rate	Interpolated
Stub isda convention	Current index
	Next index
	Previous index
	Interpolated
	Max(Next,Previous)

Interpolated index is a computed figure deduced from the interpolation of the previous and the next index (most common configuration). The system will compute the exact forward-forward rate on the archiving group rate curve.

Product #1 IRS Vanilla
Instrument EURIBOR A 6M
Maturity period 3Y8M
Adjusted maturity Fri 27/05/16
Nominal 100,000,000
Flows Open
Leg Pay Receive
Leg type Fixed Floating
Currency EUR
Index EUR EURIBOR 6M
First fixing 0.0000
Calculation frequency 1y 6m

Leg 2 All legs

Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor
27 Sep 2012	27 Nov 2012	100,000,000	25 Sep 2012	1.10896	1.10896	0.0000	1.00
27 Nov 2012	27 May 2013	100,000,000	23 Nov 2012	1.64250	1.64250	0.0000	1.00
27 May 2013	27 Nov 2013	100,000,000	23 May 2013	1.75546	1.75546	0.0000	1.00
27 Nov 2013	27 May 2014	100,000,000	25 Nov 2013	2.32638	2.32638	0.0000	1.00

Interpolated Fixing Detail

Index: EUR EURIBOR 6M
Rate: 1.1089567736
Fixing Date: 25 Sep 2012
Estimation Period Between: 27 Sep 2012 27 Nov 2012
Rounding rule: Nearest
Number of Days: 61
Decimals: 3

Number	Index	Start Date	End Date	Days	Rate	Weight
1	EUR EURIBOR 2M	27 Sep 2012	27 Nov 2012	61	1.108959	1.000000
2	EUR EURIBOR 3M	27 Sep 2012	27 Dec 2012	91	1.530000	0.000000

End-user session: Configuration/ Generators/ Swap Generators/Stub period: Details/ Stub period rate

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MUREX™ Swap Generator
Margin mode: Floating leg

Margin mode defines how the margin is “added” to the floating rate.

Margin mode Additive
Total return Additive
Rate formula In underlying
Factor

Margin mode	Description
Additive	The margin is added to the floating index at each period.
Multiplicative	The floating index is multiplied by the margin (similar to an index factor).
In the underlying (Applies to formula based indices only)	The margin is added to each underlying index of the formula based resulting index used in the swap. Example: the margin is applied to each daily published index of an OIS used before compounding.

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MUREX™ Swap Generator
Rate formula: Floating leg

Rate Formula is a way to create payments on average indices. It is applicable for Zero-coupon or Frequency ratio payments.
The average of all index fixings is applied on all periods rather than each fixing on its period

Rate formula: Average
Day count: Standard
Mid exchange: ☐

Flow Tp	Calculation Start date	Calculation End date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flows	Cur
INC	30 Aug 2006	28 Feb 2007	100,000,000.00	28 Aug 2006	4.64151	4.99112	0.0000	1.0000		2,523,289.07	EUR
INC	28 Feb 2007	30 Aug 2007	100,000,000.00	26 Feb 2007	5.31067	4.99112	0.0000	1.0000		5,124,462.08	EUR
INT	30 Aug 2007	29 Feb 2008	100,000,000.00	28 Aug 2007	5.02118	4.99112	0.0000	1.0000	29 Feb 2008	7,791,630.84	EUR
INC	29 Feb 2008	29 Aug 2008	100,000,000.00	27 Feb 2008	5.30875	5.32044	0.0000	1.0000		2,689,776.37	EUR
INC	29 Aug 2008	27 Feb 2009	100,000,000.00	27 Aug 2008	5.23444	5.32044	0.0000	1.0000		5,451,901.72	EUR
INT	27 Feb 2009	31 Aug 2009	100,000,000.00	25 Feb 2009	5.41812	5.32044	0.0000	1.0000	31 Aug 2009	8,335,076.24	EUR
INC	31 Aug 2009	26 Feb 2010	100,000,000.00	27 Aug 2009	5.45556	5.55515	0.0000	1.0000		2,762,143.48	EUR
INC	26 Feb 2010	30 Aug 2010	100,000,000.00	24 Feb 2010	5.60660	5.55515	0.0000	1.0000		5,693,728.68	EUR
INT	30 Aug 2010	28 Feb 2011	100,000,000.00	26 Aug 2010	5.60330	5.55515	0.0000	1.0000	28 Feb 2011	8,664,128.14	EUR

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MUREX™ Swap Generator
Amortization

Capital changes do not always follow calculation periods (amortising starts at roll date and follows its own dates generator).

Amortizing details

Amortizing: Linear (rate)
Frequency: None
Roll date: Linear (rate)
Rate: Constant
Constant annuities
Coupon Reinvestment
Constant annuities (r)
Linear (amount)
Real Coupon reinvestment

Amortizing details

Amortizing: Linear (rate)
Generator:
Payment shifter: +0 DAY
Roll date: ☒ 20 Jan 2012
End date: Inherited
Rate: 5.000000

nominal
interest

6M 12M 18M 24M

Start date Maturity

Example: Period 2 (between 6 and 12 months) interests are computed using the amortised nominal after period 6.

Manual adjustment of capitals can be done directly in the flows screen (customised amortising).

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ Swap Generator
Amortization: Constant annuities(r)

Constant annuities(r): Capital repayment is calculated so that (capital repayment + interest) remains constant throughout the trade's life.

☒ Pay ☐ Receive
☒ fixed rate ☐ floating rate

Currency: EUR

Start date: 03 Jan 2006
Maturity: 04 Jan 2010
Frequency: 1y
Convention: LIN ACT/360
Rate: 4.000000
Leg NPV: -9,374,738

Amortizing details

Amortizing: Constant annuities (r)
Amortizing leg: 1
Rate: 5.000000
End nominal: 0
Annuitiy: 28,251,147

Global flow schedule: Source/Targetpad

Leg 1 Leg 2 All legs

Flow Tp	Calculation Start date	Calculation End date	Remaining Capital	Rate	Payment Date	Flows	Cur
NT	03 Jan 2006	03 Jan 2007	100,000,000.00	4.00000003	03 Jan 2007	-4,055,555.56 EUR	-5,069,444.44
AMO	03 Jan 2007	03 Jan 2008	76,819,297.57	4.00000003	03 Jan 2008	-23,181,702.43 EUR	-28,251,146.87
NT	03 Jan 2007	03 Jan 2008	76,819,297.57	4.00000003	03 Jan 2008	-3,115,408.72 EUR	-3,894,260.92
AMO	03 Jan 2008	05 Jan 2009	52,461,411.62	4.00000005	03 Jan 2009	-24,356,885.95 EUR	-28,251,146.87
NT	03 Jan 2008	05 Jan 2009	52,461,411.62	4.00000005	05 Jan 2009	-2,145,038.33 EUR	-2681361.038
AMO	05 Jan 2009	04 Jan 2010	26,891,625.79	4.00000004	05 Jan 2009	-25,569,785.83 EUR	-28,251,146.87
NT	05 Jan 2009	04 Jan 2010	26,891,625.79	4.00000004	04 Jan 2010	-1,087,616.87 EUR	-1,087,616.87

Flows with r=5% annuities (5%)

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ Swap Generator
Amortization: Coupon reinvestment

Coupon reinvestment: The flows will be reinvested into the nominal but will still be paid as well.

Case: At payment date of coupon, the counterparty is "reinvesting" 25% of the fixed rate interest payments.

Amortizing details

Amortizing: Coupon Reinvestment
Fraction: 0.250000
Roll date: ☒ 20 Jan 2012

Reinvest the fraction specified.
The full coupon is paid.

Leg 1 Leg 2 All legs

Flow Tp	Start Date	End Date	Remaining Capital	Rate	Payment Date	Flow	Cur	Interest flows
INT	19 Jan 2012	19 Sep 2012	100,000,000	3.000000	19 Sep 2012	-2,000,000.00 EUR		0.00
INT	19 Sep 2012	19 Sep 2013	100,500,000	3.000000	19 Sep 2013	-3,015,000.00 EUR		0.00
INT	19 Sep 2013	19 Sep 2014	101,253,750	3.000000	19 Sep 2014	-3,037,612.50 EUR		0.00
INT	19 Sep 2014	21 Sep 2015	102,013,153	3.000000	21 Sep 2015	-3,077,396.79 EUR		0.00
AMO	19 Sep 2012	19 Sep 2013	100,500,000		19 Sep 2012	500,000.00 EUR		
AMO	19 Sep 2013	19 Sep 2014	101,253,750		19 Sep 2013	753,750.00 EUR		
AMO	19 Sep 2014	21 Sep 2015	102,013,153		19 Sep 2014	759,403.12 EUR		

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ Swap Generator
Amortization: Real coupon reinvestment

Real coupon reinvestment: The flows will be reinvested into the nominal but will not be paid.

Case: At payment date of coupon, the counterparty is only paid 75% of the rate and the counterparty is "reinvesting" 25% of the fixed rate interest payment, which we will get back only at maturity.

Amortizing details

Amortizing: Real Coupon reinvestment
 Fraction: 0.250000
 Roll date: ☒ 20 Jan 2012
 Pay Coupon: ☐

Reinvest the fraction specified. The remaining money is paid only at maturity.

Fixed interest payments EUR

Flow	Start Date	End Date	Remaining Capital	Rate	Payment Date	Flow	Cur	Interest flows	Int/flows + Int/Int	Rate Convention	Nb of Days
NT	19 Jan 2012	19 Sep 2012	100,000,000	3.00000	19 Sep 2012	-1,500,000.00	EUR	0.00	0.00 LIN 30/360		240
INT	19 Sep 2012	19 Sep 2013	100,500,000	3.00000	19 Sep 2013	-2,261,250.00	EUR	0.00	0.00 LIN 30/360		360
INT	19 Sep 2013	19 Sep 2014	101,253,750	3.00000	19 Sep 2014	-2,278,209.37	EUR	0.00	0.00 LIN 30/360		360
INT	19 Sep 2014	21 Sep 2015	102,013,153	3.00000	21 Sep 2015	-2,308,047.59	EUR	0.00	0.00 LIN 30/360		362

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ Swap Generator
Other useful settings

Indexed	Generic term (MX terminology): We can input a formula on the coupon or notional with reference to an index that can be an FX rate, an inflation rate, etc. (multi currency below is a case of indexation). Useful for inflation linked swaps, non deliverable swaps, marked-to-market swaps.
Multi Currency	Useful for non deliverable swaps of emergent markets currencies: We trade on a specific currency, interests are calculated with this currency but payments are done in USD (or another liquid currency) with the current FX rate.
Initial exchange	
Intermediate payments	Concern the exchanges of capital.
Final exchange	These settings are not used for IRS but for currency swaps or specific swap structures.
Accrual/Yield calculation	This setting does not impact the structure of coupons and cash flows of a trade. It is for the display of risk views. There are different ways to see the risk: - Standard way: Mark-to-market (PV01: sensitivity to the rate curve). - Duration: Length of the deal in years. - For a swap 5Y, the risk will be only on the 5Y pillar and the duration is 5. - For an amortized swap 5Y (the notional decreases), there are intermediary risks and a lower risk on the 5Y pillar. The duration is inferior to 5 as the flows are weighted.

Currency swaps and IR swaps share the same configuration list (swap generators) but are physically stored as two trade types (same family 'IRD' but different groups 'IRS' and 'CS').

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MUREX™ | Agenda

Introduction

Swap Generator

Currency Swap Generator

- Definition
- Marked to market
- Forward start capital

Bond Generator

Cap/Floor, FRA, Loan Generators

Practice

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MUREX™ | Currency Swap Generator
Definition

Standard currency swap

Capital of the Currency 2	Remains constant
Equal to	Swap nominal × FX rate on trade date
Capitals are exchanged	At start and end dates

Leg 1 Leg 2 All legs

Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur
NT	27 Sep 2012	27 Dec 2012	100,000,000	25 Sep 2012	1.53000	2.53000	1.0000	1.0000	27 Dec 2012	-636,537.78	EUR
NT	27 Dec 2012	27 Mar 2013	100,000,000	21 Dec 2012	1.68146	2.68146	1.0000	1.0000	27 Mar 2013	-670,364.97	EUR
NT	27 Mar 2013	27 Jun 2013	100,000,000	25 Mar 2013	1.80335	2.80335	1.0000	1.0000	27 Jun 2013	-716,412.07	EUR
NT	27 Jun 2013	27 Sep 2013	100,000,000	25 Jun 2013	1.93643	2.93643	1.0000	1.0000	27 Sep 2013	-750,421.84	EUR
NT	27 Sep 2013	27 Dec 2013	100,000,000	25 Sep 2013	2.06878	3.06878	1.0000	1.0000	27 Dec 2013	-775,718.65	EUR
NT	27 Dec 2013	27 Mar 2014	100,000,000	23 Dec 2013	2.19970	3.19970	1.0000	1.0000	27 Mar 2014	-799,925.54	EUR
NT	27 Mar 2014	27 Jun 2014	100,000,000	25 Mar 2014	2.33160	3.33160	1.0000	1.0000	27 Jun 2014	-851,409.92	EUR
NT	27 Jun 2014	29 Sep 2014	100,000,000	25 Jun 2014	2.46648	3.46648	1.0000	1.0000	29 Sep 2014	-905,136.95	EUR
PR									27 Sep 2012	100,000,000.00	EUR
PR									29 Sep 2014	-100,000,000.00	EUR

EUR EURIBOR 3M

☐ Exclude additional flows ☐ Exclude capital flows
☐ Exclude premium flows ☐ Exclude capitalized interests
☐ Exclude security flows ☒ Exclude residual flows

End-user session: Pricing/ e-Tradepad

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MUREX™ | **Currency Swap Generator**
Marked to market

Marked to market

Nominal of the Currency 2	Evolves with the FX rate of each period start date
Capitals are exchanged	Intermediately at each period

Global flow schedule Source:Notepad

Leg 1 Leg 2 All legs

Floating interest payments USD											
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur
INT	27 Sep 2012	27 Dec 2012	135,005,000	25 Sep 2012	1.65000	1.65000	0.0000	1.0000	27 Dec 2012	563,083.35	USD
INT	27 Dec 2012	27 Mar 2013	135,051,915	21 Dec 2012	3.36803	3.36803	0.0000	1.0000	27 Mar 2013	1,137,145.97	USD
INT	27 Mar 2013	27 Jun 2013	135,598,320	25 Mar 2013	3.31007	3.31007	0.0000	1.0000	27 Jun 2013	1,147,035.25	USD
INT	27 Jun 2013	27 Sep 2013	136,123,415	25 Jun 2013	3.38700	3.38700	0.0000	1.0000	27 Sep 2013	1,178,239.75	USD
INT	27 Sep 2013	27 Dec 2013	136,628,321	25 Sep 2013	3.46403	3.46403	0.0000	1.0000	27 Dec 2013	1,196,359.63	USD
INT	27 Dec 2013	27 Mar 2014	136,670,426	23 Dec 2013	3.55131	3.55131	0.0000	1.0000	27 Mar 2014	1,213,398.38	USD
INT	27 Mar 2014	27 Jun 2014	136,523,922	25 Mar 2014	3.59641	3.59641	0.0000	1.0000	27 Jun 2014	1,254,768.30	USD
INT	27 Jun 2014	29 Sep 2014	136,166,464	25 Jun 2014	3.60097	3.60097	0.0000	1.0000	29 Sep 2014	1,280,309.72	USD
PRI									27 Sep 2012	-135,005,000.00	USD
PRI									29 Sep 2014	136,166,463.57	USD
PRI	27 Dec 2012	27 Mar 2013	135,051,915						27 Dec 2012	-46,914.94	USD
PRI	27 Mar 2013	27 Jun 2013	135,598,320						27 Mar 2013	-546,404.96	USD
PRI	27 Jun 2013	27 Sep 2013	136,123,415						27 Jun 2013	-525,094.61	USD

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ | **Currency Swap Generator**
Forward start capital

Forward start capital

Swap start date	Future date
Nominal of currency 2	Swap nominal * FX rate at forward start date

Global flow schedule Source:Notepad

Leg 1 Leg 2 All legs

Floating interest payments USD											
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow	Cur
INT	27 Sep 2012	27 Dec 2012	135,005,000	25 Sep 2012	1.65000	1.65000	0.0000	1.0000	27 Dec 2012	563,083.35	USD
INT	27 Dec 2012	27 Mar 2013	135,005,000	21 Dec 2012	3.36803	3.36803	0.0000	1.0000	27 Mar 2013	1,136,750.94	USD
INT	27 Mar 2013	27 Jun 2013	135,005,000	25 Mar 2013	3.31007	3.31007	0.0000	1.0000	27 Jun 2013	1,142,016.32	USD
INT	27 Jun 2013	27 Sep 2013	135,005,000	25 Jun 2013	3.38700	3.38700	0.0000	1.0000	27 Sep 2013	1,168,559.12	USD
INT	27 Sep 2013	27 Dec 2013	135,005,000	25 Sep 2013	3.46403	3.46403	0.0000	1.0000	27 Dec 2013	1,182,145.34	USD
INT	27 Dec 2013	27 Mar 2014	135,005,000	23 Dec 2013	3.55131	3.55131	0.0000	1.0000	27 Mar 2014	1,198,612.26	USD
INT	27 Mar 2014	27 Jun 2014	135,005,000	25 Mar 2014	3.59641	3.59641	0.0000	1.0000	27 Jun 2014	1,240,808.15	USD
INT	27 Jun 2014	29 Sep 2014	135,005,000	25 Jun 2014	3.60097	3.60097	0.0000	1.0000	29 Sep 2014	1,269,389.02	USD
PRI									27 Sep 2012	-135,005,000.00	USD
PRI									29 Sep 2014	135,005,000.00	USD

End-user session: Pricing/ e-Tradepad/ Right-click Leg: detail/ Characteristics/ Amortizing details

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MUREX™ | **Agenda**

Introduction

Swap Generator

Currency Swap Generator

Bond Generator

- Bond security
- Anatomy of a bond generator
- Schedule definition
- Accrual and yield calculations

Cap/Floor, FRA, Loan Generators

Practice

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MUREX™ | **Bond Generator**
Bond security

```

graph LR
    BG[Bond Generator] --> BA[Bond A]
    BG --> BB[Bond B]
    BA --> BC1[Bond Contract 1]
    BA --> BC2[Bond Contract 2]
    BB --> BC3[Bond Contract 3]
  
```

To define a new bond in MX.3:

Specify the following information in the configuration of the security:

- >Nominal amount
- >Maturity
- >Coupon rate
- >Issue date
- >ISIN
- >Issuer ...

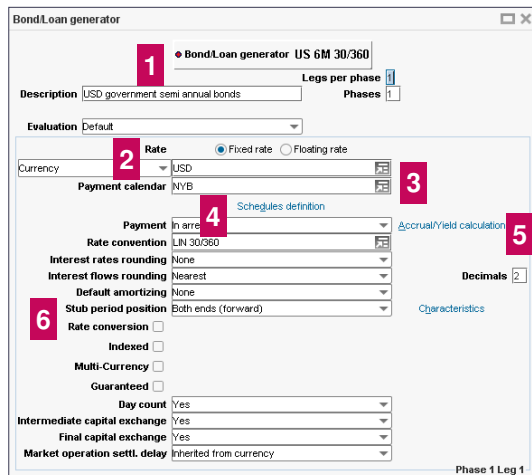
End-user session: Configuration/ Securities/ Bonds

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MUREX™ Bond Generator
Anatomy of a bond generator

A **bond generator** is viewed as a **one-leg swap generator**, with all flags available.



To define a new bond generator:

Specify the following information in the configuration of the security:

- 1) Label of Bond Generator
- 2) Select rate type
 - If floating rate, select an index
- 3) Set payment currency and calendar
- 4) Complete Schedules definition
- 5) Complete accrual/ yield calculation methods
- 6) Add features if applicable:
 - Amortization
 - Broken periods etc.

Payment currency	is the currency in which nominal and coupon payments are made
Payment calendar	used for coupon calculation period and coupon payment dates
Schedules definition	stores rules used for coupon calculation and payment dates
Rate convention	used for coupon flow calculation

End-user session: Configuration/ Generators/ Bond generators

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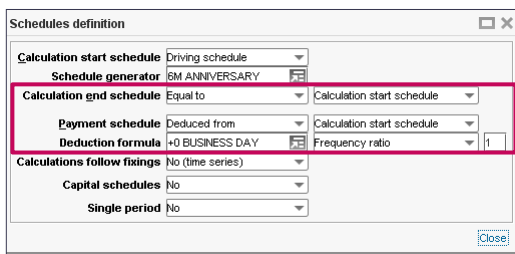
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MUREX™ Bond Generator
Schedule definition

Define calculation and payment schedules:

- ✓ One is the driving schedule
- ✓ The others are either Equal to or Deduced from the driving schedule.

If deduced, then a deduction formula is defined (e.g. + 2 open days).



End-user session: Configuration/ Generators/ Bond generators

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Bond Generator

Accrual and yield conventions

Specific conventions can be defined for accrual and yield calculations.
Yield can be based on the payment schedule or the anniversary schedule (calculation schedule).

Accrual & Yield conventions

Accrual conventions

Accrual method: Use interest convention
 Accrual rounding: None
 Accrual rounding mode: Standard

Yield conventions

Yield calculation: AIBD
 Yield compounding: at coupon frequency (standard)
 Gross to clean accrual: Standard
 Alternate yield convention: During last period
 Alternate yield schedule: Anniversary schedule
 Duration: Macaulay

Convention: LIN 30/360 X CP

Close

End-user session: Configuration/ Generators/ Bond generators

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Agenda

- Introduction
- Swap Generator
- Currency Swap Generator
- Bond Generator
- Cap/Floor, FRA, Loan Generators**
 - Overview
 - Cap-floor generator
 - Forward Rate Agreements (FRA)
- Practice

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Other Generators

Overview

Generators are necessary for complex structures, high-volume deal booking. Simple trades are often built on the selection of currency or index only.

Cap Floors (CFs), Forward Loan Agreements (FRAs) and Loan and Borrowing (LN_BR) trades can be booked with or without the use of generators.

The two types of generators are:

One leg generator	Bonds, CFs, LN_BR
Two leg generators	IRS, CS, FRAs

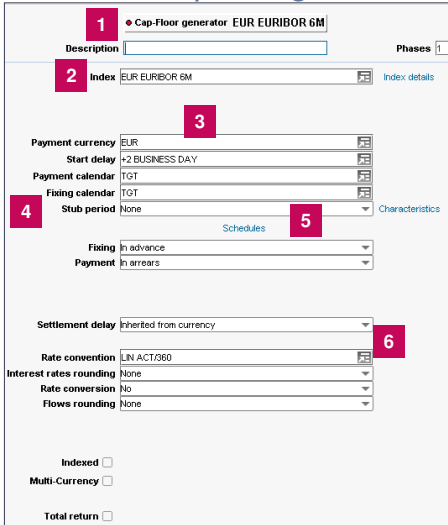
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Other Generators

Cap-floor generator



The configuration of a CF generator is similar to an index definition screen and can also be considered as a one-leg swap definition. When a CF generator is not used to book the CF trade, the application applies the index characteristics to the trade (e.g.: generic CF)

To define a new CF generator:

Specify the following information in the configuration of the CF generator:

- 1) Label of CF Generator
- 2) Select index
- 3) Set payment currency and calendar(inherited)
- 4) Broken period
- 5) Set schedules
- 6) Rate convention

End-user session: Configuration/ Rates/Cap/Floor generators

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MUREX™ | **Other Generators**
Forward Rate Agreement generator

The configuration of a FRA generator is similar to the configuration of a swap generator (two legs). Users can customize the terms of the FRA, such as the stub period, conventions etc.

End-user session: Configuration/ FRA generators

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MUREX™ | **Agenda**

- Introduction
- Swap Generator
- Currency Swap Generator
- Bond Generator
- Cap/Floor, FRA, Loan Generators
- Practice**

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 **Practice 6**
Generator Customization at trade level

 Customize a generator when booking a trade.

1. Login & password: MUREXFO into group FO
2. Go to the Pricing/ e-Tradepad
3. Insert a IRS trade by selecting a generator USD LIBOR S 3M and customize it:
 - > Receive 1.5% USD semi-annually
 - > Pay floating USD LIBOR 3M quarterly
 - > USD 250 million as nominal
 - > 5Y maturity
4. Create a 2Y customization over the last 2 years with a fixed rate of 1.6% with a margin of 10 bp.

 **Solution**

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 **Practice 7**
Generator Configuration from Scratch

 Configure a swap generator from scratch

1. Login & password: MUREXFO into group FO
2. In the end-user session, go to menu: Configuration/ Generators/ Swap generators
3. Hit "Insert" to configure a new swap generator with the following characteristics:
 - > Label = "Generator_Training"
 - > USD Libor 6M paying every 2 years
 - > Stub period: both ends backwards
 - > Fixed rate paying annually

 **Solution**

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SOLUTION TO EXERCISES

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Swap Generator
Practice I

Identify the payment schedules: which is zero-coupon and which is using a frequency ratio?

A is frequency ratio and B is zero-coupon

A

Leg 1	Leg 2	All legs								
Floating interest payments EUR										
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow
NC	08 Apr 2010	08 Jun 2010	100,000,000	06 Apr 2010	1.10000	1.10000	0.0000	1.0000		186,388.89
NT	08 Jun 2010	08 Dec 2010	100,000,000	04 Jun 2010	1.64250	1.64250	0.0000	1.0000	08 Dec 2010	1,022,882.62
NC	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.0000		505,555.56
INC	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.0000		2,183,121.34
INT	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.0000	08 Jun 2012	3,593,556.31
INC	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.0000		1,578,952.21
INC	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.0000		3,298,192.25
INT	10 Jun 2013	09 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.0000	09 Dec 2013	5,111,822.66

B

Leg 1	Leg 2	All legs								
Floating interest payments EUR										
Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate	Margin	Rate Factor	Payment Date	Flow
NC	08 Apr 2010	08 Jun 2010	100,000,000	06 Apr 2010	1.10000	1.10000	0.0000	1.0000		186,388.89
NC	08 Jun 2010	08 Dec 2010	100,000,000	04 Jun 2010	1.64250	1.64250	0.0000	1.0000		1,022,882.62
NC	08 Dec 2010	08 Jun 2011	100,000,000	06 Dec 2010	1.00000	1.00000	0.0000	1.0000		1,533,609.42
INC	08 Jun 2011	08 Dec 2011	100,000,000	06 Jun 2011	3.28353	3.28353	0.0000	1.0000		2,238,334.73
INC	08 Dec 2011	08 Jun 2012	100,000,000	06 Dec 2011	2.71535	2.71535	0.0000	1.0000		
INC	08 Jun 2012	10 Dec 2012	100,000,000	06 Jun 2012	3.07256	3.07256	0.0000	1.0000		
INC	10 Dec 2012	10 Jun 2013	100,000,000	06 Dec 2012	3.34783	3.34783	0.0000	1.0000		
INT	10 Jun 2013	09 Dec 2013	100,000,000	06 Jun 2013	3.47286	3.47286	0.0000	1.0000	09 Dec 2013	10,602,882.62

Exercise

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MUREX™ | **Swap Generator**
Practice 2

 Match the right flow schedules to the correct corresponding stub periods
Which is the right flow schedule for "in arrears" stub period?

C

Flow schedule Source:Notepad Leg:2											
20546 Floating interest payments EUR (Rec)											
Flow Tp	Calculation Start date	Calculation End date	Remaining Capital	Fixing Date	Rate	Margin	Rate Factor	Payment Date	Flows	Cur	
INT	03 Jan 2006	03 Jul 2006	100,000,000.00	30 Dec 2005	3.65856	0.0000	1.0000	03 Jul 2006	1,839,441.44	EUR	
INT	03 Jul 2006	03 Jan 2007	100,000,000.00	29 Jun 2006	4.40626	0.0000	1.0000	03 Jan 2007	2,252,088.23	EUR	
INT	03 Jan 2007	03 May 2007	100,000,000.00	29 Dec 2006	4.97030	0.0000	1.0000	03 May 2007	1,656,766.67	EUR	

 **Exercise**

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MUREX™ | **Swap Generator**
Practice 3

 Match the right flow schedules to the correct corresponding stub periods frequency ratio? Which is the right flow schedule for "up front" stub period?

B

Flow schedule Source:Notepad Leg:2											
20546 Floating interest payments EUR (Rec)											
Flow Tp	Calculation Start date	Calculation End date	Remaining Capital	Fixing Date	Rate	Margin	Rate Factor	Payment Date	Flows	Cur	
INT	03 Jan 2006	03 May 2006	100,000,000.00	30 Dec 2005	3.61324	0.0000	1.0000	03 May 2006	1,204,413.33	EUR	
INT	03 May 2006	03 Nov 2006	100,000,000.00	28 Apr 2006	4.11184	0.0000	1.0000	03 Nov 2006	2,101,605.50	EUR	
INT	03 Nov 2006	03 May 2007	100,000,000.00	01 Nov 2006	4.88141	0.0000	1.0000	03 May 2007	2,454,264.10	EUR	

 **Exercise**

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MUREX™ Swap Generator Practice 4

Match the right flow schedules to the correct corresponding stub periods

Both ends (forward) (IRS 1Y4M) Both ends (backward) (IRS 1Y4M)

I Generator: EURBOR A 6M
Start date: Spot 03 Jan 2006
Maturity: 1y4m 03 May 2007
Nominal: 100,000,000 EUR

2 Generator: EURBOR A 6M
Start date: Spot 03 Jan 2006
Maturity: 1y4m 03 May 2007
Nominal: 100,000,000 EUR

Roll date: 01 Mar 2007 Roll date: 01 Feb 2006

Payment calendar: TGT Payment calendar: TGT

A Flow schedule Source: Notepad Leg: 2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tr	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Feb 2006	100,000,000.00	30 Dec 2005	3.47346	0.0000	1.00000	01 Feb 2006	279,806.47	EUR
INT	01 Feb 2006	01 Aug 2006	100,000,000.00	30 Jan 2006	3.75339	0.0000	1.00000	01 Aug 2006	1,887,120.25	EUR
INT	01 Aug 2006	01 Feb 2007	100,000,000.00	28 Jul 2006	4.52527	0.0000	1.00000	01 Feb 2007	2,312,914.16	EUR
INT	01 Feb 2007	03 May 2007	100,000,000.00	30 Jan 2007	5.21492	0.0000	1.00000	03 May 2007	1,318,215.67	EUR

B Flow schedule Source: Notepad Leg: 2

Flow	Calculation	Calculation	Remaining	Fixing	Rate	Margin	Rate	Payment	Flows	Cur
Tr	Start date	End date	Capital	Date			Factor	Date		
INT	03 Jan 2006	01 Mar 2006	100,000,000.00	30 Dec 2005	3.52524	0.0000	1.00000	01 Mar 2006	558	EUR
INT	01 Mar 2006	01 Sep 2006	100,000,000.00	27 Feb 2006	3.81172	0.0000	1.00000	01 Sep 2006	1,948	EUR
INT	01 Sep 2006	01 Mar 2007	100,000,000.00	30 Aug 2006	4.64828	0.0000	1.00000	01 Mar 2007	2,337	EUR
INT	01 Mar 2007	03 May 2007	100,000,000.00	27 Feb 2007	5.31309	0.0000	1.00000	03 May 2007	929,790.57	EUR

Exercise

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MUREX™ Swap Generator Practice 5

Match the right flow schedules to the correct stub periods coupon.
Which is the right flow schedule for "short coupon" and "long coupon"?

A is long coupon and B is short coupon

A Global flow schedule Source: Notepad

Flow	Start	End	Remaining	Fixing	Rate	Convert	Margin	Rate	Payment	Flow	Cur
Tr	Date	Date	Capital	Date		Rate		Factor	Date		
INT	08 Apr 2010	08 Dec 2010	100,000,000.00	06 Apr 2010	1.50900	1.50900	0.0000	1.00000	08 Dec 2010	1,022,766.67	EUR
INT	08 Dec 2010	08 Jun 2011	100,000,000.00	06 Dec 2010	1.00000	1.00000	0.0000	1.00000	08 Jun 2011	505,555.56	EUR
INT	08 Jun 2011	08 Dec 2011	100,000,000.00	06 Jun 2011	3.28353	3.28353	0.0000	1.00000	08 Dec 2011	1,669,127.42	EUR
INT	08 Dec 2011	08 Jun 2012	100,000,000.00	06 Dec 2011	2.71535	2.71535	0.0000	1.00000	08 Jun 2012	1,380,301.32	EUR
INT	08 Jun 2012	10 Dec 2012	100,000,000.00	06 Jun 2012	3.07256	3.07256	0.0000	1.00000	10 Dec 2012	1,578,952.21	EUR
INT	10 Dec 2012	10 Jun 2013	100,000,000.00	06 Dec 2012	3.34783	3.34783	0.0000	1.00000	10 Jun 2013	1,692,518.02	EUR
INT	10 Jun 2013	09 Dec 2013	100,000,000.00	06 Jun 2013	3.47286	3.47286	0.0000	1.00000	09 Dec 2013	1,755,723.28	EUR

B Global flow schedule Source: Notepad

Flow	Start	End	Remaining	Fixing	Rate	Convert	Margin	Rate	Payment	Flow	Cur
Tr	Date	Date	Capital	Date		Rate		Factor	Date		
INT	08 Apr 2010	08 Jun 2010	100,000,000.00	06 Apr 2010	1.10000	1.10000	0.0000	1.00000	08 Jun 2010	186,388.89	EUR
INT	08 Jun 2010	08 Dec 2010	100,000,000.00	04 Jun 2010	1.64250	1.64250	0.0000	1.00000	08 Dec 2010	834,937.50	EUR
INT	08 Dec 2010	08 Jun 2011	100,000,000.00	06 Dec 2010	1.00000	1.00000	0.0000	1.00000	08 Jun 2011	505,555.56	EUR
INT	08 Jun 2011	08 Dec 2011	100,000,000.00	06 Jun 2011	3.28353	3.28353	0.0000	1.00000	08 Dec 2011	1,669,127.42	EUR
INT	08 Dec 2011	08 Jun 2012	100,000,000.00	06 Dec 2011	2.71535	2.71535	0.0000	1.00000	08 Jun 2012	1,380,301.32	EUR
INT	08 Jun 2012	10 Dec 2012	100,000,000.00	06 Jun 2012	3.07256	3.07256	0.0000	1.00000	10 Dec 2012	1,578,952.21	EUR
INT	10 Dec 2012	10 Jun 2013	100,000,000.00	06 Dec 2012	3.34783	3.34783	0.0000	1.00000	10 Jun 2013	1,692,518.02	EUR
INT	10 Jun 2013	09 Dec 2013	100,000,000.00	06 Jun 2013	3.47286	3.47286	0.0000	1.00000	09 Dec 2013	1,755,723.28	EUR

Exercise

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MUREX™ | Practice 6
Practice 6

 Customize a generator when booking a trade.

Product	#1 IRS Vanilla	
Instrument	USD LIBOR S 3M	
Template details	Open	
Start period	Spot	Thu 27/09/12
Maturity period	Wed 27/09/17	
Adjusted maturity	Wed 27/09/17	
Nominal	250,000,000	USD
Flows	Open	
Leg	Pay	Receive
Leg type	Fixed	Floating
Currency	USD	USD
Index	USD LIBOR 3M	
Npv	-6,065,709	USD
Bpv	-121,694	
Typology Usage	IRS	Libor
Fees	Open	
CSA Type		
Int. Counterparty	No	ABBEEY NATIONAL PI
Portfolio	MX_PTF	

 Exercise

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MUREX™ | Practice 6
Practice 6

 **Solution 6:** Customize a generator when booking a trade.

- Edit/Define Phases
- Change Rate/Margin to independent
- Create a second phase of 2Y
- 4) Change the rate and margin for the 2nd phase

Phase definition

Constraints across phases

	1	Pay	2	Receive
Rate/margin	Common		Common	
Strike	Common		Common	
Pay/Receive	Common		Common	
Cap/Floor	Common		Common	

Joint phases ☒ Yes ☐ No

Phase	Start	Length	End	Payout
1	27 Sep 2012	3Y	27 Sep 2015	Pay USD Receive USD LIBOR 3M
2	27 Sep 2015	2Y	27 Sep 2017	Pay USD Receive USD LIBOR 3M

☒ Pay ☐ Receive

☐ fixed rate ☒ floating rate

Currency: USD

Start date: 27 Sep 2012

Maturity: 28 Sep 2015

Schedules definition: Details

Convention: LIN 30/360

Rate: 1.600000

Leg NPV: -4,586,126

Leg BPV: 28,663

☐ Pay ☒ Receive

☐ fixed rate ☒ floating rate

Currency: USD

Start date: 27 Sep 2012

Maturity: 28 Sep 2015

Schedules definition: Details

Convention: LIN ACT/360

Margin: 0.100000

Leg NPV: 5,697,076

Leg BPV: 28,118

 Exercise

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MUREX™ | Practice 7
Practice 7

 Configure a swap generator from scratch

Schedules Common set different frequencies

Stub period Both ends (backward)

Market quote Automatic

Bid/Ask driving leg Automatic

Settlement delay Inherited from currency

Default amortizing None

Evaluation Default

Future Cash proceed cut-off

Main

☒ Fixed rate ☐ Floating rate

Evaluation Inherited

Currency USD

Payment frequency 1y

Payment In arrears

Main

☐ Fixed rate ☒ Floating rate

Index USD LIBOR 6M

Evaluation Inherited

Currency USD

Start delay +2 BUSINESS DAY

Payment calendar NYB

Fixing calendar LNB

Schedules definition

Fixing In advance

Payment In arrears

Compounding (t-M) at (t-M)

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 Exercise

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MUREX™ | Practice 7
Practice 7

 Configure a swap generator from scratch

Schedules definition

Calculation start schedule Driving schedule

Schedule generator 6M MODFOL

Calculation end schedule Equal to

Payment schedule Equal to

Fixing schedule Equal to

Calculations follow fixings No (time series)

Capital schedules No

Single period No

Calculation start schedule

Frequency ratio 4

Backward (broken periods included)

Calculation start schedule

Frequency ratio 1

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 Exercise

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Thank You!

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1

A presentation slide titled 'Learning Objectives Market Parameters'. The top left features the Murex logo (a stylized 'M' with a red swoosh) and the text 'MUREX™'. The title 'Learning Objectives' is in a large, bold font, and 'Market Parameters' is below it in a smaller font. The main content area has the text 'At the end of this course, you will be able to:' followed by a list of four bullet points, each preceded by a checkmark. The bottom of the slide has a small copyright notice and a page number. A large, diagonal watermark across the slide reads 'Accenture: In-house FO Modules August 2020 DO NOT COPY'.

Learning Objectives

Market Parameters

At the end of this course, you will be able to:

- ✓ Explain how market data is used in MX.3
- ✓ Explain market data sets and how they are used in the application
- ✓ Explain the different modes to update market data
- ✓ Explain the difference between "Historical data" and "Past market data prices"

2



Agenda

Introduction to Market Data
Market Data Sets
Market Data Update
Historical Data

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3



Introduction to Market Data

Market data parameters

Needed for	Pricing & deal booking Evaluating open positions: sensitivities and P&L computations Computing risk scenarios, P&L Variance
Can include	Market prices (stock, credit spreads, bond price, spot rate...) Interest rates Volatilities, Recovery rates, Correlations Rate curves and Rate curve assignments Etc...
Require the configuration of	Financial instruments (End-user session/ Configuration menu) FO desks and Processing center (Configurator session) Market data sets



In the market data sets we define:

- ✓ Which market data are used by each desk?
- ✓ How are the different data imported?
- ✓ How are the different data saved?

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 **Agenda**

Introduction to Market Data

Market Data Sets

- Definition
- Organisation
- End-user selection
- Market data screens
- Market data viewer
- Audit and purge

Market Data Update

Historical Data

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 **Market Data Sets**

Definition

A **Market data set** is a collection of market parameters values at different dates.
It is defined by:

Label	Name of market data set
Alias	Physical set of data stored in tables
Rate Curve Assignment	Specific assignment of curves to instruments, currencies etc.

For each market data type, contained in a market data set, there are 3 dimensions:
date, class, value.



Two different market data sets can share the same Alias.
In this case market data modification (e.g. Rate modification) done on one set will impact the sets sharing the same alias.

Market data set

Label:

Description:

Alias:

Rate curve assignment:

Type:

Collateral assignment:

OK Cancel

Configurator session: Infrastructure/ Organization/ Market data set

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MUREX™ | **Market Data Sets Organization**

Market data sets are assigned at different levels depending on the type of user.

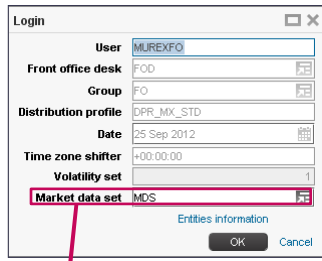
	Market data set definition	Menu
Front office	Desk Level One session refers to one desk → Only one default market data set	Configurator session: Infrastructure/ Organization/ Front Office Desk
Back office	Closing Entity Each portfolio will inherit the market data set of the closing entity it belongs to.	Configurator session: Infrastructure/ Organization/ Closing Entity Configurator session: Infrastructure/ Trade Classification/ Portfolio Tree Templates

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MUREX™ | **Market Data Sets End-user selection**

Users are able to switch between market data sets, using "ctrl+U":



LABEL	DESCRIPTION	ALIAS	RATE CURVE ASSIGNMENT	TYPE	COLLATERAL ASSIGNMENT
MDS	MARKET DATA SET	MDS	MAIN	Standard	
RISK	RISK MARKET DATA SET	JRISK	MAIN	Standard	

 Changing the market data set in a session does not change the real-time data feed coming into the session (if connected to real-time).

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MUREX™ | **Market Data Sets**
Market data screen: FX spots

Standard screen for FX spot rates

Spot rates

Display

Mode Bid/Ask

Show Low/High

Spot scenario

STD

Reference Currency USD

Market parameters

Values

Cross quotation

25 Sep 2012

Date

Global save

End-user session: Market data/ Forex/ Spots

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MUREX™ | **Market Data Sets**
Market data screen: Equity and future prices

Standard screen for equity and futures prices

Market FR EURONXT

View SECURITY MAIN

25 Sep 2012

25 Sep 2012

Main quotation

Second quotation

Main Closing Price

Main Last Price

ISIN Code

Time

End-user session: Market data/ Equities/Equity and futures prices

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MUREX™ | **Market Data Sets**
Market data screen: Bonds and future prices

Standard screen for bond prices

Currency	Security	MLabel	Main quotation	Second quotation	Main Closing Price	Main Last Price	ISIN Code	Time
EUR	DBR 1.5% 4/16		101.014	1.205360	101.029	101.046	DE0001030500	17:11:54
	DBR 5.625% 1/28		101.014	5.522015	101.066	101.067	DE0001135069	17:11:54
	OBL 2.25% 4/14		100.879	1.001963	100.948	100.973	DE0001141547	17:11:54
	OBL 1.25% 4/13		102.695	-2.563852	102.692	102.723	DE0001030918	17:11:54
	DBR 3.75% 1/19		102.493	3.301007	102.509	102.564	DE0001135374	17:11:54
	DBR 3.75% 1/17		101.045	3.479035	101.111	101.127	DE0001135317	17:11:54
	DBR 3.75% 1/15		100.947	3.304267	100.950	100.988	DE0001135267	17:11:54
	OBL 4% 10/13		101.202	2.804152	101.245	101.275	DE0001141539	17:11:54
	DBR 3% 7/20		102.301	2.060908	102.348	102.350	DE0001135408	17:11:54
	BKO 0.75% 9/13		103.260	-2.565505	103.295	103.311	DE0001137354	17:11:54
	DBR 4.5% 1/13		101.082	0.438726	101.117	101.123	DE0001135218	17:11:54
	DBR 3.75% 7/13		103.644	-0.972975	103.704	103.731	DE0001135234	17:11:54
	DBR 3.5% 1/16		101.454	3.022650	101.515	101.549	DE0001135291	17:11:54
	DBR 4.75% 7/40		103.260	4.539877	103.320	103.323	DE0001135366	17:11:54
	DBR 5.5% 7/27		102.493	6.233101	102.537	102.552	DE0001135044	17:11:54
	DBR 4.25% 7/39		102.493	4.094004	102.515	102.543	DE0001135325	17:11:54
	DBR 4.25% 7/14		102.108	2.998754	102.176	102.194	DE0001135259	17:11:54
	DBR 4.75% 7/28		101.342	4.626425	101.355	101.415	DE0001135085	17:11:54
	DBR 4.25% 7/17		100.307	4.173848	100.318	100.363	DE0001135333	17:11:54
	DBR 4.25% 7/15		100.000	4.246941	100.014	100.077	DE0001135358	17:11:54
	DBR 4.75% 7/34		100.000	4.748524	100.009	100.058	DE0001135226	17:11:54
	DBR 3.25% 7/15		101.405	2.712443	101.412	101.503	DE0001135283	17:11:54
	DBR 3.25% 1/20		100.522	3.166934	100.522	100.560	DE0001135390	17:11:54
	OBL 1.75% 10/15		100.559	1.559639	100.564	100.552	DE0001141588	17:11:54
	OBL 2.5% 2/15		101.014	2.062513	101.035	101.067	DE0001141562	17:11:54
	OBL 2.5% 10/14		101.014	1.985537	101.024	101.112	DE0001141554	17:11:54

End-user session: Market data/ Bonds/Equity and futures prices

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MUREX™ | **Market Data Sets**
Market data screen: Revaluation rate curve

Standard screen for revaluation rate curve

Label	Generator	Maturity	do factor	do rate	Market quote	Curve spread
27 Sep 2012	CASH	27 Mar 2013	6M	0.9991	0.1723	0.0000
29 Oct 2012	FRA 6M	29 Apr 2013	1M-7M	0.9978	0.3736	0.0000
27 Nov 2012	FRA 6M	27 May 2013	2M-8M	0.9978	0.3330	0.0000
27 Dec 2012	FRA 6M	27 Jun 2013	3M-9M	0.9977	0.3970	0.0000
28 Jan 2013	FRA 6M	29 Jul 2013	4M-10M	0.9976	0.2928	0.0000
27 Feb 2013	FRA 6M	27 Aug 2013	5M-11M	0.9974	0.2841	0.0000
27 Mar 2013	FRA 6M	27 Sep 2013	6M-12M	0.9972	0.2784	0.0000
29 Apr 2013	FRA 6M	29 Oct 2013	7M-13M	0.9958	0.3840	0.0000
27 May 2013	FRA 6M	27 Nov 2013	8M-14M	0.9957	0.3240	0.0000
27 Jun 2013	FRA 6M	27 Dec 2013	9M-15M	0.9956	0.3502	0.0000
29 Jul 2013	FRA 6M	27 Jan 2014	10M-16M	0.9954	0.3455	0.0000
27 Aug 2013	FRA 6M	27 Feb 2014	11M-17M	0.9952	0.3409	0.0000
27 Sep 2013	FRA 6M	27 Mar 2014	12M-18M	0.9949	0.3386	0.0000
27 Sep 2012	EURBOR A 6M	29 Sep 2014	2Y	0.9905	0.4725	0.0000
27 Sep 2012	EURBOR A 6M	28 Sep 2015	3Y	0.9822	0.5971	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2016	4Y	0.9694	0.7766	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2017	5Y	0.9519	0.9841	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2018	6Y	0.9309	1.1922	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2019	7Y	0.9075	1.3848	0.0000
27 Sep 2012	EURBOR A 6M	28 Sep 2020	8Y	0.8828	1.5559	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2021	9Y	0.8576	1.7042	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2022	10Y	0.8321	1.8360	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2023	11Y	0.8062	1.9565	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2024	12Y	0.7804	2.0638	0.0000
27 Sep 2012	EURBOR A 6M	29 Sep 2025	13Y	0.7563	2.1600	0.0000
27 Sep 2012	EURBOR A 6M	28 Sep 2026	14Y	0.7314	2.2312	0.0000
27 Sep 2012	EURBOR A 6M	27 Sep 2027	15Y	0.7080	2.2901	0.0000

End-user session: Market data/ Revaluation Rate Curve

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MUREX™ Market Data Sets

Market data viewer

The **Market data viewer** is a powerful tool covering all the aspects of the market data management and monitoring. It allows to update, publish and retrieve in real-time all the market data present into the application.

The tool is viewer based so the monitoring view is fully user-definable.

End-user session: Market data/Viewer

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MUREX™ Market Data Sets

Audit and purge

Audit of market data: Helps user to identify gaps in the P&L caused by market parameters changes.

Several filtering criteria are possible:

Time	Modification date, trading date
User	The last modifier
Market Parameter	Date, market data set, type, instrument

Purge of market data can be done per alias or per desk.

Tools

- Simulation
- Pricing
- Configuration
- Market data
- Historical data
- Processing
- Middle office
- Risk

Tools

- Copy market data set
- Perform end of day copy
- Undo end of day copy
- Market data audit
- Current Session/Subscription Rule
- SHM-Activity Feeder Monitor
- Import/Export wizard

Do not remove market data as of system date!

End-user session: Market data/Tools/Market data audit
Configurator session: Processing center/Scripts/Scripts definition

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MUREX™ | Agenda

Introduction to Market Data

Market Data Sets

Market Data Update

- Real-time update
- Update modes

Historical Data

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MUREX™ | Market Data Update
Real-time update

Market data prices for the different instruments in the Market data, Pricing and Simulation screens can be updated in Real-Time with their corresponding prices in Reuters, Bloomberg, etc. This update is handled by the RT module below.

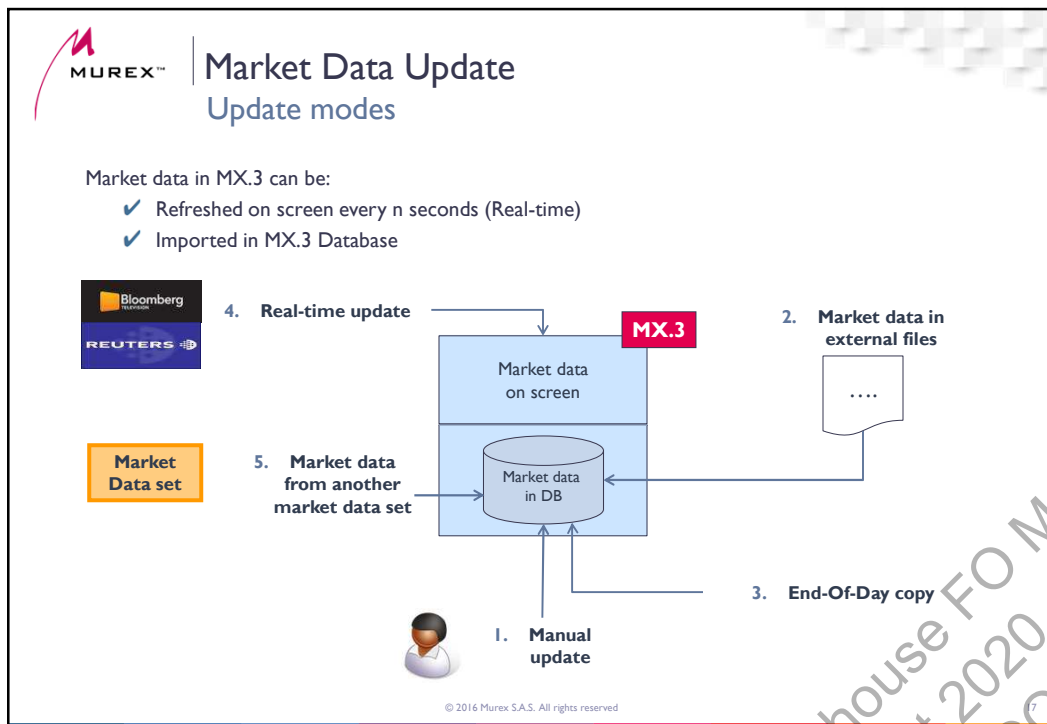
```

graph LR
    subgraph Sources
        B[Bloomberg]
        R[REUTERS]
        O[Other sources]
    end
    Sources --> RTBS[Real-time Bridging Service RTBS]
    RTBS --> MDCS[Market Data Contribution Server MDCS]
    MDCS -- "Subscription Rules" --> MS1[MX Session]
    MDCS -- "Subscription Rules" --> MS2[MX Session]
  
```

Real-time update bypasses assigned market data set.
 RT prices are refreshed only on screen and not saved in the DB (market data set).
 The type of RT refresh however is configured at the level of the Front-office desk settings.

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MUREX™ | **Market Data Update**
Update modes

	Description	How
Manual entry	To update any rate at any time. However this method is outdated for daily update of an entire set of market data.	Fill in the market quotes in the market data screen
Copy set to set	Copies market data from one set to another. For instance at the end of the day, market data of the front-office desk can be copied to the market data set used for official PL computation.	Processing script using the MARKETDATA.COPY unit
Data file import	Imports market quotes from another software in order to consolidate or reconcile data.	Import through exchange of XML files
Real-time Import	Mostly used by FO users to get the latest rates in pricing and simulation tools.	Traders can choose how often market data is refreshed.
EOD process	Shifts the system date to the next trading date. Copies current market data into next day's market data. Specific parameters are copied into Historical files (O/N financing rates, equity dividends, Futures closings).	Automated to run during the EOD procedures

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Market Data Update

Practice I

 **Verify your understanding of market data.
Answer the following questions.**

- 1. How would you describe a market data set in Murex?**
 - From a definition point of view
 - From a content point of view
 - From a usage point of view
- 2. Provide at least two methods to update market data in the system.**

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Agenda

Introduction to Market Data

Market Data Sets

Market Data Update

Historical Data

- Purpose
- Contribution to P&L
- Historical data vs. past market data
- Impacts on P&L

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MUREX™ | **Historical Data Purpose**

The **Historical data** are a set of official indices and closing values used for trade fixing.

Deals need fixings in their evaluation:	Swaps (Commodity, IRS, Variance swaps) Baskets, Asian options Dividends, Auto-callable notes, etc.	Stored in historical files: HS <file nb>_DBF or HG<file nb>_HIS
Archiving group types:	Published rates EQ, COM publications FX publications	
Fixing Procedure:	Detects all flows remaining to be fixed Secured procedure: fills FXNG_DBF and revokes rights on historical data	Fills FXNG_DBF and revokes rights on historical data
Internal fixing sets:	Overnight rates fixing set FX conversion factors (accounting rates) fixing set	

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MUREX™ | **Historical Data Contribution to P&L**

Illustration of OverNight rate (O/N) and today's rate contribution
E.g.: P&L of an option

Past cash (Historical rates used for financing)

MV (Today's rates used for MV and PV effect)

PL = Financed past cash + Discounted market value

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MUREX™ | **Historical Data**
Historical data vs. past market data

	Stored in
Entire set of market data as of revaluation date	MARKET RATE SHEETS
Selection of previous days published rates	HISTORICAL DATA TABLES

	Historical data	Past market data
Purpose	Quick access to past data required to compute P&L as of past date or today (i.e.: O/N rates necessary for cash financing, past cash flows etc.)	Entire set on a given past date to compute P&L as of a past date
Type of data	Selection of rates: ▪ O/N rates for financing ▪ Fixing rates for swaps or Asian options ▪ Past dividend rates ▪ Closing prices	Entire set of market data at a past date ▪ Interest rates ▪ Spot rates ▪ Volatilities

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MUREX™ | **Historical Data**
Historical data vs. past market data

What happens when Market data are changed on date T (Past date) ?

	Change of Market rate Sheets	Change of Historical files
Impact on Past P&L	Stored results as of date T ≠ Absolute P&L on date T	Stored results from T until Today ≠ Absolute P&L from T until Today
Impact on Today's P&L	No impact on Today's absolute P&L	Direct impact on Today's absolute P&L

 If O/N financing rates or fixing data *originated from rate sheets* should be modified, then both market data and historical data should be updated simultaneously to keep the consistency.

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MUREX™ | **Historical Data Practice 2**

 **Modify the Historical data and Market data and explain the impact on the P&L**

1. Open an end-user session (MUREXFO/MUREXFO), group FO.
2. Enter a backdated vanilla swap deal (via the e-Tradepad) with the following attributes:

Pricing date	30 Dec 2011
Instrument	EURIBOR A 6M
Maturity	30Y
Fixed rate	2.5
Portfolio & counterpart	As you wish

3. Open the flows and fix the floating rate for the first two past interest rate flows (see screenshot below):

Global flow schedule Source:Deals Trade:4802

Leg 1 Leg 2 All legs

Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert Rate
INT	03 Jan 2012	03 Jul 2012	100,000,000	30 Dec 2011	1.60000	1.60000
INT	03 Jul 2012	03 Jan 2013	100,000,000	29 Jun 2012	0.95000	0.95000
	Dec 2012				0.00000	0.00000
	Jul 2013				0.00000	0.00000

Please mark down the contract/trade number while saving the trade.

 **Solution**

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MUREX™ | **Historical Data Practice 2**

 **Modify the Historical data and Market data and explain the impact on the P&L.**

*The following actions should be done in the same session.
Use Excel to keep a copy of each price you modify so you can revert it afterwards.*

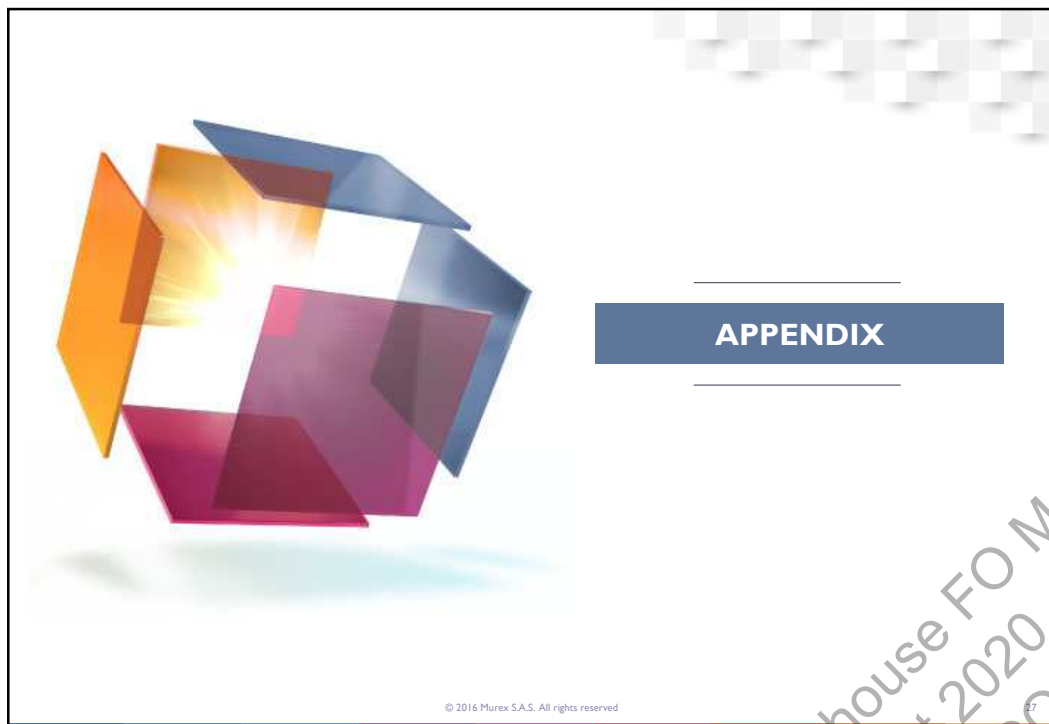
In the Trade query result screen, select View/P&L on your deal before and after each of the following actions:

3. Modify EUR O/N rate in historical data of yesterday.
 - a) Check impact on past market parameter (in the EUR market rate sheet)
 - b) Check impact on today P&L
 - c) Please revert the change
4. Modify EUR O/N rate in the EUR market rate sheet of yesterday.
 - a) Check impact on historical files (in historical data)
 - b) Check impact on today P&L
 - c) Please revert the change
5. Update yesterday's revaluation rate curve EUR EONIA. Check impact on today P&L. **Please revert the change.**
6. Update today's revaluation rate curve EUR EONIA. Check impact on today P&L. **Please revert the change.**

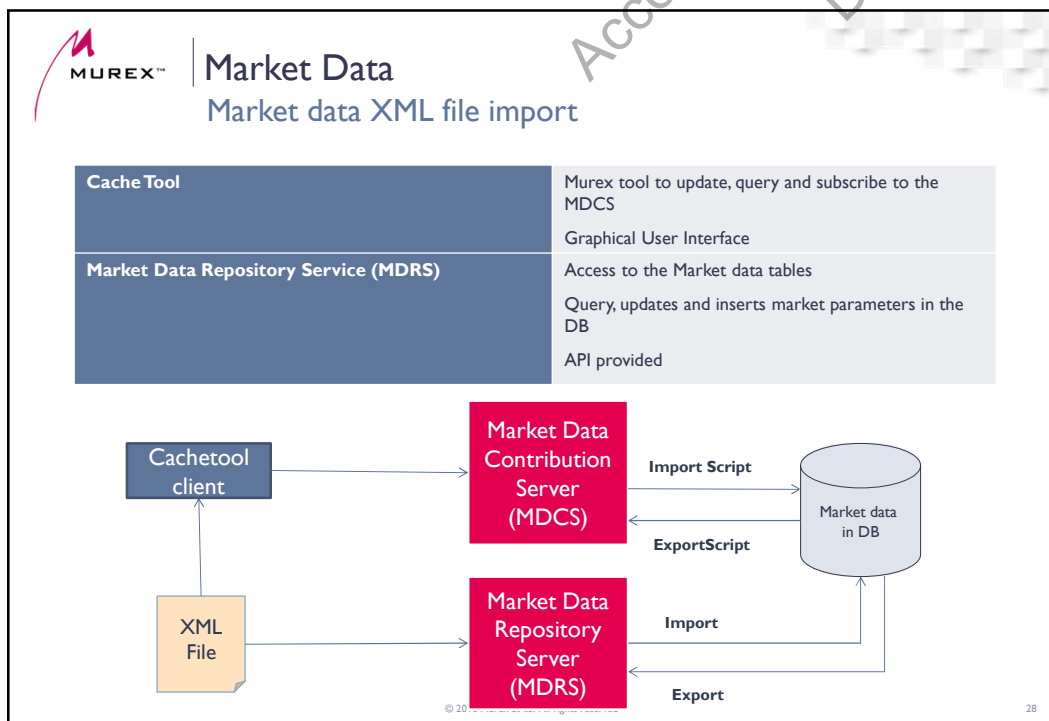
 **Solution**

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MUREX™ | **Market Data**
End of day processing script

Market Data copy	EOD.FODESK.DATE.MOVE, EOD.PCENTER.DATE.MOVE MARKETDATA.COPY MARKETDATA.IMPORT.CONTRIBUTIONSERVICE, MARKETDATA.EXPORT.CONTRIBUTIONSERVICE
Historical Data copy	FIXING.ENGINE.COPY MARKETDATA.COPY FIXING.IMPORT.CONTRIBUTIONSERVICE, FIXING.EXPORT.CONTRIBUTIONSERVICE SAVE-HS SAVE-FUT-CLOSING

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Q&A

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Thank You!

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EXERCISE SOLUTIONS

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

Enter a backdated vanilla swap deal. (via the e-Tradepad).

In the Trade query result screen, select View/P&L on your deal before and after each of the following actions:

Check the P&L of the trade by right-click/View P&L before any change:

Exercise

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

3. Modify EUR O/N rate in historical data of yesterday.

a) Check impact on past market parameter (in the EUR market rate sheet): **No impact**

Before

INTERNAL_ON_FIXING		
EUR		
	RATE	LAMBDA
24 Sep 2012	0.01000000	1.00000003
25 Sep 2012	0.00000000	1.00000000

Deposit
CASH
Floating Maturities

Maturity	Market value
O/N	-0.02000 / 0.04000
1M	-0.02000 / 0.05000
1W	-0.02000 / 0.06000
1M	-0.02000 / 0.08000
2M	0.01000 / 11000
3M	0.06000 / 16000
6M	0.12000 / 22000

After

INTERNAL_ON_FIXING		
EUR		
	RATE	LAMBDA
24 Sep 2012	0.05000000	1.00000003
25 Sep 2012	0.00000000	1.00000000

Deposit
CASH
Floating Maturities

Maturity	Market value
O/N	-0.02000 / 0.04000
1M	-0.02000 / 0.05000
1W	-0.02000 / 0.06000
1M	-0.02000 / 0.08000
2M	0.01000 / 11000
3M	0.06000 / 16000
6M	0.12000 / 22000

End-user session: Historical data/ Rates/ O/N financing rates
End-user session: Market data/ Rates/ Market rate sheets

Exercise

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

3. Modify EUR O/N rate in historical data of yesterday.
b) Check impact on today P&L: **Impacts the Financing Value**

Before

After

End-user session: Historical data/ Rates/ O/N financing rates
End-user session: Market data/ Rates/ Market rate sheets

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Exercise

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

4. Modify EUR O/N rate in the EUR market rate sheet of yesterday.
a) Check impact on historical files (in historical data): **No impact**

Before

After

End-user session: Historical data/ Rates/ O/N financing rates
End-user session: Market data/ Rates/ Market rate sheets

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Exercise

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

4. Modify EUR O/N rate in the EUR market rate sheet of yesterday.
b) Check impact on today P&L: **No impact**

Before

Results in EUR	
Theoretical	Closing
Past cash proceeds	808.889
Future cash proceeds	0
Market value	-3,875,943
Accounting P&L	-2,867,054
Economic P&L	-2,866,999

After

Results in EUR	
Theoretical	Closing
Past cash proceeds	808.889
Future cash proceeds	0
Market value	-3,875,943
Accounting P&L	-2,867,054
Economic P&L	-2,866,999

End-user session: Historical data/ Rates/ O/N financing rates
End-user session: Market data/ Rates/ Market rate sheets

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MUREX™ | **Historical Data**
Practice 2

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

5. Update yesterday's revaluation rate curve EUR EONIA: **No impact on today P&L**

Before

Results in EUR	
Theoretical	Closing
Past cash proceeds	808.889
Future cash proceeds	0
Market value	-3,875,943
Accounting P&L	-2,867,054
Economic P&L	-2,866,999

After

Results in EUR	
Theoretical	Closing
Past cash proceeds	808.889
Future cash proceeds	0
Market value	-3,875,943
Accounting P&L	-2,867,054
Economic P&L	-2,866,999

End-user session: Market data/ Rates/ Revaluation rate curves

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MUREX™ | **Historical Data Practice 2**

Practice 2: Modify the Historical data and Market data and explain the impact on the P&L

6. Update today's revaluation rate curve EUR EONIA: Impacts PVE and MV

Before

Label: EUR EONIA
Owner: INSTALL
Currency: EUR
Bid/Ask: Yes

Storage date: 25 Sep 2012
Horizon: 25 Sep 2012
Zero coupon date: 25 Sep 2012
Compute: On

Last updated: 17 dec 2014-09:55:38 (GMT +0)

Tr	Start date	Generator	Maturity	ds factor	ds rate	Market quote
Op	25 Sep 2012	CASH	25 Sep 2012	0.0000	0.4867	0.4867
Op	25 Sep 2012	CASH	27 Sep 2012	1.0000	0.4867	0.4867
Sw	27 Sep 2012	EUR EONIA	12 Oct 2012	1W	0.9998	0.5551
Sw	27 Sep 2012	EUR EONIA	19 Oct 2012	2W	0.9997	0.5476
Sw	27 Sep 2012	EUR EONIA	30 Oct 2012	3W	0.9996	0.5438
Sw	27 Sep 2012	EUR EONIA	30 Oct 2012	1M	0.9994	0.5748
Sw	27 Sep 2012	EUR EONIA	28 Nov 2012	2M	0.9989	0.5707
Sw	27 Sep 2012	EUR EONIA	28 Dec 2012	3M	0.9985	0.5644
Sw	27 Sep 2012	EUR EONIA	29 Jan 2013	4M	0.9981	0.5588

Results in EUR

Theoretical: Closing

Past cash proceeds: 806.889
Future cash proceeds: 0
Market value: -3,675.943

Accounting P&L: -2,867.054
Financing: 62
Present value effect: 3

After

Label: EUR EONIA
Owner: INSTALL
Currency: EUR
Bid/Ask: Yes

Storage date: 25 Sep 2012
Horizon: 25 Sep 2012
Zero coupon date: 25 Sep 2012
Compute: On

Last updated: 17 dec 2014-09:55:38 (GMT +0)

Tr	Start date	Generator	Maturity	ds factor	ds rate	Market quote
Op	25 Sep 2012	CASH	25 Sep 2012	0.0000	0.4867	0.4867
Op	25 Sep 2012	CASH	27 Sep 2012	1.0000	0.4867	0.4867
Sw	27 Sep 2012	EUR EONIA	12 Oct 2012	1W	0.9998	0.5551
Sw	27 Sep 2012	EUR EONIA	19 Oct 2012	2W	0.9997	0.5476
Sw	27 Sep 2012	EUR EONIA	30 Oct 2012	3W	0.9996	0.5438
Sw	27 Sep 2012	EUR EONIA	30 Oct 2012	1M	0.9994	0.5748
Sw	27 Sep 2012	EUR EONIA	28 Nov 2012	2M	0.9989	0.5707
Sw	27 Sep 2012	EUR EONIA	28 Dec 2012	3M	0.9985	0.5644
Sw	27 Sep 2012	EUR EONIA	29 Jan 2013	4M	0.9981	0.5588

Results in EUR

Theoretical: Closing

Past cash proceeds: 806.889
Future cash proceeds: 0
Market value: -3,624.006

Accounting P&L: -2,816.917
Financing: 62
Present value effect: 104

End-user session: Market data/ Rates/ Revaluation rate curves

Exercise

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1

This slide is titled 'Learning Objectives' and 'Rate Curves Overview'. It features the Murex logo in the top left corner. The main heading is 'Learning Objectives' with 'Rate Curves Overview' as a subtitle. Below this, the text 'At the end of this course, you will be able to:' is followed by a bulleted list of three objectives. At the bottom of the slide, there is a small copyright notice and a page number. A large, diagonal watermark across the right side of the slide reads 'Accenture: In-house FO Modules August 2020 DO NOT COPY'.

Learning Objectives

Rate Curves Overview

At the end of this course, you will be able to:

- ✓ Identify which rate curves are used
- ✓ Build a rate curve from scratch
- ✓ Use the different settings and parameters of rate curves

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2

2



Agenda

Introduction to Rate Curves	• Usage • Pricing screen • Using several rate curves • Assignment template
Market Rate Sheets	
Rate Curve Construction	
Configuration and Tools	
Practice	

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3



Introduction to Rate Curves

Usage

The IR curve (yield curve) is the relation between the interest rate (or cost of borrowing) and the time to maturity known as the "term" of a debt for a given borrower in a given currency.

Traders use rate curves to:	Compute zero-coupon rates for any date. Compute discount factors for any payment date.
Rate curves structure:	Rate curves are built mainly upon IR instruments and Bonds: • Loan/Deposits, Swaps, FRAs, Short term Futures • Bonds, Long term Futures • Basis swaps, FX Forwards, Contango, etc.
Other usage:	Used for discounting and estimation: • Discounting curves for all asset classes: at least one rate curve per currency. • Forward curves for instruments based on IR indices (e.g. calculate the forward value of the EURIBOR 3M Index).

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MUREX™ | Introduction to Rate Curves
Pricing screen

Illustration with an IRS pricing (fixed vs floating swap):

Product: 01 IRS Vanilla

Instrument: EURIBOR A 6M

Template details: Open

Start period: Spot Thu 19/01/12

Maturity period: 3Y8M Sat 19/09/15

Adjusted maturity: Mon 21/09/15

Nominal: 100,000,000 EUR

Flows: Open

Leg: Pay Receive

Leg type: Fixed Floating

Currency: EUR EUR

Index: EUR EURIBOR 6M

First fixing: 0.0000

Calculation frequency: 1y 6m

Rate Spread: 2.5828 0.0000

Discount curve: EUR EONIA EUR EONIA

Estimation curve: EUR EURIBOR 6M

Leg Nbr: -9,266,896 9,266,896

Legs details: No

Amortizing: None

Additional flows: Open

Customized flows: No

Mutual break: No

Define phases: Open

Not: -9 -35,879 EUR

Typology / Usage: IRS Open

CSA Type: No

Int. / Counterparty: ABBEY NATIONAL PL

Portfolio: MEX_PTF

Fixed Leg: Discounting

Floating leg: Discounting + Estimation of the floating index
(EURIBOR 6M)

Global flow schedule Source: Notepad

Flow	Start Date	End Date	Remaining Capital	Fixing Date	Rate	Convert	Margin	Rate Factor	Payment Date	Flow	Cur	Interest flows	Int/flows + index
INT	19 Jan 2012	19 Sep 2012	100,000,000	19 Jan 2012	1.11170	1.11170	0.0000	1.0000	19 Sep 2012	105,203,448.00	EUR	0.00	0.00
INT	19 Mar 2012	19 Sep 2012	100,000,000	19 Mar 2012	1.64250	1.64250	0.0000	1.0000	19 Sep 2012	836,500,000.00	EUR	0.00	0.00
INT	19 Sep 2012	19 Mar 2013	100,000,000	19 Sep 2012	1.60000	1.60000	0.0000	1.0000	19 Mar 2013	820,777,778.00	EUR	0.00	0.00
INT	19 Mar 2013	19 Sep 2013	100,000,000	19 Mar 2013	3.06569	3.06569	0.0000	1.0000	19 Sep 2013	1,566,603,128.00	EUR	0.00	0.00
INT	19 Sep 2013	19 Mar 2014	100,000,000	19 Sep 2013	2.72395	2.72395	0.0000	1.0000	19 Mar 2014	1,371,551,048.00	EUR	0.00	0.00
INT	19 Mar 2014	19 Sep 2014	100,000,000	19 Mar 2014	3.06211	3.06211	0.0000	1.0000	19 Sep 2014	1,565,080,038.00	EUR	0.00	0.00
INT	19 Sep 2014	19 Mar 2015	100,000,000	19 Sep 2014	3.33722	3.33722	0.0000	1.0000	19 Mar 2015	1,677,816,148.00	EUR	0.00	0.00
INT	19 Mar 2015	21 Sep 2015	100,000,000	19 Mar 2015	3.45467	3.45467	0.0000	1.0000	21 Sep 2015	1,785,916,038.00	EUR	0.00	0.00

End-user session: Pricing/ e-Tradepad

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MUREX™ | Introduction to Rate Curves
Using several rate curves

Several rate curves can be defined per currency, e.g.:

- Discounting and estimation curves
- Different indices, classes of indices to forecast
- Different activities: MM, FX, Rate, Inflation...
- Mid vs. Bid-Ask curves

Rate curves assignments:
Templates defined at end-user level
Assignments to FO Desks done through Market Data Set definition:

Market data set

Label: MDS

Description: Market Data Set

Alias: MDS

Rate curve assignment: MAIN

Type: Standard

Collateral assignment:

OK Cancel

Configurator session: Infrastructure/ Organization/ Market data set

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MUREX™ | Introduction to Rate Curves
Assignment template

User can assign specific rate curves following various keys such as type, calculation, index, etc. MX will then assign the rate curve following a best matching process.

Assignments are historically stored

The environment column allows to assign different curves for simulation, pricing and reporting.

End-user session : Market data/ Rates/ Rate curve assignments

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MUREX™ | Introduction to Rate Curves
Assignment template

The assignments indicated in this template drive the curves selection in the different pricing screens, e.g. in the e-Tradepad:

Example: a USD IR swap

- > paying fixed on an annual basis
- > floating on a semi-annual basis

Discount curve (for both legs)	USD LIBOR 3M
Forward curve (for floating leg)	USD LIBOR 6M

Without assignment both curves would have been USD:STD

Product: USD LIBOR A 6M

Spot: 20y

Maturity: 27/09/32

Rate/Swap: 2.3112

Discount curve: USD LIBOR 3M

Index curve: USD LIBOR 6M

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MUREX™ | Introduction to Rate Curves
Assignment template

Users can change the rate curve assignment locally.

e-Tradepad	Market data>Used Market Data>Market data set → Right click on Rate curve assignment
Pricing	View>PopUp Market Data>Market data set → Right click on Rate curve assignment
Simulation	View>PopUp Market Data>Market data set → Right click on Rate curve assignment



The screenshot shows the Murex interface. On the left, a 'Marketdata' panel with 'Screen' and 'Navigation' tabs. The 'Navigation' tab is active, showing a list of actions: View, Refresh (database) (Ctrl+Alt+F11), Used market data (Ctrl+Alt+F11), Viewer (Ctrl+Alt+W), and Realtime. On the right, a 'Market Parameters' tree is visible. The tree structure includes: Market Parameters, Market data set: MDS, Rate curve assignment: M, Prices, CME EURUSD 3M, Rate Curves, USD: STD, USD LIBOR 3M, Rates, and USD. A context menu is open over the 'Rate curve assignment: M' node, showing options: Change rate curve assignment template, Modify current rate curve assignment, Modify current rate curve assignment (equities), and Exit.

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MUREX™ | Agenda

Introduction to Rate Curves

Market Rate Sheets	• Purpose • Standard screen • Inserting instruments • Financing • Link types • Spreads • Displays
---------------------------	---

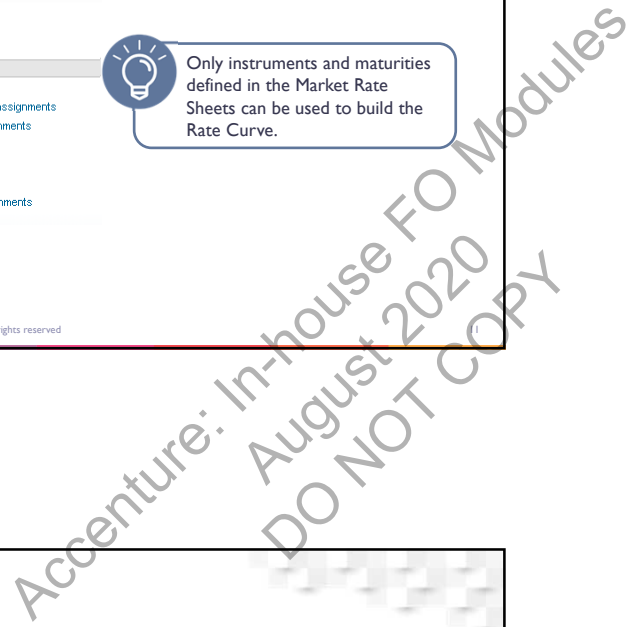
Rate Curve Construction

Configuration and Tools

Practice

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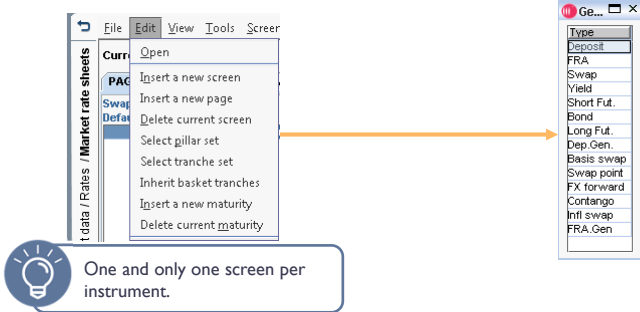
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MUREX™ | **Market Rate Sheets**
Insert instruments

All types of instruments and their underlying maturities can be inserted in the market rate sheets.



One and only one screen per instrument.

Market rate sheets are updated like any market data by:

- ✓ Manual entry
- ✓ Real-time import from the cache
- ✓ Import via processing scripts or files

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MUREX™ | **Market Rate Sheets**
Financing

P&L financing is computed using the historical O/N rates.

The source data for the historical O/N rate must be defined in the Market Rate Sheet for each currency.

The rate is then marked in red.



Maturity	Market value
O/N	0.60000 / 1.10000
1M	0.70000 / 1.20000
1W	0.60000 / 90000
1M	0.95000 / 1.45000
2M	0.85000 / 1.35000
3M	1.43000 / 63000
6M	1.61000 / 81000

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Market Rate Sheets

Link types



Links can be defined between the instruments pillars or maturities.

File Edit View **Tools** Screen Help UI Tools

Currency USD Apply a scenario

PAGE 1 PAGE 2 Mark as historical O/N source

Swap Unlink a cell

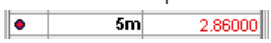
Default Link a cell to another one

Maturity Link a cell to 2 other cells

Link a cell to a block

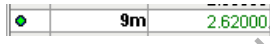
How to link one cell to another one in the market rate sheets screen:

- Put the cursor on the cell I.
- Go to Tools>Link a cell to another one.
- Select the corresponding value.
- A red ball is displayed on the original cell to indicate that it has been linked. The original cell value will be equal to the linked cell.



How to link one cell to two other cells in the market rate sheets screen:

- Put the cursor on the cell I.
- Go to Tools>Link a cell to 2 other cells.
- Select the corresponding values.
- A green ball is displayed on the original cell to indicate it has been linked to 2 other cells. The new cell value will be linearly interpolated on the two other cells.



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Market Rate Sheets

Spreads

Additional Spreads can be applied **only to cells** linked with other cells.

For example:
 $\text{USD LIB 6M}_I\text{Y} = \text{USD LIB 6M}_M + \text{spread}$

The sum of the “link spread” and the market value is displayed in the linked cells.

To add a spread to the pillar 5m:

1. Rates without Spreads

Maturity	Market value
0M	3.25000
1w	3.25000
1m	3.30000
2m	3.10000
3m	3.05000
4m	3.00000
5m	2.86000
6m	2.86000
9m	2.62000
1y	2.88000

2. View>Additional Spreads

Maturity	Spread
0M	0.00000
1w	0.00000
1m	0.00000
2m	0.00000
3m	0.00000
4m	0.00000
5m	0.00277
6m	0.00000
9m	0.00000
1y	0.00000

3. View>Rates

Maturity	Market value
0M	3.25000
1w	3.25000
1m	3.30000
2m	3.10000
3m	3.05000
4m	3.00000
5m	2.86277
6m	2.86000
9m	2.61169
1y	2.88000

View Tools Screen H

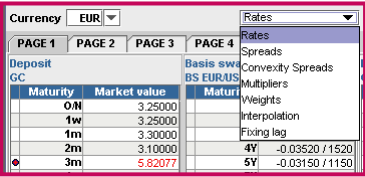
- Rates
- Additional Spreads
- Multipliers
- Convexity Spreads
- Weights
- Interpolation Method
- Fixing lag
- Reference rate
- Nth default
- Delta atm 1
- Delta atm 2
- Display links tree

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 **Market Rate Sheets Displays**

Market rate sheets display modes:



Display Mode	Description
Rates	Standard market rate sheet
Spreads	Displays spreads defined for linked cells
Convexity spreads	Spreads that are added to the prices or rates of the corresponding instruments during rate curve calculation (typically used for short futures)
Multipliers	Displays multipliers applied on linked cells
Weights	Display weights applied to linked cells, if any
Interpolation	Display interpolation method
Fixing lag	Used for inflation swaps

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 **Agenda**

Introduction to Rate Curves

Market Rate Sheets

Rate Curve Construction

- Per currency
- Pillar selection
- Standard screens
- Spread displays

Configuration and Tools

Practice

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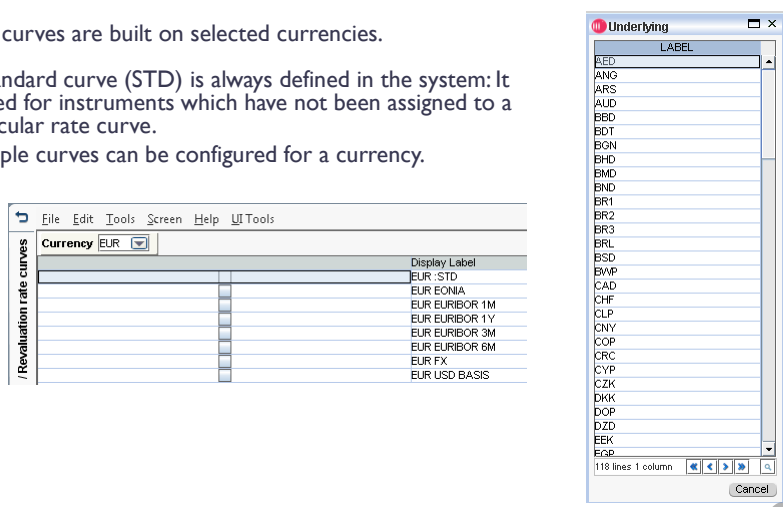
18

MUREX™ | **Rate Curve Construction**
Per currency

Rate curves are built on selected currencies.

A standard curve (STD) is always defined in the system: It is used for instruments which have not been assigned to a particular rate curve.

Multiple curves can be configured for a currency.



The 'Underlying' dialog box shows a list of currencies and their corresponding labels. The 'Revaluation rate curves' table shows the configuration for the EUR currency.

Currency	Display Label
EUR	EUR - STD
	EUR EONIA
	EUR EURIBOR 1M
	EUR EURIBOR 1Y
	EUR EURIBOR 3M
	EUR EURIBOR 6M
	EUR FX
	EUR USD BASIS

End-user session: Market data/ Rates/ Revaluation rate curves

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MUREX™ | **Rate Curve Construction**
Pillar selection

Link with market rate sheet:

- ✓ Market value changes in the market rate sheets are reflected back on the rate curve market values, recalibrating the curve.

Instrument Selection:

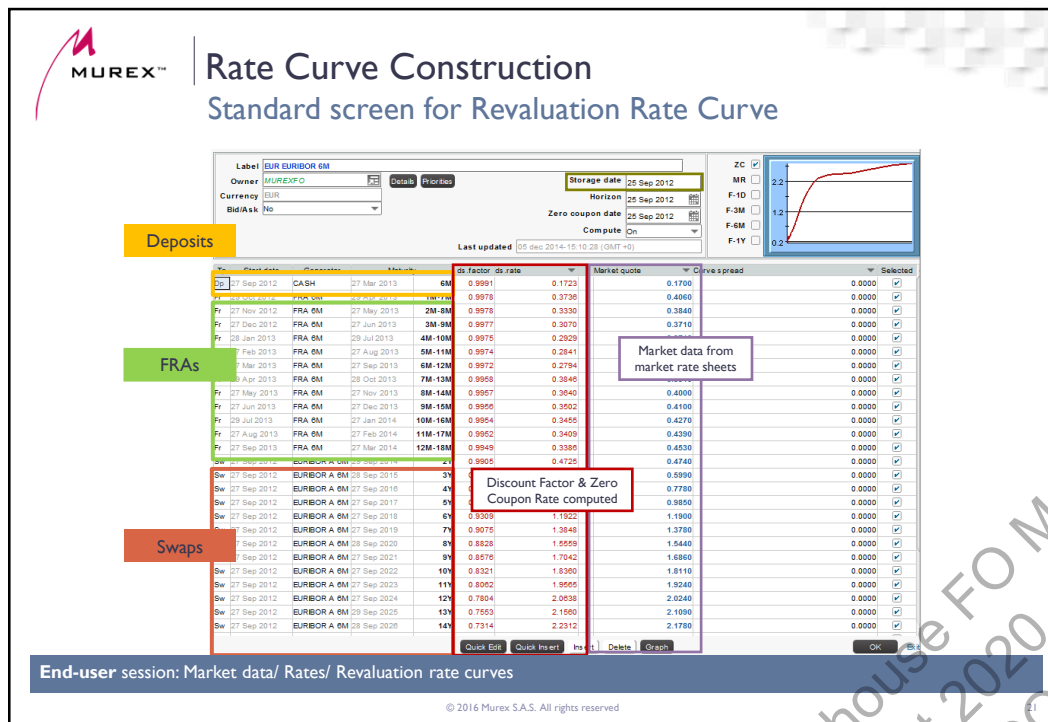
- ✓ Any or all pillars of instruments inserted in the market rate sheets may be selected to build the curve.

Discount factors and zero-coupon computation:

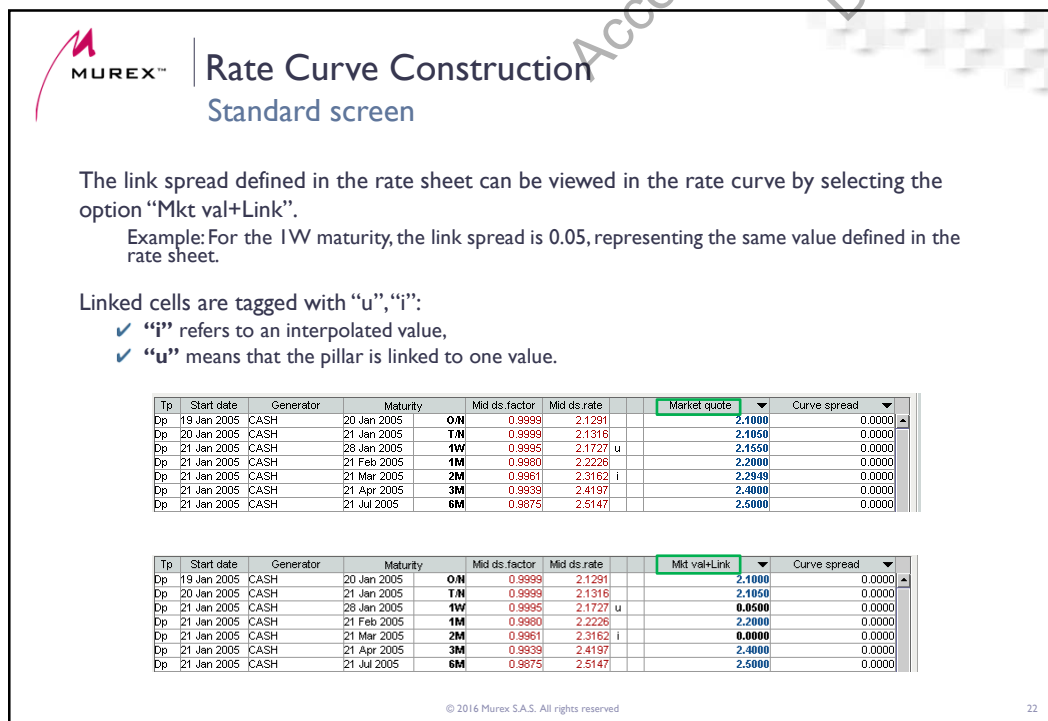
- ✓ Computed only for pillars with non-zero value.
- ✓ All instruments belonging to the curve have a 0 value if priced on the curve.

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MUREX™ | **Rate Curve Construction**
Spreads displays

Spreads can be defined for all instrument cells

Label: **EUR:STD**

Owner: **INSTALL**

Currency: **EUR**

Bid/Ask: **No**

Details: Priorities

Storage date: 25 Sep 2012

Horizon: 25 Sep 2012

Zero coupon date: 25 Sep 2012

Compute: On

Last updated: 15 mar 2013-20:17:17 (GMT +0)

ZC ☒

MR ☒

F-1D ☐

F-3M ☐

F-6M ☐

F-1Y ☐

Tr	Start date	Generator	Maturity	ds factor	ds rate	Market quote	Curve spread	Selected
Dp	25 Sep 2012	CASH	26 Sep 2012	0.01	1.0000	0.8618	0.8500	0.0000 ☒
Dp	26 Sep 2012	CASH	27 Sep 2012	0.01	1.0000	0.9125	0.9000	0.0000 ☒
Dp	27 Sep 2012	CASH	04 Oct 2012	0.01	0.9997	1.2941	0.8714	0.5000 ☒
Dp	27 Sep 2012	CASH	29 Oct 2012	0.01	0.9984	1.7000	1.2266	0.5000 ☒
Dp	27 Sep 2012	CASH	27 Nov 2012	0.01	0.9981	1.1215	1.1140	0.0000 ☒
Dp	27 Sep 2012	CASH	27 Dec 2012	0.01	0.9981	1.5439		
Sf	19 Dec 2012	EUREX EURBOR 3M	19 Mar 2013	DEC 12	0.9929	1.4829		
Sf	20 Mar 2013	EUREX EURBOR 3M	20 Jun 2013	MAR 13	0.9898	1.3902		
Sf	19 Jun 2013	EUREX EURBOR 3M	19 Sep 2013	JUN 13	0.9852	1.5142		
Sf	18 Sep 2013	EUREX EURBOR 3M	18 Dec 2013	SEP 13	0.9789	1.7327		
Sf	18 Dec 2013	EUREX EURBOR 3M	18 Mar 2014	DEC 13	0.9686	2.1624		
Sw	27 Sep 2012	EURBOR A 6M	26 Sep 2015	3Y	0.9136	3.0029		
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2016	4Y	0.8986	2.8687		
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2017	5Y	0.8847	2.9027		
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2018	6Y	0.8364	2.9729	3.0004	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2019	7Y	0.7345	3.2628	3.2904	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	28 Sep 2020	8Y	0.7501	3.4222	3.4233	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2021	9Y	0.7263	3.5486	3.5393	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2022	10Y	0.6288	4.6377	3.6423	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2023	11Y	0.6627	3.7370	3.7383	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	27 Sep 2024	12Y	0.6306	3.6379	3.6242	0.0000 ☒
Sw	27 Sep 2012	EURBOR A 6M	29 Sep 2025	13Y	0.5993	3.9321	3.9062	0.0000 ☒

Tag 1: A curve spread was defined

Tag 2: A convexity spread was defined

Tag 3: Both spreads were defined

Convexity spreads apply only to short futures.

End-user session: Market data/ Rates/ Revaluation rate curves

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MUREX™ | **Agenda**

Introduction to Rate Curves

Market Rate Sheets

Rate Curve Construction

Configuration and Tools

- Details box
- Interpolator
- Graphs

Practice

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Configuration and Tools

Details box



Docid:
925

Details

Model type:

Interpolation formula:

Value to interpolate:

Interpolation before first pillar:

Interpolation after last pillar:

Zero rate convention:

Futures:

Blocks consistency:

Forwards convention:

Curve type:

Ignore fixing:

Calibration:

Business days:

Spread curve:

Hedge curve:

Default entry:

Maturity step and gap:

Snapshot generation:

Classification:

Details

Model type:

Interpolation formula:

Value to interpolate:

Interpolation before first pillar:

Interpolation after last pillar:

Zero rate convention:

Futures:

Blocks consistency:

Forwards convention:

Murex Model	Represents interpolation model offered by Murex other models (require DLL interface).
Interpolation formula	Spline 3rd degree polynomial resolution Linear 1st degree polynomial resolution
Value to interpolate	Rate, Rate * time, Factor, Log (factor).
Zero rate convention	Specific rate convention of revaluation curve, used as specified in the flag Convention.
Futures	Standard interpolation, or no intermediate interpolation.
Block consistency	Checking Disabled, or According to priorities.
Forward convention	Generally inherited from currency.

End-user session: Market data/ Rates/ Revaluation rate curves/ Details

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Configuration and Tools

Details box



Docid:
925

Details

Model type:

Interpolation formula:

Value to interpolate:

Interpolation before first pillar:

Interpolation after last pillar:

Zero rate convention:

Futures:

Blocks consistency:

Forwards convention:

Curve type:

Ignore fixing:

Calibration:

Business days:

Spread curve:

Hedge curve:

Default entry:

Maturity step and gap:

Snapshot generation:

Classification:

Calibration method:

Calibration tolerance:

Calibration iterations:

Accept zero rates:

Zero coupon spread

Ignore fixing

Calibration

Business days

Spread curve

Hedge curve

Zero coupon spread	Curve spreads can be input as percentage or absolute value.
Ignore fixing	Instruments based on floating rate index can take into account the official reset rate in the zero curve building. The reset date can be "Today" or any date prior to Today .
Calibration	Standard / Estimation / Curve assignments
Business days	Used in the Brazilian market.
Spread curve	Relevant to basis swap curves, BMA swaps, Brazilian bonds, and USD swaps quoted in spreads over the Treasury Yield.
Hedge curve	Possibility to define a hedge curve.

End-user session: Market data/ Rates/ Revaluation rate curves/ Details

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MUREX™ Configuration and Tools Details box

Docid: 925

Field	Value
Default entry	Redefined
Maturity step out gap	Inherited
Snapshot generation	Inherited
Classification	Inherited
Calibration method	Inherited
Calibration tolerance	Inherited
Calibration iterations	Inherited
Accept zero rates	Inherited

Field	Description
Default entry	Import market rate or swap points.
Maturity step out gap	Futures are automatically deleted from the rate curve when they arrive at Maturity + Maturity step out gap.
Snapshot generation	If set to "After save", each time the rate curve is modified and saved, it is saved in a file at the level of the application directory.
Classification	If set to hedge, the curve is considered to be a hedge curve and you can specify the native curve to be used to calibrate it.
Calibration method	Calibrate each pillar separately or all together (minimisation of the absolute error of the NPV).
Accept zero rates	To ignore or take into account pillars with zero rates.

End-user session: Market data/ Rates/ Revaluation rate curves/ Details

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MUREX™ Configuration and Tools Interpolator

Allows the user to compute capitalization factor, discount factor and discount rate between any two dates based on current curve.

End-user session: Market data/ Rates/ Revaluation rate curves/ Tools/ Interpolator

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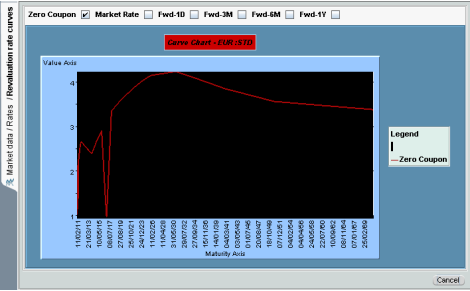
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MUREX™ | Configuration and Tools
Graphs

Tools/Graph:
View rate curve

Tools Settings Screen UI Tools

- Interpolator
- Export daily rates
- Graph
- Quick pricer
- Compute convexity adjustment
- Apply a scenario
- Create a scenario from curve



End-user session: Market data/ Rates/ Revaluation rate curves/ Tools

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MUREX™ | Agenda

Introduction to Rate Curves

Market Rate Sheets

Rate Curve Construction

Configuration and Tools

Practice

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MUREX™ | **Practice**
Practice 1: Build links in market rate sheets

 Build links in the market rate sheets screen.

1. Open an end-user session (MUREXFO/MUREXFO), group FO.
2. Go to Market data/ Rates/ Market rate sheets for the EUR currency.
3. Use the *FX EUR (USD) Dep. Gen.* instrument to build links on the *Default deposit* for the following pillars: O/N, T/N, 1M & 3M.
4. Add a +10 basis points spread on all linked pillars.

Maturity	Market value
O/N	-0.06324 / 0770
T/N	-0.17453 / 0.04784
5M	-0.09101 / 3547
1M	0.04938 / 5809
2M	0.09502 / 10613
3M	0.17433 / 865
4M	-0.02858 / 171
5M	-0.01060 / 0.02637
6M	0.05176 / 6855
7M	-0.05443 / 0889
8M	-0.02491 / 0092
9M	0.01539 / 3878
10M	-0.04374 / 2165
11M	-0.02723 / 0602
1Y	0.00867 / 1636
2Y	0.01665 / 4045

Maturity	Market value
O/N	0.00000
T/N	0.00000
1M	0.00000
3M	0.00000

 **Solution**

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MUREX™ | **Practice**
Practice 2: Build a rate curve

 Build a new rate curve for EUR and assign it to discount EUR transactions.

1. Using the same session as in practice 1, go to Market data/ Rates/ Revaluation rate curves.
2. Insert a new rate curve "EUR CURVE TRAINING" for the EUR currency.
3. Insert the following instruments into a new rate curve:
 - > Deposit Default: O/N, T/N, 1M, 3M
 - > Short Futures EUREX EURIBOR 3M: DEC 12, MAR 13, JUN 13, SEP 13, DEC 13
 - > Swap EURIBOR A 1M: 2Y, 3Y, 5Y, 7Y, 8Y, 10Y, 12Y
4. Assign the rate curve to EUR trades, in the MAIN template, for discounting only (under Market data/ Rates/ Rate curve assignments).

 **Solution**

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**MUREX™**

Practice Quiz

1. What are the steps to follow to configure a rate curve starting from a database where no rate sheet is configured?
2. Where can we modify the interpolation method for a given rate curve?
3. Why do clients have several rate curves for a given currency?

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Q&A

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Thank You!

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EXERCISE SOLUTIONS

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MUREX™ | **Practice**
Practice 1: Build links in market rate sheets

Practice: Build links in the market rate sheets screen.

1. Open an end-user session (MUREXFO/MUREXFO), group FO.
2. Go to Market data/ Rates/ Market rate sheets for the EUR currency.
3. Use the FX EUR (USD) Dep. Gen. instrument to build links on the Default deposit for the following pillars: O/N, T/N, 1M & 3M.
4. Add a +10 basis points spread on all linked pillars.

Linking

Maturity	Market value
O/N	-0.06324 / 0.0770
T/N	-0.17433 / 0.04784
S/N	-0.09101 / 0.3547
1M	0.04938 / 5806
2M	0.09502 / 10613
3M	0.17433 / 866
4M	-0.02858 / 171
5M	-0.01060 / 0.02637
6M	0.05176 / 6855
7M	-0.05443 / 0869
8M	-0.02481 / 0082
9M	0.01539 / 3878
10M	-0.04374 / 2165

Adding spreads

Maturity	Spread
O/N	0.10000
T/N	0.10000
1M	0.10000
3M	0.10000

Exercise

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MUREX™ | **Practice**
Practice 2: Build a rate curve

Practice: Build a new rate curve for EUR and assign it to discount EUR transactions.

Label: EUR CURVE TRAINING
Owner: MUREXFO
Currency: EUR
Bid/Ask: No
Storage date: 25 Sep 2012
Horizon: 25 Sep 2012
Zero coupon date: 25 Sep 2012
Compute: On
Last updated: 17 dec 2014:17:07:38 (GMT +0)

ZC ☒
MR ☐
F-1D ☐
F-3M ☐
F-6M ☐
F-1Y ☐

Tr	Start date	Generator	Maturity	ds factor	ds rate	Market quote	Curve spread
Dp	25 Sep 2012	Default	26 Sep 2012	O/N	1.0000	0.0654 u	0.0645
Dp	26 Sep 2012	Default	27 Sep 2012	T/N	1.0000	0.0513 u	0.0368
Dp	27 Sep 2012	Default	29 Oct 2012	1M	0.9999	0.1497 u	0.1537
Dp	27 Sep 2012	Default	27 Dec 2012	3M	0.9996	0.1536 u	0.1537
Sf	19 Dec 2012	EURX EURBOR 3M	19 Mar 2013	DEC 12	0.9982	0.1773	99.8025
Sf	20 Mar 2013	EURX EURBOR 3M	20 Jun 2013	MAR 13	0.9985	0.2001	99.7600
Sf	19 Jun 2013	EURX EURBOR 3M	19 Sep 2013	JUN 13	0.9977	0.2325	99.6772
Sf	18 Sep 2013	EURX EURBOR 3M	18 Dec 2013	SEP 13	0.9968	0.2582	99.6441

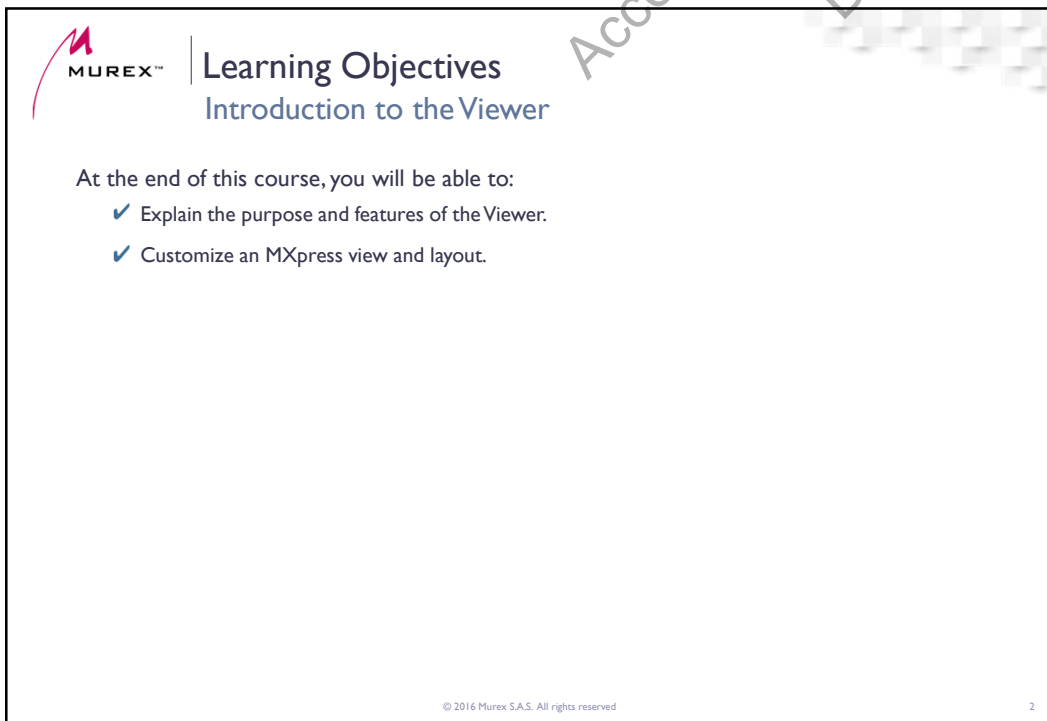
Exercise

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Agenda

Introduction to the Viewer

- What is the Viewer?
- Where is the Viewer used?
- What are the basic components of the Viewer?
 - > Viewer components
 - > Single View components
 - > Edit View mode
 - > Edit Layout mode
- Hands on

View Customization

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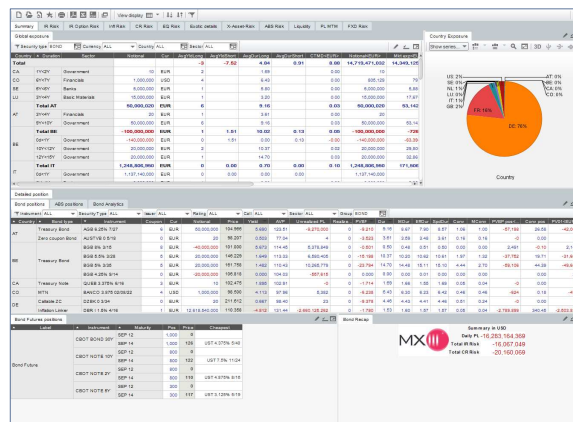
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Introduction to the Viewer

What is the Viewer? (1/3)

The Viewer is an **interactive data visualization tool** used in main MX.3 business screens.



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Introduction to the Viewer

What is the Viewer? (2/3)

It displays **aggregated results** with possibility to **drill down on details**.

Country	Notional	Country	Bond type	Notional	Country	Bond type	Instrument	Notional
AT	50,000,020	AT	Treasury Bond	50,000,000	AT	Treasury Bond	AGB 6.25% 7/27	50,000,000
BE	-20,000,000		Zero coupon Bond	20		Zero coupon Bond	AUSTVB 0 5/18	20
CA	10	BE	Treasury Bond	-20,000,000	BE	Treasury Bond	BGB 8% 3/15	-40,000,000
CO	1,000,000	CA	Treasury Note	10		Treasury Bond	BGB 5.5% 3/28	20,000,000
DE	12,948,540,140	CO	MTN	1,000,000		Treasury Bond	BGB 5% 3/35	20,000,000
FR	-3,004,420,000		Callable ZC	20		Treasury Bond	BGB 4.25% 9/14	-20,000,000
GB	0		Inflation Linker	12,618,540,000	CA	Treasury Note	QUEB 3.375% 6/16	10
IT	111,666,950	DE	MTN	120	CO	MTN	BANCO 3.875 02/08/22	1,000,000
LU	15,000,000		Treasury Bill	30,000,000		Callable ZC	DZBK 0 3/34	20
NL	201,000,020		Treasury Bond	280,000,000		Inflation Linker	DBR 1.15% 4/16	12,618,540,000
SE	5,000,000			20,000,000		MTN	KFW 3.5% 7/15	120
US	0	FR	Bullet Bond	305,000,000		Treasury Bill	OBL 2.5% 2/15	20,000,000
			Callable Fixed	1,000,000			OBL 2.5% 10/14	10,000,000
			Inflation Linker	-3,441,420,000			DBR 6.5% 7/27	-50,000,000
		FR	MTN	31,000,000			DBR 5.625% 1/28	80,000,000
			Treasury Bond	100,000,000			DBR 5.5% 1/31	20,000,000
		GB	MTN	0			DBR 4.75% 7/40	10,000,000
			Treasury Bond	100,000,000			DBR 4.75% 7/34	30,000,000
			Inflation Linker	21,667,000			DBR 4.75% 7/28	40,000,000

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Introduction to the Viewer

What is the Viewer? (3/3)

It offers many **customization** tools: format, user actions, calculation settings...

Region/Country	Sector	Rating	Currency/Maturity	Mkt value	Exp %	Duration	Rebalance by exposure
Country	Notional<EUR>	Mkt exp<EUR>	PVBP<EUR>	PV01<EUR>	ZC risk<EUR>	Aswpt risk<EUR>	
Total	53,050,033,872	45,277,842,118	-19,093,674	443,184	424,700	-18,324,911	
AR	70,605	67,121	-19	-20	-19	-21	
DE	15,000,000	16,686,557	-29,007	-30,199	-30,186	-24,495	
FR	1,031,942	981,167	-392	-412	-411	-458	
QA	-529,611,065	-505,841,879	435,112	456,287	455,584	695,727	
US	535,502	508,815	-48	-50	-50	-53	
QA	9,639,028	2,620,217	-397	-137	-386	-396	
	-535,502	-509,597	440	461	461	703	
	53,551,656,654	45,761,217,053	-19,498,839	17,738		-18,995,594	
	1,149,702	1,069,714	-226	-183			
AR	1,097,096	1,042,950	-297	-303	-302	-325	

Gradient coloring : 3-color scale

First point Transition point End point

Color palette :

☒ Automatic scale Minimum 0 Maximum

Gradient coloring : 2-color scale

First point End point

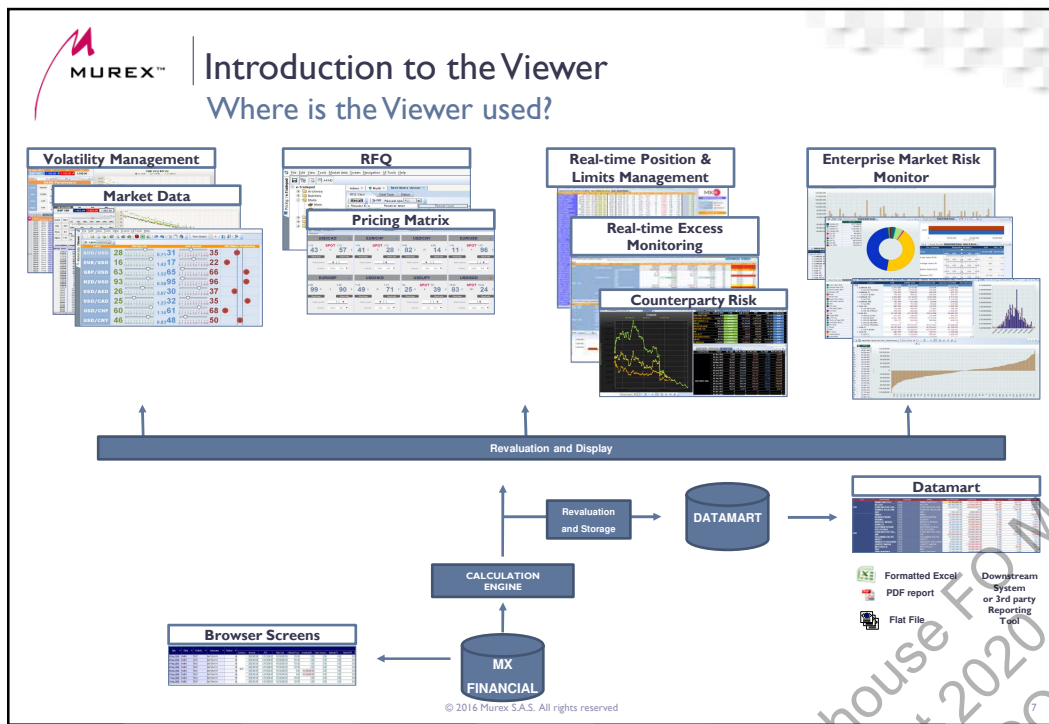
Color palette :

☒ Automatic scale Minimum Maximum

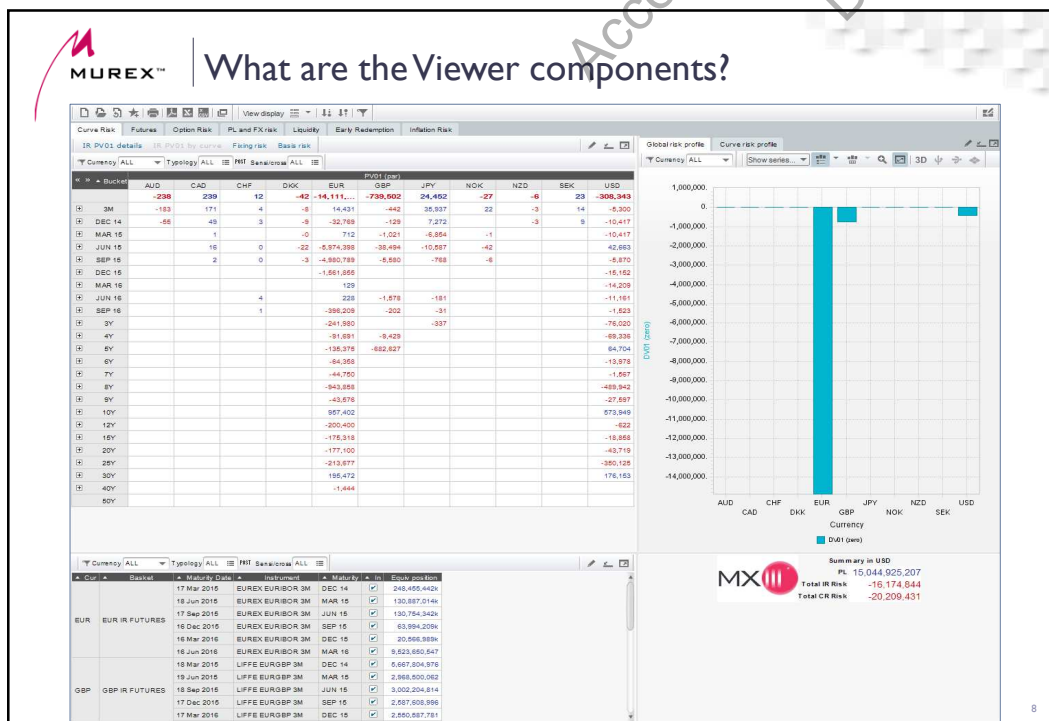
Region/Country	Sector	Rating	Currency/Maturity	Mkt value	Exp %	Duration	Rebalance by exposure
AR	70,605	67,121	-19	-20	-19	-21	
DE	15,000,000	16,686,557	-29,007	-30,199	-30,186	-24,495	
FR	1,031,942	981,167	-392	-412	-411	-458	
QA	-529,611,065	-505,841,879	435,112	456,287	455,584	695,727	
US	535,502	508,815	-48	-50	-50	-53	
QA	9,639,028	2,620,217	-397	-137	-386	-396	
	-535,502	-509,597	440	461	461	703	
	53,551,656,654	45,761,217,053	-19,498,839	17,738		-18,995,594	
	1,149,702	1,069,714	-226	-183			
AR	1,097,096	1,042,950	-297	-303	-302	-325	

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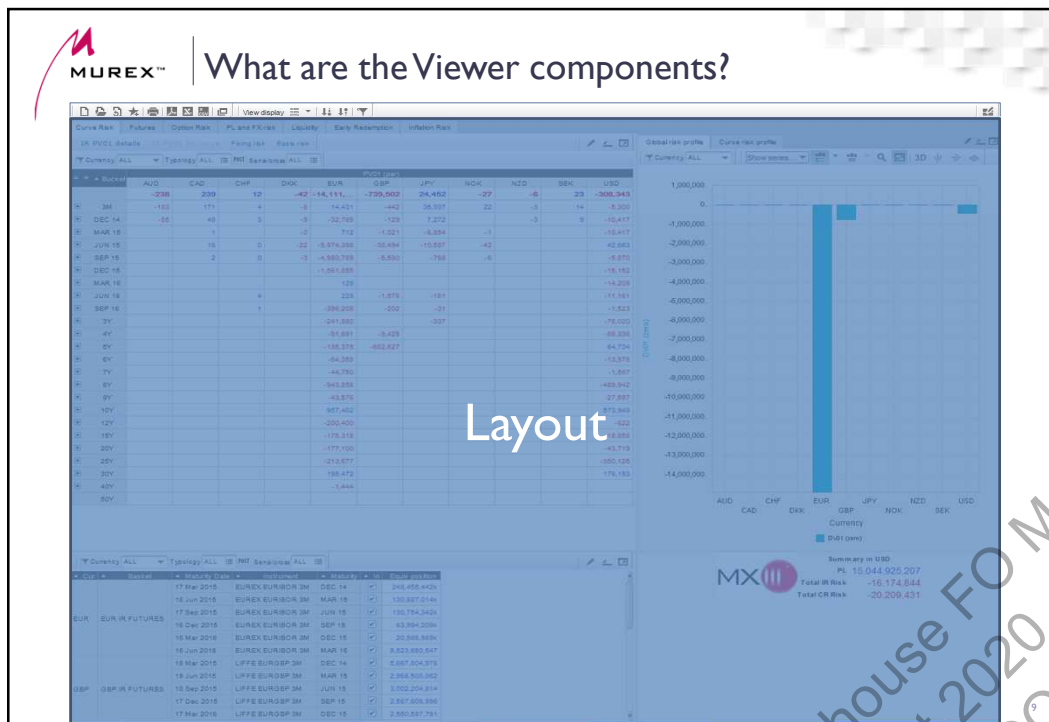
6



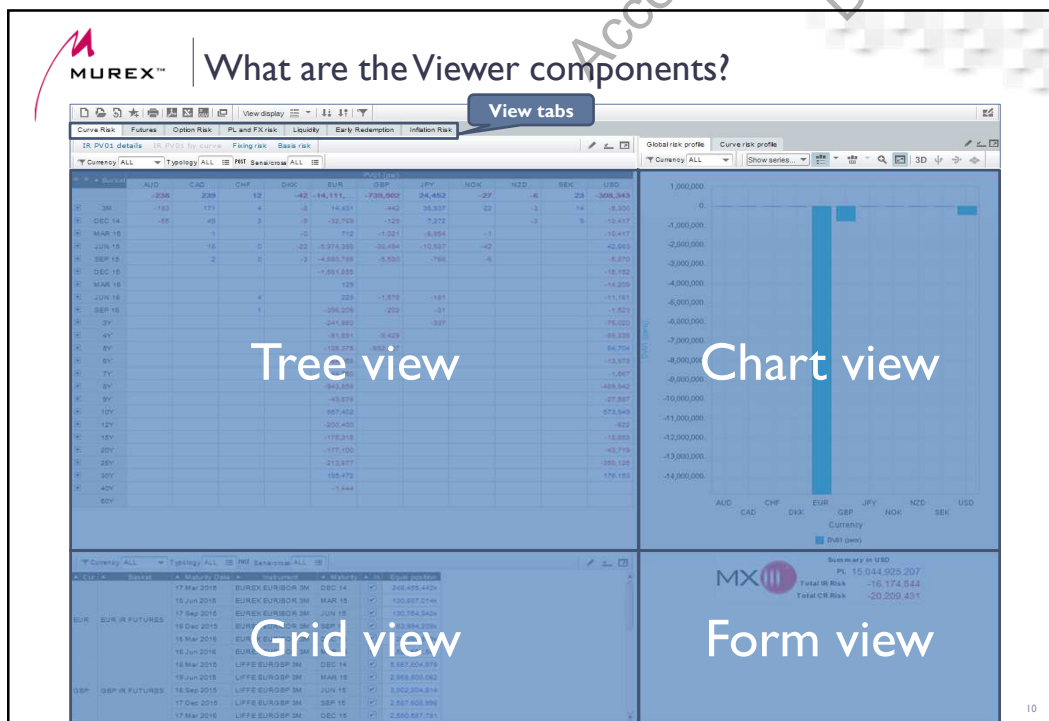
7



8



9



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MUREX™ | What are the Viewer components?

Standard toolbar for standard actions
(New, Reload, Print, ...)

View display toolbar
(change display type, sort, filter)

The screenshot displays the Murex Viewer interface. On the left, a table lists currency risk data for various currencies (AUD, CAD, CHF, CNY, EUR, GBP, JPY, NOK, NZD, SEK, USD) across different time periods (SEP 16, DEC 16, MAR 16, JUN 16, SEP 16, 3Y, 4Y, 5Y, 6Y, 7Y, 8Y, 9Y, 10Y, 12Y, 15Y, 20Y, 30Y, 40Y, 50Y). The table includes columns for 'Cur', 'Basis', 'Maturity', 'Instrument', 'MTM', 'PL', and 'CR'. On the right, a bar chart titled 'Currency risk profile' shows the risk profile for each currency. The chart has a vertical axis for 'Currency risk' ranging from -14,000,000 to 1,000,000. The bars are colored blue and red, indicating positive and negative risk respectively. A summary box at the bottom right shows 'Summary in USD' with values for 'PL', 'Total R Risk', and 'Total CR Risk'.

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MUREX™ | What are the Viewer components?

Single View Components

The screenshot displays the Murex Viewer interface showing a detailed table of P&L components. The table is titled 'P&L' and has columns for 'Portfolio', 'Cur', 'Typology', 'Instrument', 'Daily PL < EUR >', 'MTD PL < EUR >', and 'YTD PL < EUR >'. The data is organized into a hierarchical structure, starting with 'Total' and then breaking down into 'EQD_NYC', 'Total FI_BND_CORP_EU', and 'FI_BND_CORP_EU'. The 'FI_BND_CORP_EU' section further breaks down into 'Total EUR', 'Asset Swap IRS', and 'Bond'. The 'Bond' section lists various bond instruments with their respective P&L values.

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What are the Viewer components?

Single View Components

Edit view and Edit layout buttons
(more on this in the next slide!)

Portfolio	Cur	Typology	Instrument	Date			
Total							
EQD_NYC	GBP	Equity	BARC.GB				
Total F1_BND_CORP_EU							
		Total EUR			-98,387,906	-787,103,249	-1,180,684,883.15
		Total Asset Swap IRS			-3,527,801	-28,222,408	-42,363,622.30
		Asset Swap IRS	AXASA VAR 160440		-970,091	-7,760,726	-11,671,098.80
			BXP 5.875 151019		-2,557,710	-20,461,682	-30,722,533.50
		Total Bond			-104,632,175	-837,057,400	-1,255,616,110.67
			ABNANV FRN 25 310518		18,981	151,851	197,765.93
			ALOPF 3.625 051018		603,400	4,827,199	7,210,788.16
			ALVGR VAR 080741		10,512,270	84,098,160	126,117,230.01
			AUSTVB 0 5/18		-205	-1,641	-32,471.02
			AXASA VAR 160440		1,629,026	13,032,207	19,518,299.99
			BPCEGP 9 25 291049		1,666,584	13,332,672	19,968,997.87
			DZBK 0 3/34		-187	-1,493	-32,250.09
			EADS 5.5% 9/18		-206	-1,651	-32,486.68
			FRTEL 5.625 22MAY18		15,355,345	122,842,760	184,234,129.42
			ISPM 4.125 140420		516,295	4,130,358	6,165,526.67
			ISPM 5.15 160720		1,275,632	10,205,055	15,277,573.19
			ISPM 5.625 080518		1,387,984	11,103,875	16,625,902.25
			KFW 3.5% 7/15		-308	-2,466	-33,709.25
			MCD 4 170221		39,900	319,197	448,755.66
			MTNA STEP 290318 S		2,220,642	17,765,136	26,617,639.50
			NDASS 4 290321		277,823	2,222,567	3,303,879.43
			QUEB 3.375% 6/16		-104	-832	-31,255.42

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What are the Viewer components?

Use mode versus Edit modes

The Viewer has three operation modes:

Use (default mode)

Restricted to navigation, filtering, sorting and business-related actions.

Edit View

Zoom and design one view at a time.

Edit Layout

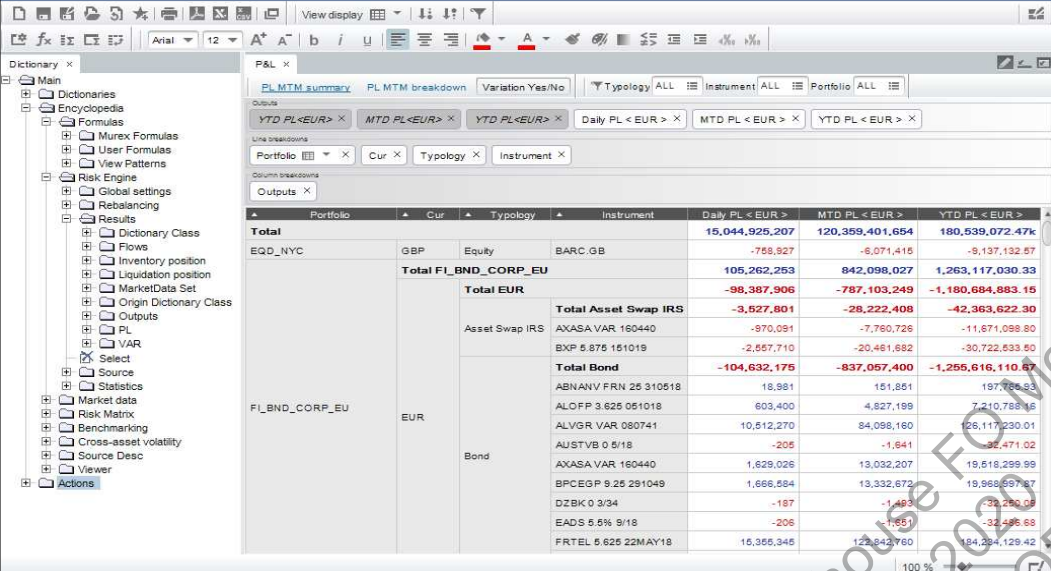
Shape the views arrangement (split the screen, add new views, reorder views position in the screen).

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What are the Viewer components?

Edit View Mode

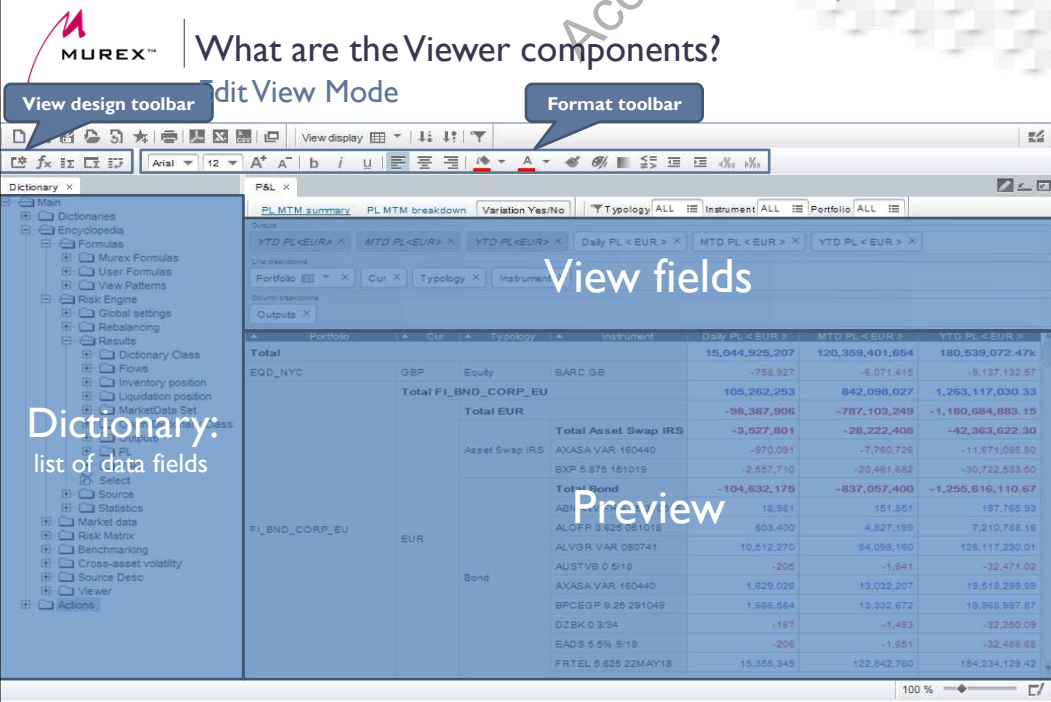


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What are the Viewer components?

Edit View Mode

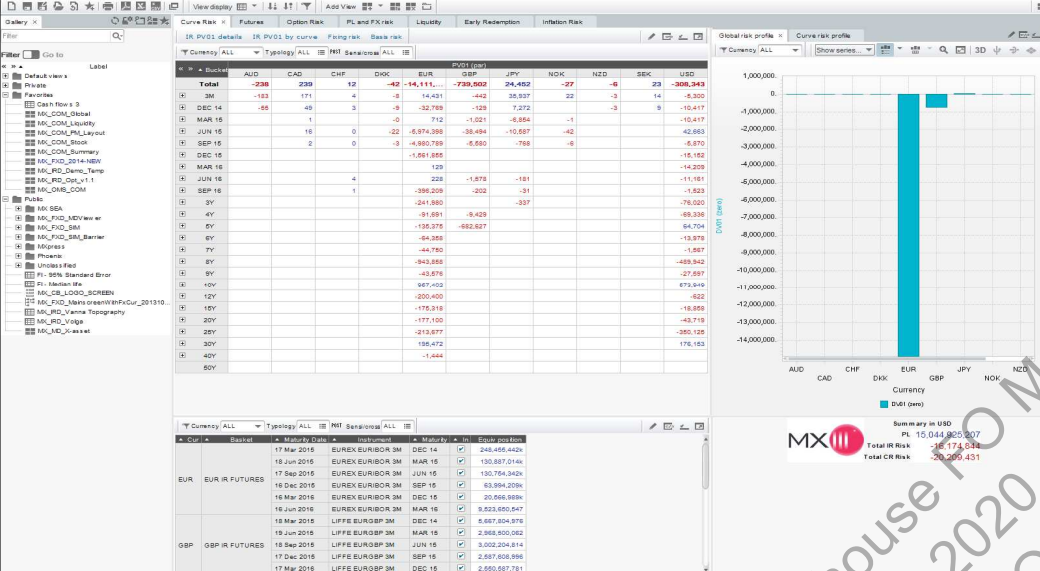


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What are the Viewer components?

Edit Layout Mode



The screenshot displays the Murex Viewer interface in 'Edit Layout Mode'. The interface is divided into several panels:

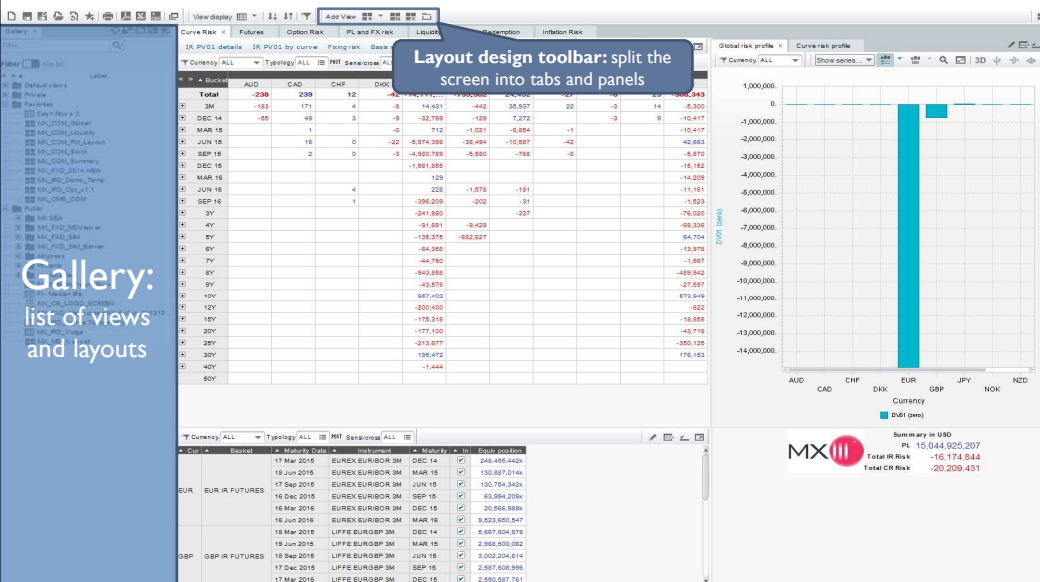
- Gallery:** A list of views and layouts on the left side, including 'Default view', 'Private', 'Cash flow', 'Risk', 'Summary', and 'Charts'.
- Data Table:** A central table showing risk data for various currencies (AUD, CAD, CHF, DKK, EUR, GBP, JPY, NOK, NZD, SEK, USD) across different dates (e.g., 17 Mar 2016, 19 Jun 2016, 17 Sep 2016).
- Global risk profile:** A bar chart on the right showing the risk exposure across different currencies, with a summary in USD at the bottom right.

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What are the Viewer components?

Edit Layout Mode



The screenshot displays the Murex Viewer interface in 'Edit Layout Mode', similar to the previous one but with additional annotations:

- Layout design toolbar:** A callout box points to the toolbar with the text "split the screen into tabs and panels".
- Gallery:** A callout box points to the gallery with the text "list of views and layouts".

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Introduction to the Viewer

Practice I: View and Layout manipulation



Manipulate a view.

1. Log into MX with user MUREXFO/MUREXFO.
2. Select the group FO.
3. Load a consolidated simulation (by position) on portfolio FI_BND_ALL.
4. Split the "PL breakdown" tab horizontally.
5. Rename the tab "PL MTM".
6. Add the view "MX_PL_MTM summary (c)" to the bottom of the tab.
7. Add the field "Risk Engine.Results.PL.P&L (Abs).Cash/MV" to the outputs of the view "MX_PL_MTM summary (c)".
8. Change its view display to Tree.
9. Save the layout as "<your name>_BOND".

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Agenda

Introduction to the Viewer

View Customization

- Filters
 - > Pre-filters
 - > Post-filters
 - > Differences
- Field settings
- Format
- Viewer actions

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View Customization

Comparison between Pre-filters and Post-filters

Pre-filters (data source filters)	Post-filters (ad hoc filters)
Filter out data from the data source, before the view is built.	Hide or show rows of data, after the view is built.
Limited to non-calculated fields.	Can be applied on any fields displayed in the view (calculated fields, totals and formulas).
Accelerate initial load and market data refresh. Changing a pre-filter value is not very fast, because it rebuilds the view.	Do not accelerate initial load nor market data refresh. Changing a post-filter value is fast.
Should be pre-configured and hidden. Not meant to be modified by the user.	Can be added and used on the fly by the user.
Impact totals.	By default, do not impact totals. Setting available to impact totals: View Format > Post filters : display like pre filters

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View Customization

Practice 2: Filters



Add filters and observe difference between pre and post filters.

1. In the bottom view of the "Bond and Futures position" tab, remove the filters "Security Type" and "Typology".
2. Add a pre-filter and a post-filter on the field "Sec code".
3. Display these filters as combo boxes.
4. With the pre-filter: filter on a single Sec code, then filter on ALL Sec code.
5. Do the same with the post-filter and notice the difference in performance.

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Agenda

Introduction to the Viewer

View Customization

- Filters
- Field settings
 - > Viewer generic properties
 - > Specific FO settings
- Format
- Viewer actions

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View Customization Generic Viewer Properties

In Edit view mode, right click > Properties
These settings are common to all fields.

The screenshot shows the 'Properties' dialog box with the 'General' tab selected. The 'Name' field is set to 'PL Instrument' and the 'Path' is 'Formulas.PL.Key.PL Instrument'. The 'Custom Instance' checkbox is unchecked, and the 'Instance Number' is '-1'. The 'Case Sensitive' checkbox is checked, and the 'Hide zero values' checkbox is unchecked. Under the 'Totals settings' section, the 'Enable totals' checkbox is checked, and the 'Display mode' is set to 'Automatic'. The 'Place totals at bottom' checkbox is unchecked, and the 'Customize Sub Total Label' field is empty. There are 'More', 'Proceed', and 'Cancel' buttons at the bottom.

Breakdown properties

The screenshot shows the 'Properties' dialog box with the 'Format' tab selected. The 'Name' field is set to 'P&L (Abs)' and the 'Path' is 'Risk Engine.Results.PL.P&L (Abs)'. The 'Allow Vertical Span' checkbox is unchecked, and the 'Custom Instance' checkbox is unchecked, with the 'Instance Number' set to '-1'. The 'Hide empty column' checkbox is unchecked, and the 'Hide zero values' checkbox is unchecked. The 'Hide null values' checkbox is checked, and the 'Filter rows with zero values' checkbox is unchecked. Under the 'Totals settings' section, the 'Enable totals' checkbox is checked, and the 'All totals' dropdown is set to 'All totals'. There are 'More', 'Proceed', and 'Cancel' buttons at the bottom.

Output properties

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View Customization Specific FO Settings

In Edit view mode, right click > Breakdown settings... or right click > Output settings...
These settings are specific to the field and to the context (e.g. Simulation).

Set date fields to bucketed/pillar set mode:

▲ Bucket	DV01 (zero)
26 Sep 2014	9
27 Sep 2014	0
29 Sep 2014	10
30 Sep 2014	0
01 Oct 2014	-2
02 Oct 2014	0
03 Oct 2014	-2
04 Oct 2014	-0
05 Oct 2014	0
06 Oct 2014	0

▲ Bucket	DV01 (zero)
1M	-76
3M	-143
6M	114
9M	84
1Y	-9,749
2Y	-5
3Y	-5
5Y	-0

Settings

Default

Mat. Source: Maturity Set | Disp. maturity set: LONG TERM | By Reference

Bucket type: Buckets | Reduced label: Yes | Standard label: Yes

Display Mode: Label

All Pillars: ☐

Split Mode: Next pillar

Shifter:

Truncate Flows: ☐

Apply

Convert values to a specific currency:

▲ PL Instrument	P&L (Abs)
Total	
AUD	2,804,602,511
AUD/USD	242,766
EUR	9,300,539,932
EUR/JPY	872,514
EUR/USD	-1,067,845
GBP	2,500,889,192
GBP/JPY	-4,956,294

▲ PL Instrument	P&L (Abs)<USD>
Total	39,059,695,289
AUD	2,416,943,430
AUD/USD	209,210
EUR	11,551,612,060
EUR/JPY	1,083,695
EUR/USD	-1,326,303
GBP	3,961,882,619
GBP/JPY	-7,851,709

Settings

Statistics Settings | Snapshot settings

Express in: Specific unit

Currency: USD

Display with title: Yes

Risk:

Apply haircut: No

Apply

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View Customization Practice 3: FO Settings



Customize a view.

1. In the top view of the "Bond and Futures position" tab, change the "Duration" display to "All data" instead of buckets.
2. Change the "Notional<USD>" currency to EUR instead of USD.

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Agenda

Introduction to the Viewer

View Customization

- Filters
- Field settings
- Format
 - > View Format
 - > Field Format
 - > Conditional Format
- Viewer actions

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View Customization Format

Conditional
Format

Field Format

View Format

Instrument	Type	Bar type	End Date	Knock	Distance	Delta Gap EUR
USD/JPY	BAR2	IN	22 Feb 2015	<i>KNOCKED</i>	-0.29	0.00
		OUT	29 Nov 2014	<i>KNOCKED</i>	-0.23	0.00
		OUT	24 Jan 2015	<i>KNOCKED</i>	-0.23	0.00
GBP/USD	BAR	DI	25 Sep 2015	<i>NEAR</i>	-0.02	661,170,976.40
		UI	22 Nov 2014	<i>NEAR</i>	0.01	-2,775,046.84
		UI	28 Sep 2014	<i>NEAR</i>	0.01	98.57
		UI	08 Oct 2014	<i>NEAR</i>	0.00	13,470.08
		UI	31 Oct 2014	<i>NEAR</i>	0.03	-23,574,678.45
		UI	25 Mar 2015	<i>KNOCKED</i>	0.00	0.00
		UI	25 Mar 2015	<i>KNOCKED</i>	0.00	0.00
		UI	24 Oct 2014	<i>NEAR</i>	0.01	82,864.85
		DI	31 Oct 2014	<i>NEAR</i>	-0.01	-6,303,373.37
		DI	31 Oct 2014	<i>NEAR</i>	-0.01	-6,303,373.37
EUR/USD	BAR2	IN	24 Jan 2015	<i>KNOCKED</i>	0.00	0.00
		IN	24 Sep 2017	<i>KNOCKED</i>	0.00	0.00
		IN	24 Sep 2017	<i>KNOCKED</i>	0.00	0.00
		OUT	24 Sep 2017	<i>KNOCKED</i>	0.00	0.00
		IN	25 Sep 2018	<i>KNOCKED</i>	0.00	0.00
		OUT	25 Sep 2018	<i>KNOCKED</i>	0.00	0.00
		OUT	25 Sep 2018	<i>KNOCKED</i>	0.00	0.00
		OUT	25 Sep 2018	<i>KNOCKED</i>	0.00	0.00
		IN	02 Oct 2014	<i>KNOCKED</i>	0.00	0.00
		OUT	02 Oct 2014	<i>KNOCKED</i>	0.00	0.00

Default format

Change the appearance of your view with Formatting.

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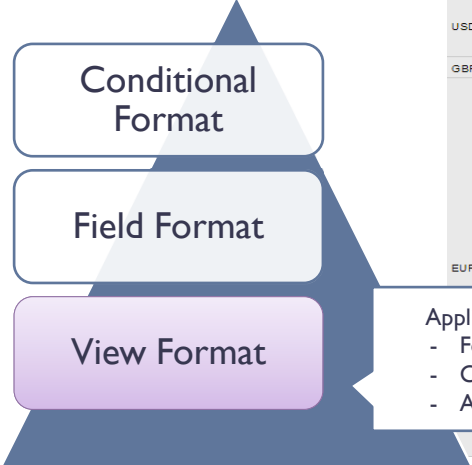
30

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View Customization

View Format



Conditional Format

Field Format

View Format

Instrument	Type	Bar type	End Date	Knock	Distance	Delta Gap EUR
USD/JPY	BAR2	IN	22 Feb 2015	KNOCKED	-0.29	0.00
		OUT	29 Nov 2014	KNOCKED	-0.23	0.00
		OUT	24 Jan 2015	KNOCKED	-0.23	0.00
GBP/USD	BAR	DI	25 Sep 2015	NEAR	-0.02	661,170,976.40
		UI	22 Nov 2014	NEAR	0.01	-2,775,046.84
		UI	28 Sep 2014	NEAR	0.01	98.57
		UI	08 Oct 2014	NEAR	0.00	13,470.08
		UI	31 Oct 2014	NEAR	0.03	-23,574,678.45
		UI	25 Mar 2015	KNOCKED	0.00	0.00
		UO	25 Mar 2015	KNOCKED	0.00	0.00
		UI	24 Oct 2014	NEAR	0.01	82,864.85
		DI	31 Oct 2014	NEAR	-0.01	-6,303,373.37
		IN	24 Jan 2015	KNOCKED	0.00	0.00
EUR/USD	BAR	IN	24 Sep 2017	KNOCKED	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00
OUT	02 Oct 2014	KNOCKED	0.00	0.00		

Applies to all fields in the view:

- Font size
- Cell colors
- Action button format...

View format

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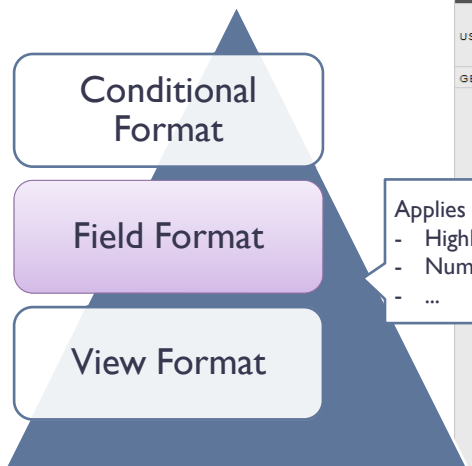
31

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View Customization

Field Format



Conditional Format

Field Format

View Format

Instrument	Type	Bar type	End Date	Knock	Distance	Delta Gap EUR	
USD/JPY	BAR2	IN	22 Feb 2015	KNOCKED	-0.29	0.00	
		OUT	29 Nov 2014	KNOCKED	-0.23	0.00	
		OUT	24 Jan 2015	KNOCKED	-0.23	0.00	
GBP/USD	BAR	DI	25 Sep 2015	NEAR	-0.02	661,170,976.40	
		UI	22 Nov 2014	NEAR	0.01	-2,775,046.84	
		UI	28 Sep 2014	NEAR	0.01	98.57	
		UI	08 Oct 2014	NEAR	0.00	13,470.08	
		UI	31 Oct 2014	NEAR	0.03	-23,574,678.45	
		0.00	0.00	0.00	0.00	0.00	
		0.01	82,864.85	-0.01	-6,303,373.37	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00
		0.00	0.00	0.00	0.00	0.00	0.00
EUR/USD	BAR2	OUT	24 Sep 2017	KNOCKED	0.00	0.00	
		IN	25 Sep 2016	KNOCKED	0.00	0.00	
		OUT	25 Sep 2016	KNOCKED	0.00	0.00	
		OUT	25 Sep 2016	KNOCKED	0.00	0.00	
		OUT	25 Sep 2016	KNOCKED	0.00	0.00	
		IN	02 Oct 2014	KNOCKED	0.00	0.00	
		OUT	02 Oct 2014	KNOCKED	0.00	0.00	

Applies to individual fields:

- Highlight color
- Number of decimals
- ...

Field format

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MUREX™ | View Customization
Conditional Format

Conditional Format

Format based on a condition, e.g.:
knocked barriers become red, etc.

Field Format

View Format

Instrument	Type	Bar type	End Date	Knock	Distance	Delta Gap EUR
USD/JPY	IN	22 Feb 2015	KNOCKED	-0.29	0.00	
USD/JPY	BAR2	OUT	29 Nov 2014	KNOCKED	-0.23	0.00
EUR/USD	UI	24 Oct 2014	NEAR	-0.02	661,170,976.40	
EUR/USD	DI	31 Oct 2014	NEAR	0.01	-2,775,046.84	
EUR/USD	IN	24 Jan 2015	KNOCKED	0.01	98.57	
EUR/USD	IN	24 Sep 2017	KNOCKED	0.00	13,470.08	
EUR/USD	IN	24 Sep 2017	KNOCKED	0.03	-23,574,678.45	
EUR/USD	OUT	24 Sep 2017	KNOCKED	0.00	0.00	
EUR/USD	IN	25 Sep 2016	KNOCKED	0.01	82,864.85	
EUR/USD	OUT	25 Sep 2016	KNOCKED	-0.01	-6,303,373.37	
EUR/USD	IN	24 Jan 2015	KNOCKED	0.00	0.00	
EUR/USD	IN	24 Sep 2017	KNOCKED	0.00	0.00	
EUR/USD	IN	24 Sep 2017	KNOCKED	0.00	0.00	
EUR/USD	OUT	24 Sep 2017	KNOCKED	0.00	0.00	
EUR/USD	IN	25 Sep 2016	KNOCKED	0.00	0.00	
EUR/USD	OUT	25 Sep 2016	KNOCKED	0.00	0.00	
EUR/USD	OUT	25 Sep 2016	KNOCKED	0.00	0.00	
EUR/USD	OUT	25 Sep 2016	KNOCKED	0.00	0.00	
EUR/USD	IN	02 Oct 2014	KNOCKED	0.00	0.00	
EUR/USD	OUT	02 Oct 2014	KNOCKED	0.00	0.00	

Conditional format

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MUREX™ | View Customization
Practice 4: Formatting

Modify View formats.

The MXpress formatting template should be already unzipped in the cfg folder of your client directory.

1. Reset all field formatting from your view.
2. Import and apply the template to your view.
3. Highlight an output by setting its background color to RGB(242,225,255).

Check design recommendations in the MXpress views design document (docid:6407).

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Agenda

Introduction to the Viewer

View Customization

- Filters
- Field settings
- Format
- **Viewer actions**

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Viewer Actions

Actions are buttons or links that can be added to a view.

They have various effects:

- ✓ Viewer-specific actions (open a new view, input parameter...)
- ✓ Functional actions (build a smile, insert what-if orders...)

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View Customization

Practice 5: Actions



Create a new action in a view.

1. Add a view action button to the “MX_PL_MTM summary (c)” view (dictionary path: Actions.Default Actions.View).
2. Configure the new action to open the “MX_BOND_Position” view in a new window.

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APPENDIX

Viewer e-learning
Viewer documentation
Viewer features
MXpress views

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Viewer E-learning

MUREX KB: Getting started > MX.3 Platform Overview > Navigation and rendering tools > Viewer > Viewer starting v3.1.23

Section	Details
Viewer for Designer	Tutorial 1: Introduction Tutorial 2: First Steps Tutorial 3: Format Tutorial 4: Sorting Tutorial 5: Filters (Part1) Tutorial 5: Filters (Part2) Tutorial 6: Action Tutorial 7: Formulas Tutorial 8: Totals Tutorial 9: Layout Tutorial 10: Form Tutorial 11: Charts Tutorial 12: Data Export
Viewer for End Users	Tutorial 1: Introduction Tutorial 2: Viewer for End User Tutorial 3: Gallery Tutorial 4: Data Export

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Viewer Documentation

Doc title	Doc ID	Preview
Viewer FAQ	12074	Short questions & answers on the viewer, related to installation, upgrade, layout & view construction, menus, use & design modes, find & filter, toolbars, data display, printing and rights management.
MX Viewer Technical guide	7037	Detailed technical doc on the viewer features and settings
Data Visualization Catalog	19521	Description of the available charts in MX in order to make users aware of the many possibilities of data visualization in the Viewer and help them find the most appropriate representation for their data.
Viewer Configuration Management	14366	Viewer import/export through the Import/export wizard
Views & Layouts assignment	10302	General views & layout assignment (excluding Simulation views)
Simulation screen assignments	12990	Simulation views & layout assignment
Linked views	17066	How to create linked views
Viewer Rights management	17065	How to set rights on views at view level or using Rights templates
Viewer Forms	16545	How to design forms
Viewer 3D Charts	17747	How to set up a view in 3D
Viewer Data Export & Printing	17100	How to export data to Excel, CSV, PDF or send it to a printer
Viewer User Guide	7037	Getting started with the viewer

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Viewer Features

Data Selection

Comprehensive Data Dictionary	Choose the data fields to feed your view from the data dictionary.
Breakdown Grouping and Columns Arrangement	Group your data fields by adding, reordering, hiding columns.
Aggregated Calculations	Display summed totals, average or standard deviation for each grouping level.

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Viewer Features

Data Display

Layout Organization	Split the screen into different panels and tabs to design the most efficient data dashboard.
Cells Format	Resize cells, set their background/foreground/border colors, choose the text font and alignment.
Conditional Format	Update the cells format depending on a certain condition.
Wide Range of Data Visualization Types	Choose the best suited data visualization type to display your data (grids, charts...).

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Viewer Features

Data Interaction

Drill-down Navigation	Expand/Collapse nodes to investigate data in increasing detail.
Filtering and Sorting	Individual or shared filters across views. Absolute sorting or within grouping levels.
User-defined Actions	Add action buttons to trigger specific commands.

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Viewer Features

Other Distinctive Features

View and Layout Assignment	Assign a default view or layout to load automatically for particular users.
Reusable Objects	Viewer objects can be exported and reused in different environments (formatting templates, view an layout definitions, gallery organization...).
Exportable Data	Data can be exported to CSV, Excel and PDF files.

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The MXpress package is a set of out-of-the-box views.
Use MXpress preconfigured views as a building base to get recommended Viewer settings.
The following list is a sample of MXpress views categories.

P&L, liquidity, e-Tradepad blotter for all assets
IRD blotter, PV01, position
CRD blotter, CR01 risk and exposure
EQD sensitivities and risk
FXMM position, blotter, trade contribution
Security finance inventory, delivery
COM risk and hedge position
FXD risk and position

[illegible]

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MUREX™

Thank You!

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twitter.com/Work_at_Murex





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contact@murex.com

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A presentation slide titled 'Learning Objectives' for 'P&L Variance'. The slide features the Murex logo in the top left corner. The main content is a list of four learning objectives, each preceded by a blue checkmark. The slide is framed by a thin black border. A large, diagonal watermark reading 'Accenture: In-house FO Modules August 2020 DO NOT COPY' is visible across the right side of the slide.

MUREX™ | Learning Objectives
P&L Variance

At the end of this course, you will be able to:

- ✓ Explain how the P&L Variation Analysis module is used
- ✓ Configure a P&L Variance Scenario and View
- ✓ Drill down through the results of the P&L Variance view
- ✓ Explain the differences between the P&L Explain and P&L predict figures

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Agenda

- Introduction**
- Definition of a Scenario
- Viewing the Results

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Introduction

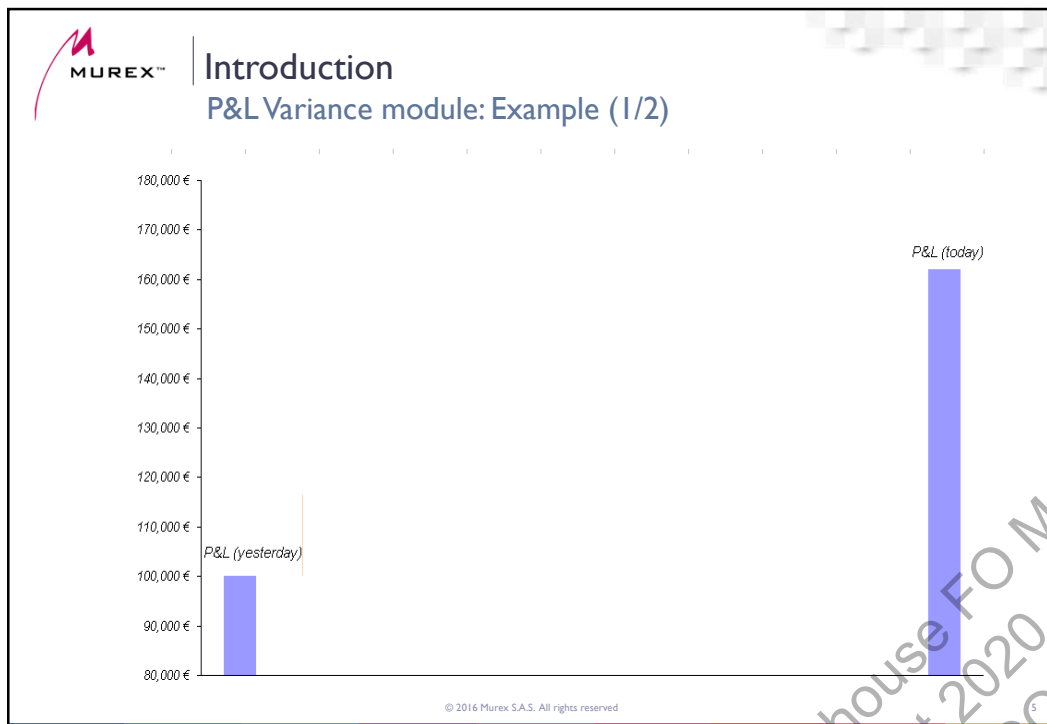
P&L Variance module

P&L Variance module is a cross-asset tool enabling analysis of P&L variations between two dates. The P&L Variance module:

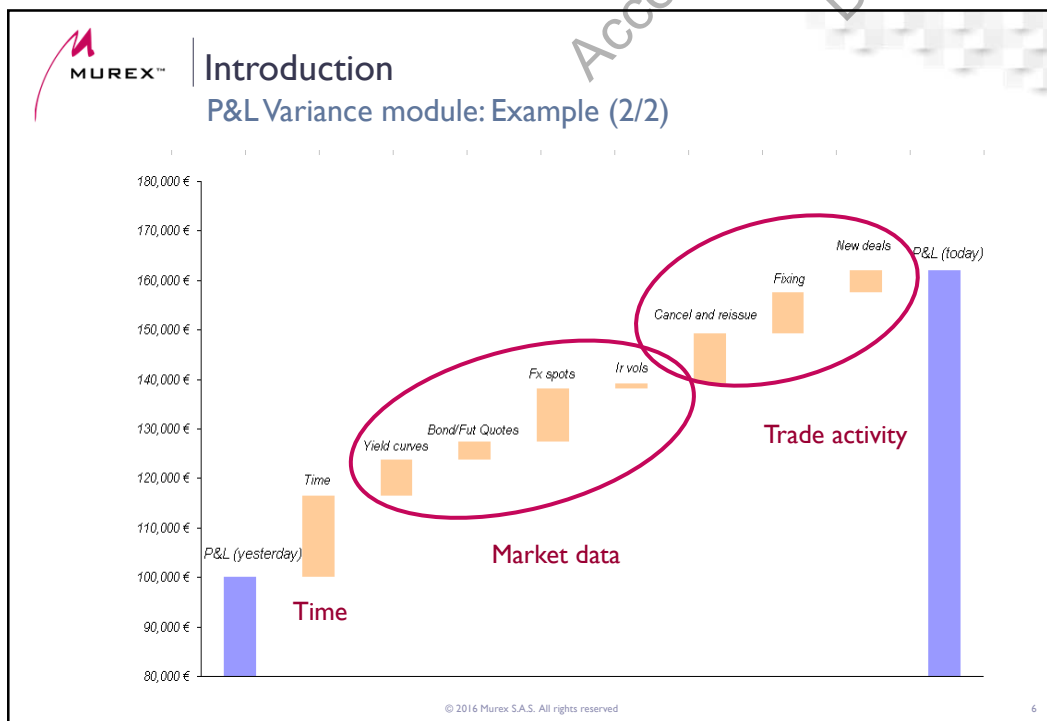
Provides detailed breakdown of the P&L change	Between the closing P&L of the previous open day and today's current P&L due to: ✓ Time decay ✓ All market data variations ✓ All market operations on deals
Quantifies	The impacts on the P&L of each effect individually
Zooms	As far as deal level to understand a PL change
Computes	In addition to the P&L Explain, it also computes the P&L Predict values based on sensitivities.

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Agenda

Introduction

Definition of a Scenario

- Scenario
- Scenario header
- List of available effects
- P&L predict

Viewing the Results

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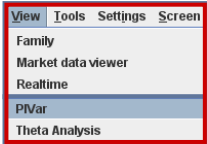


Definition of a Scenario

Scenario

A scenario is a set of overall settings and of ordered effects to be processed by the tool. The scenario is configured via the simulation module.

View/ PIVar	List of available scenarios
Space bar	To open an existing one
Insert	To create a new one





Complex and comprehensive scenarios can be created by combining different effects.

End-user session: Simulation/ Simulation

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MUREX™ | Definition of a Scenario
Scenario header (1/3)

The Scenario definition screen is divided into three parts:

Scenario header

Selection of effects

List of available effects

End user session: Simulation/ Simulation/ Tools/ PIVar

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The screenshot shows the 'Scenario Edit' window. At the top, there's a header section with 'User MUREXFO Group FO Visible by' and 'Currency EUR'. Below this is a table with columns for 'Family 1', 'Family 2', 'Step Name', 'Description', 'Restore', 'By Curve', 'First Order', 'Second Order', 'Explain', and 'Start predict level'. The table lists various effects like 'Theta analysis', 'Time Decay', 'PL variance', 'Market data variation', and 'Trades'. A list of available effects is shown on the right, including 'Yield without prop', 'Basis curves', 'Credit curves', 'Credit without propagation', 'FX butterflies', 'FX risk reversals', 'FX smiles', 'Eq vols', 'Eq smiles', 'FX vols', 'FX spots', 'Eq/Fut prices', 'Bond/Fut prices', 'Correlations', 'Repo spreads', 'Dividends', 'Bond/Fut spreads', 'Recovery rate assignments', 'Commodity curves', and 'Rates assignments'.

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MUREX™ | Definition of a Scenario
Scenario header (2/3)

The Scenario definition screen is divided into three parts:

Scenario header: Label, overall settings

Selection of effects: Initial State is mandatory

- "No effect" is selected by default
- Effects selected using '+' button in 'Action' column
- Order can be changed with drag & drop function

End user session: Simulation/ Simulation/ Tools/ PIVar

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This screenshot focuses on the 'Selection of effects' section of the 'Scenario Edit' window. It shows the table with columns for 'Family 1', 'Family 2', 'Step Name', 'Description', 'Restore', 'By Curve', 'First Order', 'Second Order', 'Explain', and 'Start predict level'. The table lists various effects like 'Theta analysis', 'Time Decay', 'PL variance', 'Market data variation', and 'Trades'. The 'Initial State' is highlighted as mandatory. The 'Action' column contains '+' buttons for selecting effects. The 'Order' column shows the sequence of effects, which can be changed using drag & drop.

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MUREX™ | **Definition of a Scenario**
Scenario header (3/3)

The Scenario definition screen is divided into three parts:

List of available effects:

- ✓ Categories:
 - > Time decay
 - > Market data variation
 - > Trades
- ✓ User-defined step: combination of effects

End user session: Simulation/ Simulation/ Tools/ PIVar

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MUREX™ | **Definition of a Scenario**
Scenario header

The following fields are set in the scenario header screen:

Label	Description
Label	Label of the scenario
Display setting	Default layout considered to display results
Calendar	A specific calendar can be defined
Currency	A specific accounting currency can be defined
Additional Outputs View	Defines outputs in addition to the P&L (clean P&L for example)
MarketdataSet	A specific MDS can be defined
Show zero variation	Yes/No

End user session: Simulation/ Simulation/ Tools/ PIVar

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MUREX™ | **Definition of a Scenario**
List of available effects

The P&L scenario provides detailed breakdown of the P&L change due to a wide range of effects that are classified into three categories:

- ✓ Time decay
- ✓ Market data variation
- ✓ Trades

Category	Effect	Effect	Effect
Time Decay	Time Decay	Time	Pure time decay
	Yield curves	Yield curves	Yield curves
	Yield without prop	Yield without propagation	Yield without propagation
	Basis curves	Basis curves	Basis curves
	Inflation curves	Inflation curves	Inflation curves
	Credit curves	Credit curves	Credit curves
	Credit without prop	Credit without propagation	Credit without propagation
	Fx vols	Fx vols	Fx vols
	Fx butterflies	Fx butterflies	Fx butterflies
	Fx risk reversals	Fx risk reversals	Fx risk reversals
Market data variation	Fx smiles	Fx smiles	Fx smiles
	Eq vols	Eq vols	Eq vols
	Eq smiles	Eq smiles	Eq smiles
	Ir vols	Ir vols	Ir vols
	Ir smiles	Ir smiles	Ir smiles
	Fx spots	Fx spots	Fx spots
	EqFut prices	EqFut prices	EqFut prices
	BondFut prices	BondFut prices	BondFut prices
	Correlations	Correlations	Correlations
	Repo spreads	Repo spreads	Repo spreads
Trades	BondFut spreads	BondFut spreads	BondFut spreads
	Recovery rate	Recovery rate	Recovery rate
	Commodity curves	Commodity curves	Commodity curves
	Rates assignments	Rates assignments	Rates assignments
	Commodity vols	Commodity vols	Commodity vols
	Commodity smiles	Commodity smiles	Commodity smiles
	Option prices	Option prices	Option prices
	Base Correlation curves	Base Correlation curves	Base Correlation curves
	Credit Entry event	Credit Entry event	Credit Entry event
	Substitution event	Substitution event	Substitution event
Other	Credit vols	Credit vols	Credit vols
	Credit Curve Assignment	Credit Curve Assignment	Credit Curve Assignment
	Fx Future prices	Fx Future prices	Fx Future prices
	Eq Vol-Smile	Eq Vol-Smile	Eq Vol-Smile
	Fx vanna	Fx vanna	Fx vanna
	Dividends yield	Dividends yield	Dividends yield
	Com. trans. smil	Com. trans. smil	Com. trans. smil
	All Market Data	All Market Data	All Market Data
	Floats	Floats	Floats
	Market Operation	Market Operation	Market Operation
Other	Cmp curves	Cmp curves	Cmp curves
	Models assignment	Models assignment	Models assignment
	Closing	Closing	Closing
	Security Modification	Security Modification	Security Modification
	Global other	Global other	Global other
	Other (Time)	Other (Time)	Other (Time)
	Spot PL Conversion	Spot PL Conversion	Spot PL Conversion

End user session: Simulation/ Simulation/ Tools/ PIVar

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MUREX™ | **Definition of a Scenario**
List of available effects

For each effect, the user has the choice between two methods:

	Definition	Advantage	Disadvantage
Cumulative mode (default mode):	Moving the parameters in a specific order, one after the other, keeping the already shifted parameters in their final value.	The sum of all effects matches the daily P&L (cross effects are taken into account)	Contribution of each effect may vary depending on the scenario order
Restore mode	Each parameter is shifted to new value, then reset to its initial value before any other parameter is moved.	It gives the exact contribution of each parameter independently of the other effects.	The sum of all effects does not match the daily P&L

PLVar module assumes several computation settings (which are hard-coded for the module) under Configuration/ Settings/ Simulation settings (Simulation setup):

- Rate configuration>Reaction to time decay>Rate curve: Keep market rates constant.
- Rate configuration>Reaction to time decay>Volatilities: EOD mode.
- Fx configuration>Volatility>ATM Volatility Shift mode: Roll pillar dates and keep volatility curve constant.
- Fx configuration>Evaluation>Pricing configuration>Roll spot on forward curve: Off

Without those settings in simulation, it will not be possible to match the figures.

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MUREX™ | **Definition of a Scenario**
P&L predict

The P&L predict shows the variations of P&L based on a Taylor expansion:

To Activate the Predict mode:

- 1) Check the “Start predict level” flag.
- 2) Check the First Order column for the desired effects.



Family 1	Family 2	Step Name	Description	Restore	By Curve	By Curr. Pair	By Index	By Asset Class	First Order	Second Order	Explain	Start predict level
Theta analysis	Initial State	Initial state	Initial state									
	Time Decay	Time	Pure time decay	☐								
		Yield curves	Yield curves	☐	☐				☒	☐	☒	☐
		Basis curves	Basis curves	☐							☒	☐
		Initiation curves	Initiation curves	☐							☒	☐
		Rates assignments	Rates assignments	☐							☒	☐
		Recovery rate	Recovery rate	☐					☐		☒	☐
		Credit curves	Credit curves	☐	☐				☒		☒	☐

Computations are done using a first order Taylor expansion.

Sensitivities used in P&L predict are the sensitivities of the day i.e. after the time effect movement.

End user session: Simulation/ Simulation/ Tools/ PIVar

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MUREX™ | **Definition of a Scenario**
Practice I: Configure a P&L Variance scenario

Configure a P&L Var scenario

1. Open an end-user session (MUREXFO/MUREXFO), group FO.
2. Go to Simulation/ Portfolio simulation by trade.
3. Select portfolio US_P_FLOW_ST.
4. Go to View/ PIVar
5. Create a PL Var scenario labeled “PLVAR_EX” with layout *MX_ALL_PLVariance_ljt*, calendar *GLOBAL*, currency *USD*, market data set *MDS*, Show zero variation activated and with the following steps:
 - ✓ Time
 - ✓ Yield curves
 - ✓ Repo spreads
 - ✓ Eq/Fut prices
 - ✓ Bond/Fut prices
 - ✓ Fx spots
 - ✓ Eq vols
 - ✓ Eq smiles
 - ✓ Dividends
 - ✓ Dividends yield
 - ✓ Correlations
 - ✓ Option prices
 - ✓ Fixings
 - ✓ Closing
 - ✓ Gmp curves
 - ✓ Spot PL Conversion
 - ✓ Market Operation
 - ✓ Global other

 **Solution**

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MUREX™ | Agenda

Introduction

Definition of a Scenario

Viewing the Results

- Results in Viewer
- Viewer configuration
- Troubleshooting tools

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MUREX™ | Viewing the results

Results in Viewer

Default layout: Viewer layout to be considered by default to display results

Scenario Edit

MX_PLVA_ALL

Display Setting: ALL_PL_Vari

Currency: USD

Marketdata Set: MDS

Additional Outputs View: Show zero variation

Portfolio	Effect	Effect detail	Typology	Instrument	PL variation<USD>
Total US_P_FLOW_ST			ETO EQD	DJIA 30	89,806
				Total Equity	-312
				BAY.DE	-3
				BMW.DE	0
				BNP.FR	0
				C.US	-6
				CBK.DE	-6
				FP.FR	0
				GE.US	1
				KO.US	-2
				MSFT.US	-1
				OR.FR	0
				SAP.DE	-6
				T.US	1
				VOW.DE	0
				XOM.US	-3
				Total Vanilla O.	10
				BMW.DE	-6
				C.US	6
				CBK.DE	-6
				FP.FR	0
				GE.US	-2
				KO.US	4
				OR.FR	-1
				SAP.DE	-6
				T.US	2
				VOW.DE	1
				Total Equity	-6
				BNP.FR	0
				C.US	0
				GE.US	-6
				KO.US	1
				MSFT.US	0
				OR.FR	-6

Aggregation/ breakdown keys

Results

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MUREX™ | Viewing the results
Viewer configuration

PLVar fields are available under:
Risk Engine/ Results/ PL/ PL Explain and Predicted P&L

Main output	P&L variation
The contribution of each effect	Step label
Sub-effects	Underlying
Other standard P&L breakdown keys can be added	Portfolio, Group, Contract Number, etc.

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MUREX™ | Viewing the results
Troubleshooting tools

What could go wrong?

- ✓ Different market data sets
- ✓ Native currency for a multi-currency portfolio
- ✓ Missing effects
- ✓ Change of configuration between yesterday and today

PLVAR log	Help/ Monitor/ Rates library “Display Theta Analysis Flows” It produces a log file rt.<NPID>.trc showing each trade evaluation
PLVAR debug mode	Help/ Monitor/ PIVariance Debug and check the “Debug” box on the right hand side for each effect requiring troubleshooting. ▪ It opens a simulation at each step ▪ It allows to check the trade pricing and market data after each step ▪ As such, it is best used in parallel with a simulation as of today

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Viewing the results

Practice 2: Explain the P&L Variance effects



Explain the P&L Variance effects

1. In same end-user session as practice 1, open another MX tab so to have 2 screens simultaneously:

- ✓ Portfolio simulation as of past date by trade
 - > Select the date 24/09
 - > Select the portfolio US_P_FLOW_ST
 - ✓ Portfolio simulation by trade on the portfolio US_P_FLOW_ST
2. Compute the PL of trade number 29 (NB=29) as of 24/09/2012 & 25/09/2012. You may have to add the "Trade number" field to your view.
3. Launch the PIVar on portfolio US_P_FLOW_ST with the scenario defined in practice 1.
4. Check the PL Variance is in line with the expected total variation.



Use the PL Breakdown tab (MX_PL_MTM breakdown daily (c)) to monitor the P&L



Solution

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Thank You!

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MUREX™ | Definition of a Scenario
Practice I: Configure a P&L Variance scenario

Configure a P&L Var scenario

P&L Variance Scenario "PLVAR_EX"

Scenario Edit

User: MUREXFO Group FO Visible by: Everybody Editable by: Everybody

Display Setting: MX_ALL_PL Var Currency: USD Marketdata Set: MDS
Calendar: GLOBAL Additional Outputs View: Show zero variation: ☒

Family 1	Family 2	Step Name	Description	Restore	By Curve	By Curr. Pair	By Index	By Asset Class	First Order	Second Order	Explain	Start predict level
Theta analysis	Initial State	Initial state	Initial state	☐							☒	☒
	Time Decay	Time	Pure time decay	☐	☐				☐	☐	☒	☐
		Yield curves	Yield curves	☐					☐	☐	☒	☐
		Repo spreads	Repo spreads	☐					☐	☐	☒	☐
		Eq/Fut prices	Eq/Fut prices	☐					☐	☐	☒	☐
		Bond/Fut prices	Bond/Fut prices	☐					☐	☐	☒	☐
		Fx spots	Fx spots	☐		☐			☐	☐	☒	☐
		Eq vols	Eq vols	☐					☐	☐	☒	☐
		Eq smiles	Eq smiles	☐					☐	☐	☒	☐
		Dividends	Dividends	☐					☐	☐	☒	☐
		Dividends yield	Dividends yield	☐					☐	☐	☒	☐
		Correlations	Correlations	☐				☐	☐	☐	☒	☐
		Option prices	Option prices	☐				☐	☐	☐	☒	☐
		Fixings	Fixings	☐			☐		☐	☐	☒	☐
		Closing	Closing	☐					☐	☐	☒	☐
		Gmp curves	Gmp curves	☐	☐				☐	☐	☒	☐
Other		Spot PL Conversion	Spot PL Conversion						☐	☐	☒	☐
PL variance	Trades	Market Operation	Market Operation								☒	☐
Other		Global other	Global other								☒	☐

Exercise

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Viewing the results

Practice 2: Explain the P&L Variance effects

**Solution:** Explain the P&L Variance effects
Please refer to the MTEK_PLVar_exercise.doc file for details on reconciliation

**Exercise**

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Accenture: In-house FO Modules
August 2020
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FO Module training

PL Variance: Practice 2: Solution

Note: There might be slight differences in the numbers with the numbers on your environment due to changes made on the market data during the training.

Consider the trade number **29** which is an Equity Deal on the BMW.DE Equity. Portfolio **US_P_FLOW_ST**.

Market data for the **BMW.DE**:

Market		24 Sep 2012
View	SECURITY MAIN	

Currency	Security	M.Label	Main quotation	Second qu...	Main Closing P...	Main Last Price	ISIN Code	Time
EUR	BMW.DE		25.00	0	0.00	0.00	DE0005190003	15:12:52

Market		25 Sep 2012
View	SECURITY MAIN	

Currency	Security	M.Label	Main quotation	Second qu...	Main Closing P...	Main Last Price	ISIN Code	Time
EUR	BMW.DE		27.34	0	0.00	0.00	DE0005190003	20:15:24

Open 2 screens simultaneously.

1) Portfolio Simulation As of Past date by trade.

- Select the portfolio **US_P_FLOW_ST**.
- Select the date **24/09**

2) Portfolio Simulation by Trade.

- Select the portfolio **US_P_FLOW_ST**.

3) Go to the View **PL Breakdown** in both screens and add the field **Trade number** to the breakdown columns.

4) Look at the **PL <USD>** column in both screens for the trade **29**. Record the values.

For 24/09 the value is \$ 23,318

US_P_FLOW_ST Connect to realtime Calculate

Real time

Bond and Futures position Curve risk Yield risk Eq pos by country PL breakdown n

PL MTM summary PL MTM breakdown Variation Yes/No Typology ALL Instrument ALL Portfolio ALL

Portfolio	Cur	Typology	Instrument	Trade Number	PL<USD>	Market value	Cash	Capital gain	Revenue	PV effect	Financing	Brk/adj fees
Total US_P_FLOW_ST					-35,014,738	25,150,867	-59,294,695	-34,143,721		-107	-870,910	
Total EUR					5,715,231	21,765,136	-15,904,002	5,861,155		-21	-145,903	
Total Equity					-63,673	-1,849,864	1,767,698	-82,166		0	18,493	
VOW.DE					-106,735	-1,067,562	950,874	-116,688		0	9,953	
SAP.DE					-6,051	144,278	-148,772	-4,494		0	-1,557	
OR.FR					16,306	-430,291	441,976	11,685		0	4,821	
FP.FR					3,470	-107,188	109,513	2,325		0	1,145	
CBK.DE					-5,350	102,177	-106,413	-4,236		0	-1,114	
BNP.FR					12,168	-315,107	323,889	8,782		0	3,386	
Total BMW.DE					28,652	-96,857	124,209	27,352		-0	1,300	
28					-8,341	67,800	-74,580	-6,780	0	-0	-1,561	0
29					23,318	-116,228	136,684	20,456	0	0	2,861	0
3,440					-6,780	67,800	-74,580	-6,780	0	0	0	0
3,441					20,456	-116,228	136,684	20,456	0	-0	0	0
BAY.DE					-6,133	-79,314	72,423	-6,891		0	756	
Future EQD					4,397,293	4,397,293	0	4,397,295		-1		

For 25/09 the value is \$ 12,486

US_P_FLOW_ST Synchronized Calculate

Real time

Bond and Futures position Curve risk Yield risk Eq pos by country PL breakdown n

PL MTM summary PL MTM breakdown Variation Yes/No Typology ALL Instrument ALL Portfolio ALL

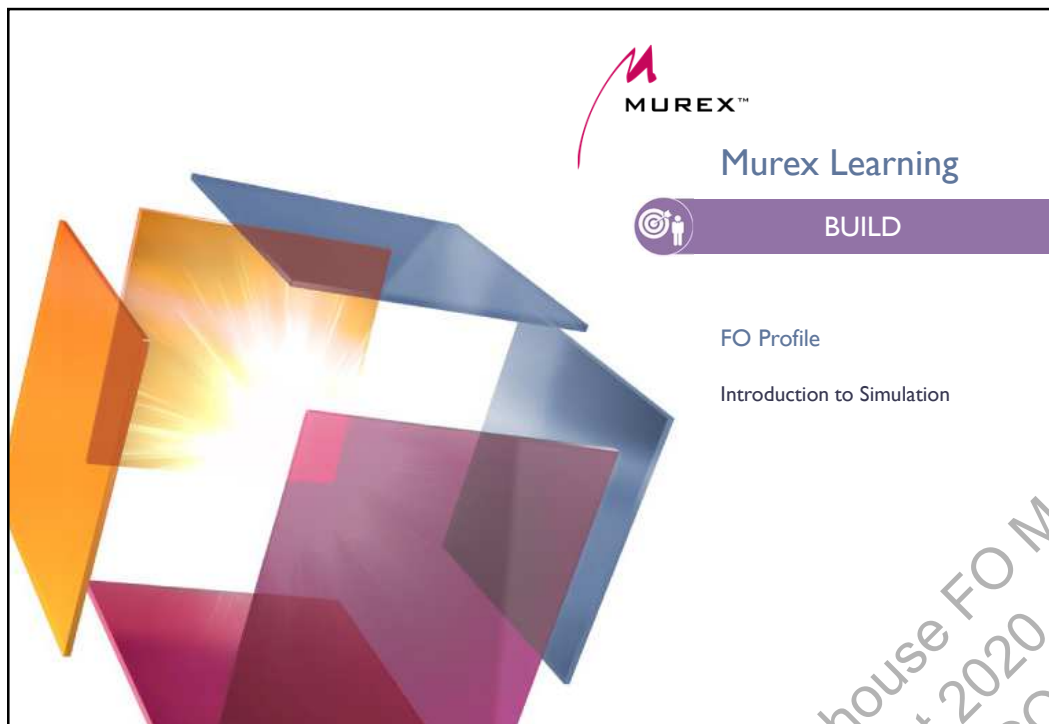
Portfolio	Cur	Typology	Instrument	Trade Number	PL<USD>	Market value	Cash	Capital gain	Revenue	PV effect	Financing	Brk/adj fees
Total US_P_FLOW_ST					-34,962,783	25,211,807	-59,303,522	-34,091,622		-93	-871,068	
Total EUR					5,767,602	21,826,061	-15,912,827	5,913,254		-19	-145,732	
Total Equity					-85,640	-1,878,615	1,774,411	-104,204		1	18,563	
VOW.DE					-107,140	-1,071,616	956,435	-117,131		0	9,991	
SAP.DE					-6,074	144,826	-149,937	-4,511		0	-1,563	
OR.FR					16,368	-431,925	443,655	11,729		0	4,639	
FP.FR					3,483	-107,595	109,928	2,333		0	1,149	
CBK.DE					-5,370	102,565	-106,817	-4,252		0	-1,118	
BNP.FR					-412	-326,930	325,119	-3,811		0	3,399	
Total BMW.DE					19,661	-105,925	124,681	18,556		0	1,305	
28					-2,003	74,427	-74,863	-436	0	-0	-1,567	0
29					12,486	-127,590	137,203	9,614	0	0	2,872	0
3,440					-436	74,427	-74,863	-436	0	0	0	0
3,441					9,614	-127,590	137,203	9,614	0	-0	0	0
BAY.DE					-6,156	-79,615	72,698	-6,917		0	761	
Future EQD					4,413,991	4,413,991	0	4,413,993		-1		

The PL Variance should be equal to 23,318 – 12,486 = - \$ 10,382

- From the **portfolio simulation by trade** screen, hit **View-> PIVar**.
- Select the scenario configured in exercise1.
- Drag and Drop the **Trade number** field to filter on the trade number 29.
- Let's reconcile to numbers which contribute to this PL Variance for the trade 29 (do not consider the variation indicated on the "Total US_P_FLOW_ST" line since it corresponds to the PL variation for the entire portfolio):

Instrument	ALL	Effect	ALL	Portfolio	ALL	Typology	ALL	POST	Trade Number	29.00
Portfolio	Effect	Effect detail	Typology	Instrument	Currency	Trade Number	PL variation<USD>			
US_P_FLOW_ST	Total US_P_FLOW_ST						120,263			
	Time		Equity	Total Equity		-3				
				BMW.DE	EUR	Total EUR	0			
			29	0						
	Yield curves		Equity	Total Equity		1				
				BMW.DE	EUR	Total EUR	0			
			29	0						
	Repo spreads									
	Eq/Fut prices		Equity	Total Equity		-21,644				
				BMW.DE	EUR	Total EUR	-9,066			
			29	-10,879						
	Bond/Fut prices									
	Fx spots									
	Eq vols									
	Eq smiles									
	Dividends									
	Dividends yield									
	Correlations									
	Option prices									
	Fixings									
	Closing									
	Gmp curves									
	Spot PL Conversion		Equity	Total Equity		-324				
				BMW.DE	EUR	Total EUR	74			
			29	47						
	Market Operation									
	Global other						-0			

Step	PL in USD	Step effect
Initial PL (on 24/09)	23,318	
Shift horizon to 25/09	12,486	-10,831.6642
Made of these steps		
Time	0.0387626881129108	
Yield curves	0	
Eq/Fut prices	-10878.9378180884	The biggest effect
Spot PL Conversion	47.2348515235935	The underlying is in EUR while the PL is displayed in USD, therefore, there is an exogenous FX effect
Total	-10,831.6642	



1

The slide features the Murex logo in the top left corner. To its right, the text "Learning Objectives" and "Simulation Introduction" is displayed. Below this, the text "At the end of this course, you will be able to:" is followed by a list of three bullet points, each preceded by a checkmark. At the bottom of the slide, there is a small copyright notice and a page number.

At the end of this course, you will be able to:

- ✓ Launch a simulation session with different choices according to your needs
- ✓ Manage the settings in the simulation screen and explain their impacts
- ✓ Present basic concepts of all existing screens and their adequacy to each financial product

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Agenda

- Simulation Overview**
- Simulation Screen Types**
- Simulation Behavior**
- P&L**
- Examples of Simulation Screens**
- Practice**

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Simulation Overview

Definition

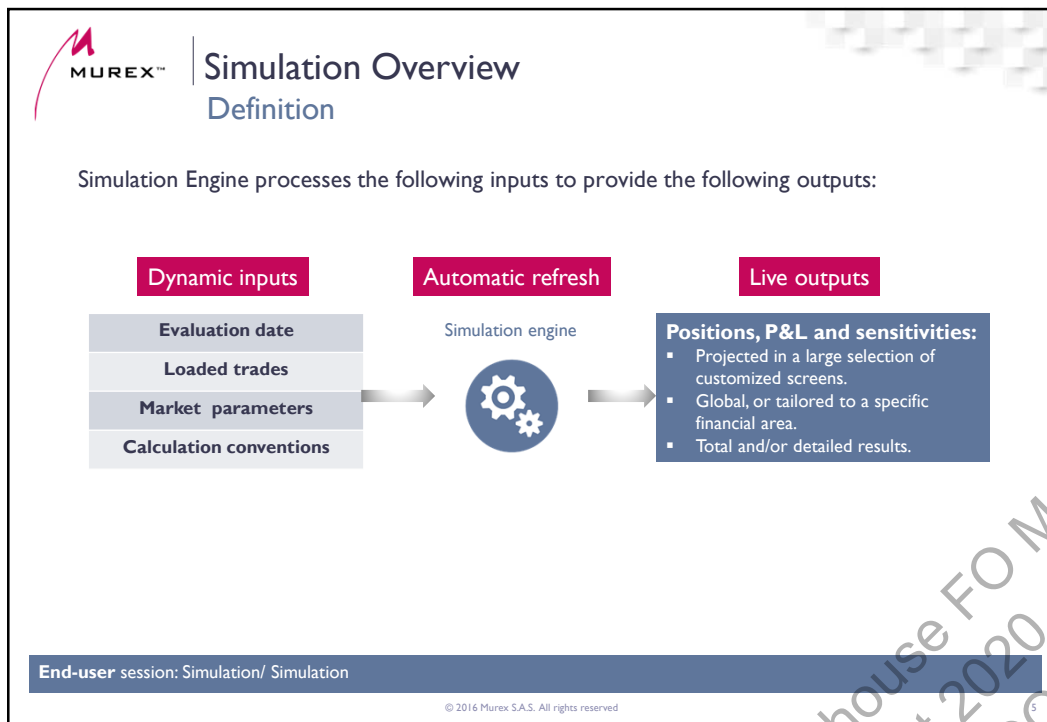
Simulation is a tool that shows the portfolio positions. It is a core Front Office module and is used across asset classes.

Used by	Traders, sales and risk managers.
Purpose	- Follow online positions, P&L and sensitivities. - Run what-if scenarios. See immediate impact on current positions, P&L and sensitivities. - Check changes due to new trades or to market data variations. - Analyse and hedge risk.
Results	Delivered with high performance.

End-user session: Simulation/ Simulation

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MUREX™ Simulation Overview Definition

From the simulation screen, end-users can **access** positions, P&L and sensitivities **for all asset classes** using **Viewer** tool.

IR PV01 (zero) by currency | **IR PV01 details** | **IR PV01 by curve** | **Fixing risk** | **Basis risk**

Curve name	Yield par rate	IR risk (par)			IR risk (zero)		
		Label/ Total	IR risk (par)	IR Future	Outright	Label/ Total	IR risk (zero)
Total		2,484	2,462	21	2,500	2,479	21
EUR EURIBOR 3M		2,462	2,462	21	2,479	2,479	21
3M	1.5300	376	376				
DEC 12	98.3364	395	395		781	781	
3Y	2.3100	1,696	1,696		1,697	1,697	
4Y	2.6100	-5	-5				
EUR FX		21	-5	21	21		21
0/N	1.4300	0	0	0			
1/N	1.4308	0	0	0	-0		-0
5/N	1.4316	0	0	0	0		0
1W	1.4348	-0	-0	-0	-0		-0
2W	1.4407	-0	-0	-0	-0		-0
3W	1.4466	-0	-0	-0	-0		-0
1M	1.4539	1	1	1	1		1
2M	1.4795	0	0	0	1		1
3M	1.5120	0	0	0	0		0
4M	1.5298	0	0	0	0		0

End-user session: Simulation/ Simulation

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Agenda

Simulation Overview

Simulation Screen Types

- How to access simulation
- Simulation by position
- Simulation by trade
- Simulation as of past date
- Other simulation screens

Simulation Behaviour

P&L

Examples of Simulation Screens

Practice

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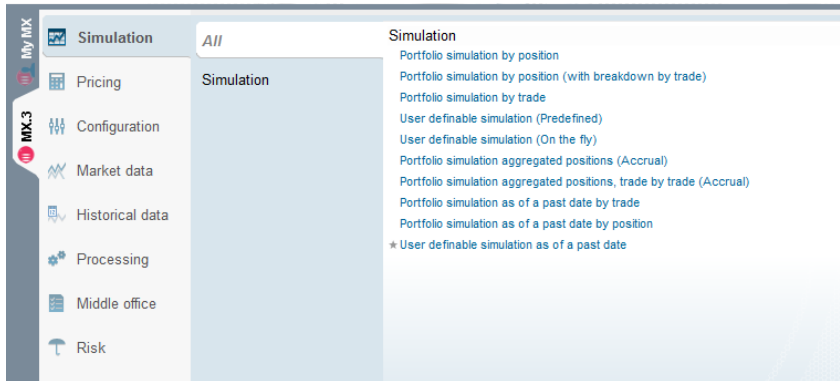
7



Simulation Screen Types

How to access simulation

To launch the Simulation, the user has several choices:



Simulation

- Portfolio simulation by position
- Portfolio simulation by position (with breakdown by trade)
- Portfolio simulation by trade
- User definable simulation (Predefined)
- User definable simulation (On the fly)
- Portfolio simulation aggregated positions (Accrual)
- Portfolio simulation aggregated positions, trade by trade (Accrual)
- Portfolio simulation as of a past date by trade
- Portfolio simulation as of a past date by position
- * User definable simulation as of a past date

End-user session: Simulation/ Simulation

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 **Simulation Screen Types**
Simulation by position: Warehouse

The **Warehouse** is a position server that aggregates and stores financial figures for trades and their events according to some given aggregation keys.

Advantage	To increase the consolidated simulation performance.
Computation	The warehouse service maintains computation/aggregation in real-time.
Warehouse rebuild	The following processing scripts should be executed in order to rebuild the consolidated data model based only on the trade repository (i.e. detailed data model) data: 1. Stop Engine 2. Warehouse Rebuild 3. Maintenance 4. Start Engine

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 **Simulation Screen Types**
Simulation by position

These are the fastest simulations:

Portfolio simulation by position	Results are consolidated, trades accessible via trade contribution.
Portfolio simulation by position (with breakdown by trade)	Results are read from the warehouse non aggregated tables (providing access to trade number).
Portfolio simulation aggregated positions (Accrual)	Results are consolidated, but MM trades are loaded with a lower level of aggregation to compute accrual P&L figures.
Portfolio simulation aggregated positions, trade by trade (Accrual)	Results are read from the warehouse non aggregated tables, and MM trades are loaded with a lower level of aggregation to compute accrual P&L figures.



When this screen is loaded:
 $\text{simulation date} \geq \text{consolidation date}$
 otherwise nothing is loaded.

End-user session: Simulation/ Simulation/ Portfolio simulation by position

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MUREX™ | **Simulation Screen Types**
Simulation by trade

The calculations are done on each deal using detailed simulation screens.

Portfolio simulation by trade	Results are detailed, by trade
	Loading by portfolio

The user definable simulation screen require the user to enter a filter criteria to apply on the calculations.

User definable simulation (Predefined)	Predefined filter, selected from the list
User definable simulation (On the fly)	Filter criteria selected on the fly Similar to trade query screen

 The user definable simulations are not refreshed by new trades/events.

 This simulation is slower than aggregated simulations.

End-user session: Simulation/ Simulation/ Portfolio simulation by trade

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MUREX™ | **Simulation Screen Types**
Practice I (1/2)

 Compare portfolio simulation by position, by position (with breakdown by trade) and portfolio simulation by trade.

1. Open 3 end-user sessions with the login MUREXFO/MUREXFO.
2. Open the group FO in each session.
3. Open one type of simulation screen in each of the different sessions.
4. Load in each simulation screen all FI, FWD and MMA portfolios.
5. Load the view MX_PL_MTM breakdown daily (c) (or (d)).
6. Compare the 3 sessions.
7. Add a breakdown by trade number by adding the field « Trade number » from the Data Dictionary.
8. Compare the 3 sessions. Why are there differences in the same view in the different sessions?

 **Solution**

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Simulation Screen Types

Practice I (2/2)



Compare portfolio simulation by position, by position (with breakdown by trade) and portfolio simulation by trade.

9. In the portfolio simulation by position, select a position to view the trades contributing to it. To do so, right click on the position and select the « Trade contribution/ Explain position » menu.
10. Once the screen is open, select the view REF_PL_TRAINING_SIMULATION.
11. Check the P&L fields by trade in the trade contribution and compare it to the portfolio simulation by trade.
12. Open another similar end-user session and load a user definable simulation (on the fly) on a set of the trades that you see in the trade contribution.
13. Compare the P&L to the portfolio simulation by position (breakdown by trade).



Solution

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Simulation Screen Types

Simulation as of past date

Simulation as of past date prompts the user to select the past date, then any filter, and finally select a portfolio.

Loads trades that correspond to status of database as of selected past date.

Bases the revaluation past date market data of the corresponding data set.

Portfolio simulation as of past date by trade	Results are detailed, by trade, at past date Loading by portfolio
Portfolio simulation as of past date by position	Results are consolidated, by position, at past date Loading by portfolio
User definable simulation as of past date	Results are detailed, by trade, at past date Filter on the fly



Simulation as of past date can be by position, trade or user definable.

End-user session: Simulation/ Simulation/ Portfolio simulation as of a past date by trade/position

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Simulation Screen Types

Other simulation screens

Other simulation screens:

Today Portfolio simulation by trade	Filtered on today's activity (deals & events).
Counterpart side simulation	Filtered on one counterpart, the buy/sell is reversed compared to portfolio simulations.
Fund simulation by trade	Filtered on a fund in detailed mode – asset management specific simulation.
Fund simulation by position	Filtered on a fund in consolidated mode – asset management specific simulation.

End-user session: Simulation/ Simulation/

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Agenda

Simulation Overview

Simulation Screen Types

Simulation Behavior	• Trade selection • Market parameters • Evaluation date • Calculation assumptions • Common features
----------------------------	---

P&L

Examples of Simulation Screens

Practice

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Simulation Behavior

Trade selection

Simulation by portfolio can be dynamically refreshed when these operations are performed:

- ✓ New trades are inserted into the loaded portfolios.
- ✓ Events (market operations) are performed on existing trades of the loaded portfolios.

Rights to the automatic refresh can be set from an end-user session under:

End-user session: Simulation/ .../ Settings> Simulation screen settings> Simulation server

All users	Automatic refresh.
By user	Only trades or events inserted by current user and attached to loaded portfolio impact the simulation.
Inactive	Only trades or events inserted from current session and attached to loaded portfolio impact the simulation.



"Autotrade capture" flag available in simulation screen.

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Simulation Behavior

Market parameters



The market data viewer connected to RT can be used to both publish and retrieve market data.

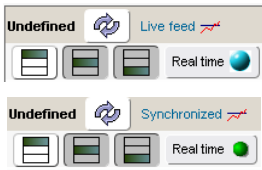
The market data loaded when the user enters the simulation screen are based on the default market data set assigned to the user's FOD.

The simulation screen can be connected to real-time feed.

Real-time activated	Will refresh the market data in the simulation screen. The P&L and different figures on screen are recomputed using the real-time market data.
Real-time deactivated	The DB market data is loaded on screen. The P&L and different figures are computed using the DB market data.

Two real-time connectivity modes are available:

Live Feed	Market data are refreshed automatically according to a user defined refresh cycle.
On demand	User is prompted when new market data becomes available. Market data are refreshed only on when requested by user.



End-user session: Simulation/ .../ File> Realtime

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MUREX™ | **Simulation Behavior**
Market parameters

Users connected to MX with different sessions can update the market data simultaneously.

Active session can be protected from changes made to other sessions unless:

1. A refresh is requested (top toolbar) File> Market data> Reload stored market data or click on 
2. Simulation is re-launched.

The user can do two types of changes on the market data in the simulation screen:

Local changes	Performed on market data (loaded in memory) from within simulation screens: • Directly on-screen • Zooming on Loaded Market Data, but without global save
Global changes	Performed on « official » market data set by saving Loaded Market Data.

 Zooming on the individual screens from the Market data tree in View> Popup market data can also allow performing live changes or saving them to the database

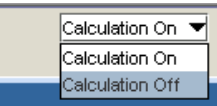
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MUREX™ | **Simulation Behavior**
Market parameters

Two calculation modes are available to the users in the simulation screen:

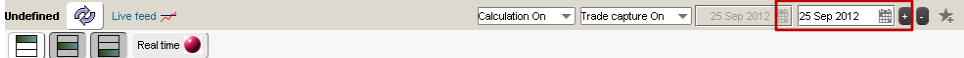
ON	Any change in market data or horizon date re-computes the position.
OFF	Single changes do not imply a recalculation, this allows several changes before the system re-computes the position.



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MUREX™ | **Simulation Behavior**
Evaluation date



The simulation can run on two dates:

Current trading date	Default date It can be shifted forward.
Past dates	1. Trades loaded with their status as of selected date. 2. Trades loaded from detailed tables. 3. Revaluation uses market data as of selected date.

Theta & horizon date: From simulation screen, forward move of horizon date shows actual time decay effect (Theta).

Theta as of past date: When simulation is loaded as of past date and horizon is moved forward, system considers loaded deals only.

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MUREX™ | **Simulation Behavior**
Calculation assumptions

The different calculation settings in simulation can customize the display choices.

They are applied on three levels:

	Scope	Set under the screen	Example
General settings and pricing settings	Shared by all users	Configuration/ Settings/ General settings	Gamma calculation: bucketed
Current session settings	Applied in the scope of current session only	Configuration/ Settings/ Current session settings	Same as general settings
Simulation settings	Used exclusively in simulation screens	Configuration/ Settings/ Simulation settings	Real-time status: Connected (Automatic retrieve)

 Some flags impact calculations only if market conditions change.
Example: When moving horizon, Rates may remain constant or equal Fwd-Fwd rates.

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Simulation Behavior

Practice 2



View the market data changes effects in the simulation.

1. Using two of the sessions of practice 1, launch a portfolio simulation by trade in these separate sessions on the portfolio FWD_EUR_LN, view MX_PL_MTM summary (c).
2. Stop the real time feed in both sessions. How would you do that?
3. Reload stored market data:
 - a) How?
 - b) Has anything changed in current session?
4. Record the P&L figures.
5. Change the EUR/USD spot rate in the loaded market data:
 - a) How?
 - b) Do not save and exit => Has anything changed in current session? In second session? Why?
6. Restore the previous values.
7. Change the EUR/USD spot rate but save globally this time:
 - a) Has anything changed in current session? In second session? Why?
 - b) Update the second session's P&L.
8. Compare the two sessions.


Solution

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Simulation Behavior

Common features

The main common tool across products is the viewer, which allows the building of customized views to fit each asset class needs.

Access to general menu

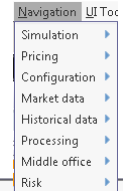
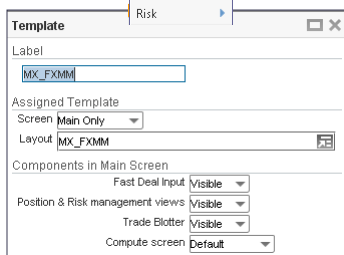
The user does not need to exit from the simulation screen to access the different screens in MX.

All the screens are accessible from the Navigation menu in Simulation.

Configurable access to the different screens in simulation

Access to the various screens can be set in a configurator session:

- Main only
- Specific screens
- Include Fast deal input & trade blotter

End-user session: Simulation/ Simulation/ Portfolio simulation by position
Configurator session: Infrastructure/ FO desk/ Settings/ Simulation screen assignments

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Agenda

- Simulation Overview
- Simulation Screen Types
- Simulation Behaviour
- P&L**
 - Absolute P&L vs. relative P&L
 - Stored P&L
 - Debugging tips
- Examples of Simulation Screens
- Practice

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P&L

Absolute P&L vs. relative P&L

P&L in simulation screens can be displayed in two modes:

Relative P&L can be displayed in the views by changing the field settings to read from the Datamart table. The origin date is specified at this level.

Absolute	Since inception (evaluation date).
Relative	Between an origin date and evaluation date.



For optimization reasons, P&L as of origin date is not recomputed but read from a stored P&L table in the DataMart.



The PL Variance module can be used to explain the P&L movement between yesterday and today.

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MUREX™ | **P&L**
Stored P&L

The Stored P&L is the absolute P&L of a position or a trade, copied and saved in the Datamart table. In the simulation, it is used when there is a need to display a relative P&L e.g. daily P&L, monthly P&L etc.

Stored P&L settings	Reads the “reporting setup” settings (in opposition to Simulation settings).
Stored P&L Storage	In the Datamart by a feeder during the EOD based on a simulation view.

 Note that it is possible to use stored fields and display them in absolute or in variations is available for any output of the risk viewer.

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MUREX™ | **P&L**
Stored P&L

Feeding view

Trade Number	P&L (Abs)
Total	
20,493	7,820,786
20,494	-1,402,464
20,495	47,017,863
20,496	2,817,813
20,498	537,286,372

User view

Portfolio	Family	Grp	Cu	Trade No	P&L (Abs)	PL_yesterday
PL_TRAINING_IRD	BOND	USD		20,493	7,820,786	80,630,146
				20,495	47,017,863	484,741,172
				20,496	2,817,813	29,050,876
				20,498	537,286,372	
	LN_BR	USD	20,494	-1,402,464	-14,459,013	

Feeder

Reporting Database

MX FINANCIAL

Loading

Official P&L retrieved from the DataMart without computation.

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MUREX™ | **P&L**

Debugging tips

 **P&L is wrong:**

Daily P&L = Today's absolute P&L – Yesterday's absolute P&L
The incorrect value can come from today's evaluated P&L or yesterday's stored P&L.

 **Stored P&L is incorrect:**

It can be recomputed manually by loading a simulation as of past date and looking at the view used for P&L storage.

 **The Stored P&L is different from the recomputed P&L:**

This happens when changes affecting past dates P&L are made, such as:

- ☒ Modification of past market data.
- ☒ Insertion of a back-traded deal.

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MUREX™ | **Agenda**

Simulation Overview

Simulation Screen Types

Simulation Behaviour

P&L

Examples of Simulation Screens

- FXMM
- Simulation viewer

Practice

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MUREX™ | Examples of Simulation screens
FXMM

The screenshot displays the Murex FXMM interface with the following sections:

- Fast Deal Input:** A table for entering trade details.

Counterparty	Nature	Lend/Borrow	Instrument	Start	Maturity	Nominal	Rate/Margin
	Fixed rate	Borrow	Spot			1,000,000	
- Position keeping and P&L monitoring:** A section for monitoring positions and P&L.

Curve name	Yield par rate	IR risk (par)	IR risk (zero)
Total		2,462	2,479
EUR EURIBOR 3M	1.5300	376	376
DEC 12	96.3364	395	395
3Y	2.3100	1,686	1,686
4Y	2.6100	-5	-5
EUR FX		21	21
- Deal Blotter:** A table showing a list of deals.

Contract nb	Trade nb	Status	Typology	Instrument	Amount	Face cur	Price	Margin	Start date	Maturity	Counterparty	Portfolio
3,455	4,797	Ins	IRS	EURBOR 3M	85,000,000	EUR	1.8500	03 Jan 2011	03 Jan 2014	DEXIA BANK BELGIUM S.A.	IR_SWAPS_EU	
4,796	4,795	Ins	Spot	EURUSD	1,000,000	EUR	1.3500	27 Sep 2012	27 Sep 2012	ABN AMRO AMS	MX_PTF3	
4,795	4,794	Ins	Depo	USD	85,500	USD	0.0000	27 Sep 2012	27 Sep 2012	DEXIA GROUP	MMA_USD_LN	
4,794	4,793	Ins	Depo	USD	85,500	USD	0.0000	27 Sep 2012	27 Sep 2012	DEUTSCHE BANK	MMA_USD_LN	
4,793	4,792	Ins	Depo	USD	89,000	USD	0.0000	27 Sep 2012	28 Sep 2020	GENERAL MOTORS	MMA_USD_LN	
4,792	4,791	Ins	Depo	USD	91,000	USD	0.0000	27 Sep 2012	27 Sep 2019	AXA	MMA_USD_LN	

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MUREX™ | Examples of Simulation screens
Simulation viewer

Views are built for each asset class. The different views are customizable and flexible.

The screenshot displays the Murex Simulation viewer interface with the following sections:

- Simulation:** A section for viewing simulation results.

Curve name	Total	IR risk (par)	IR risk (zero)
Total	2,496	2,474	2,474
EUR EURIBOR 3M	0	384	384
DEC 12	0	404	404
3Y	0	1,691	1,691
4Y	0	-5	-5
EUR FX	0	22	22
0/N	-0.0355	0	0
1/N	-0.0632	0	0
1.5/N	-0.0632	0	0
1W	-0.0570	-0	-0
1M	-0.0572	-0	-0
2M	0.0154	-0	-0
3M	0.0537	1	1
4M	0.1006	0	0
5M	0.1765	0	0
6M	-0.0251	0	0
- Portfolio:** A table showing portfolio details.

Instrument	Amount	Face cur	Price	Margin	Start date	Maturity	Counterparty	Portfolio
EUR	-480,643	USD	359,302	0.7485	1.2963	-433,643	-202,863	-262,982
USD	-191,000	USD	113,350	0.7500	1.2963	-433,643	-202,863	-262,982
EUR	5,200	USD	-3,900	0.7501	1.2963	-433,643	-202,863	-262,982
USD	-7,700	USD	5,776	0.7501	1.2964	-433,643	-202,863	-262,982
EUR	46,071	USD	-39,804	0.7500	1.2964	-433,643	-202,863	-262,982
USD	-3,298	USD	2,480	0.7504	1.2965	-433,643	-202,863	-262,982
EUR	51,896	USD	-46,923	0.7500	1.2965	-433,643	-202,863	-262,982
USD	-117,386	USD	88,033	0.7489	1.2967	-433,643	-202,863	-262,982
EUR	-40,330	USD	30,253	0.7501	1.2971	-433,643	-202,863	-262,982
USD	-14,218	USD	10,862	0.7489	1.2975	-433,643	-202,863	-262,982
EUR	-4,849	USD	3,638	0.7504	1.2987	-433,643	-202,863	-262,982
USD	85,168	USD	-64,627	0.7500	1.2988	-433,643	-202,863	-262,982
EUR	23,110	USD	-24,658	0.7508	1.2987	-433,643	-202,863	-262,982
USD	-360,674	USD	263,034	0.7501	1.2988	-433,643	-202,863	-262,982
EUR	5,600	USD	-4,200	0.7500	1.3001	-433,643	-202,863	-262,982
USD	15,400	USD	-11,550	0.7500	1.3001	-433,643	-202,863	-262,982
EUR	-793	USD	606	0.7500	1.3011	-433,643	-202,863	-262,982

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Agenda

- Simulation Overview
- Simulation Screen Types
- Simulation Behaviour
- P&L
- Examples of Simulation Screens
- Practice**

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Practice

Practice 3

 Test the behavior of a simulation as of past date.

1. Log in an end-user session using MUREXFO/MUREXFO and the group FO_IRD.
2. Launch the portfolio CLIENTI_PAR in a portfolio simulation by trade. Select the view MX_PL_MTM summary (d).
3. Enter a plain vanilla IRS as of today using the fast deal input (top section of the screen).
Hint: you can use **Tools> New deal entry template**
4. Open another end-user session with the same credentials and group and launch a simulation as of past date (yesterday) with the same view. Check the P&L.
5. Shift the horizon forward by 1 day.
6. Add a breakdown by trade number in your view in both sessions. Can you identify the trade you inserted?

 **Solution**

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Practice

Practice 4



Practice 4 – hands on: Customize the Simulation screen settings

1. In a Configurator session, change the screen Simulation by trade for the FO_IRD group so the default layout is MX_IRD_Market quotes and the trade blotter is no longer visible.
2. Change the « Trade capture » setting for your current user only and check the impacts on the simulation when entering a trade. You may want to open another session to insert a trade to test this setting (do not close your current session).
3. Change the aggregation currency to EUR for your current user only and check the impact in the P&L view MX_PL_MTM summary (c).



Solution

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EXERCISE SOLUTIONS

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Simulation Screen Types

Practice | Solution

Compare portfolio simulation by position, by position (with breakdown by trade) and portfolio simulation by trade.

6. Compare the three sessions.

The three sessions should display the same results.

8. Compare the three sessions. Why are there differences in the same view in the different sessions?

Trade number is available in the dictionary under Murex Formulas/ Trade number.

The portfolio simulation by position does not load the information deal by deal. It loads aggregated positions from the warehouse.

11. Check the P&L fields by trade in the trade contribution and compare it to the portfolio simulation by trade.

Open it on the total P&L figure.

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Simulation Screen Types

Practice | Solution

Compare portfolio simulation by position, by position (with breakdown by trade) and portfolio simulation by trade.

The P&L by trade in the trade contribution (simulation by position) should be the same as the portfolio simulation by trade. This shows that the consolidated simulation offers very similar functionalities to the detailed one but with much better performances.

13. Compare the P&L to the portfolio simulation by position (breakdown by trade).

You can take one trade only. The P&L should be the same.

Exercise

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Simulation Behavior

Practice 2 Solution



View the market data changes effects in the simulation

1. Once in the simulation, open the views gallery and select the view *MX_PL_MTM summary (c)*.
2. Go to File> Real time> Disconnect or click on the Real time button.
3. Reference market data:
 - a) Click on « Reload stored market data ».
 - b) Yes, the P&L is different because it is now calculated from the stored market data (MDS).
4. Copy the P&L for each type of trade.
5. Change of EUR/USD spot:
 - a) Go to View> Pop-up market data> FX spot and change the spot.
 - b) The P&L is changed in the view of this session. The 2nd session is not impacted.
6. Click on « Reload stored market data ».
7. Click on "Save global".
 - a) Current session's P&L has changed. Second session P&L has not changed since the real-time feed is disconnected.
 - b) To update it, click on "Reload reference market data".
8. → Both sessions should be identical.


Exercise

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Practice

Practice 3 Solution



Test the behavior of a simulation as of past date

3. Enter a plain vanilla IRS as of today using the fast deal input (top section of the screen). Hint: you can use Tools> New deal entry template
4. Open another end-user session with the same credentials and group and launch a simulation as of past date (yesterday) with the same view. Check the P&L.

→ The new inserted trade does not contribute to the P&L.

5. Shift the horizon forward by one day.
6. Add a breakdown by trade number in your view in both sessions. Can you identify the trade you inserted?

→ The new inserted trade still does not contribute to the position. Shifting the horizon just shows the real theta but does not show the simulation as of today.


Exercise

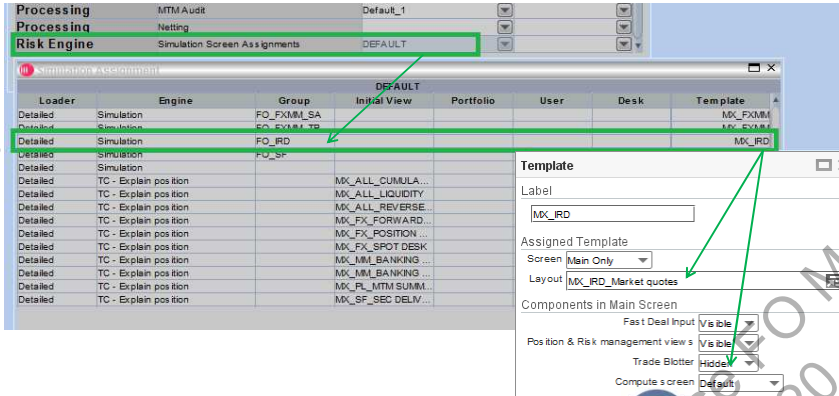
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MUREX™ | Practice
Practice 4 Solution

 Customize the Simulation screen settings

1. In a Configurator session, change the screen Simulation by trade for the FO_IRD group so the default layout is MX_IRD_Market quotes and the trade blotter is no longer visible.



Configurator session: Infrastructure/ FO desk/ Settings/ Simulation screen assignments

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Exercise

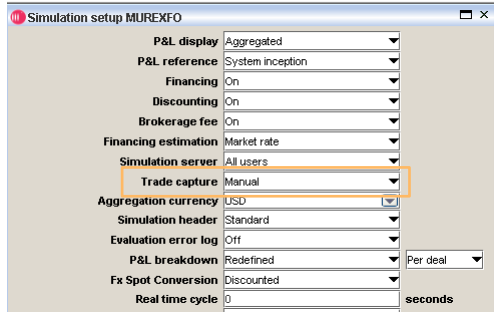
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MUREX™ | Practice
Practice 4 Solution

 Customize the Simulation screen settings

2. Change the « Trade capture » setting for your current user only and check the impacts on the simulation when entering a trade. You may want to open another session to insert a trade to test this setting (do not close your current session):

Inside the simulation, go to Settings/ Simulation screen settings. Change the «Trade capture» flag from Automatic to Manual. When entering a trade, it does not impact the positions immediately.



Exercise

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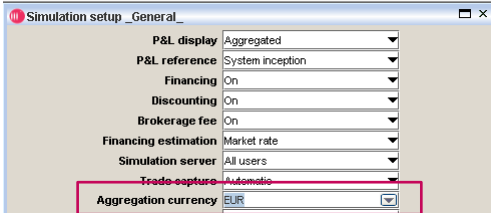
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MUREX™ | **Practice**
Practice 4 (Solution)

 **Customize the Simulation screen settings**

3. Change the aggregation currency to EUR for all users and check the impact in the P&L view MX_PL_MTM summary (c):

From outside the simulation, go to Configuration/ Settings/ Simulation screen settings and change the "Aggregation currency" to EUR. Load the simulation afterwards → the P&L is expressed in EUR.




 **Exercise**

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MUREX™

Thank You!



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Post-Training Reference List

FO Configuration

Title	docid	Content	Key concepts
Interest Rate Swaps Generator Definition	748	This document presents in detail the definition of a standard swap, as the elementary blocks have been defined elsewhere.	Introduction to IRS generator, General flags, Currency and Rates components, Capital exchange, Rate conversion, Accruals /Yield, Schedules and dates components, Stub period, Rate factor applied, Margin mode, Rate formula, Total return, Day count, Exotic features with phases and legs
FX Currency Definition	1039	This document describes how currencies are configured within the system.	Naming conventions, Default conventions, Rate conventions, Definition of Forex parameters, User definable fields
FX Swap	9448	This document describes how FX Swaps, Split FX Swaps, Multi Leg Swaps (MLS) and Split Multi Leg Swaps are handled in MX.3.	Business definition, Configuration settings, Details of FX Swaps, Details of Split FX Swaps, Details of Multi Leg Swaps, Details of Split Multi Leg Swaps
FX Spot Rates	478	FX Spot price is a conversion rate for a transaction involving two flows in two different currencies.	Financial definitions, Spot rates screen, More details on the Spot rates
Currency Pair Configuration	1060	This document describes how currency pairs are configured in the application for handling the FX Cash and FX Derivatives businesses.	FX OTC contract, OTC contract description, User definable fields
FX Settings	1050	FX Settings screens deal with user definable assumptions on FX market parameters and FX deals pricing, revaluation rules and display of results.	General organization, FX General settings, FX Pricing and FX User Pricing settings, Simulation settings, Reporting Set-up

Post-Training Reference List

FO Configuration

Title	docid	Content	Key concepts
Profit and Loss Analysis	288	This article presents the building blocks of profit and loss (P&L) analysis in MX.	P&L Breakdown: Nomenclature, P&L Analysis by Deal, Consolidated vs. Detailed Database
P&L Cycle and End of Day Cycle	1325	The current paper documents the role of the middle-office, which is focused on the generation of an official P&L per entity, for all the entities it is responsible for. It entails a typical cycle made of successive tasks and procedures achieving the generation, the validation and the distribution of official figures.	Position of the Middle-Office: On the Front-Office side, Sales or traders people, Traders using global funding tools, On the Processing-centre side, Middle-Office, Processing Center, Accounting, FO desk EOD cycle, Processing center EOD cycle, Interaction Front Office / Processing : EOD Workflow Summary
Market Interest Rate Sheets	924	Market Parameters / Rates /Market Rate sheets store the interest rate market parameters that can be used to build a zero coupon curve.	Rate sheet description, Rate sheet functionalities, Market Rate Sheet Data model
Generic Market Parameters	2047	This document describes how Generic Market Parameters (GMP) are handled in the application. The first part describes how to create a set of GMP and the second part describes how to retrieve them in a formula.	GMP Type Definition, GMP Group Definition, Retrieve through formula
Basis Curve Building and Risk	12947	This document describes how to build the curves needed to price basis swaps.	Multi curve environment, Basis swap quotes and basis risk, Implementation in MX of Basis Swap Spread Curve for a separate Basis Risk versus Interest Rate Risk, Implementation in MX of curves mixing Basis swaps and other quotes for a Risk by Curve based on quotes, Summary of typical configuration cases

Post-Training Reference List

FO Configuration

Title	docid	Content	Key concepts
MX Viewer Technical User Guide	7037	The aim of this document is to describe the generic functionality available in the MX Viewer tool. It complements other functional documents available for MX Viewer.	Introduction, Native view, Dictionary, Gallery, Formatting, Settings, Actions, Filters, Formulas, Layouts, Loader viewer, Stored data, Visibility, Clipboard, Best practices, Form Mode
Simulation Feeder	1359	This document describes how to set up Real time on a v3.1 environment in a simple and quick way.	Overview, General configuration when starting from scratch, Creation of a reference XML input file, SimulationFeeder
Simulation Viewer Overview	2075	This document, for internal use only, targets transversal teams requiring an overview of the data usually produced in simulation views and the meaning of the commonly used data dictionary fields.	Types of views and basic principles, Usage per asset class, Documentation
P&L Variation Practical Analysis	1897	The function P&L explains the profit and loss (P&L) variation of the book between yesterday (closing time) and now. This document presents a practical overview of this function, and details some effects that can be used to zoom on the P&L variation.	P&L explain overview, Details on effects computation
P&L Variance: P&L Explain Validation Guidelines	2186	The purpose of this document is to provide guidelines for testing the new P&LVAR tool. It details different testing strategies used to reconcile the figures given by the P&LVAR with the main simulation screen.	General Principals: Launcher Configuration, P&LVAR Scenario Setup, P&LVAR View Setup, Simulation Settings, Simulation Setup, FX Simulation Setup, Rates Setup; Validation of P&LVAR: P&LVAR due to the Movement of the Market Parameters, Spot P&L Conversion, Time Effect, Yield Curve Effect, FX Spot Effect, FX Volatility Effect, FX Smile Effect, P&LVAR due to the Market Operations

Term	Definition
------	------------

A

Active contract	Contract available/unavailable for selection (in pricing, trade entry...)
Adjustment before/after shift	A date adjustment rule may apply on the original date / on the shifted date. Not generally used in the shifter definition.
Algorithm	Determines how to shift the dates: Union or parallel.
Asian Past Average Rounding	Impacts the behavior of the application when an FX Asian option is exercised.

B

Backward shifter	For FX Derivatives only. Computes the Expiry date from the Delivery date and to compute the Fixing date from the Delivery date (NDF). It relies on the Date to value date calendar.
Bond	A bond is a debt security where the issuer owes the investor a debt, where the issuer pays an interest (a coupon) and repays the principal at the maturity of the Bond.

C

Calendar checking	Check whether the generated date is Holiday.
Capital Gain	Today's value - Initial value.
Cash	All past payments (from Capital or Revenue source) & future payments not included in the market value (e.g. premium for an option).
Closest index	The closest index stored in the system in terms of maturity.
CLS eligible	Continuous linked settlement eligible.
CLS start date	Start date
Contract active	If not ticked, the contract cannot be booked.
Cost of carry	See "Financing"
Cross swap mode	Ability to see cross swap points in the simulation zoom screen.
Cumulative mode (default mode)	For P&L Variance. Moving the parameters in a specific order, one after the other, keeping the already shifted parameters in their final value.

Term	Definition
Currency	Unit of exchange
Current session settings	Current session settings apply to the current session only. If the user opens a new session, he will inherit the settings from the general settings.
CUR traded quotation	Default quotation mode displayed when CUR is traded as nominal amount.

D

Data Dictionary	A "list" of available fields arranged by categories.
Date to value date calendar	Computes the Spot date from the Horizon date or the Delivery date from the Expiry date (or the other way around).
Default quotation	Quotation mode. E.g. futures but not only. Futures are the most used quotation mode for a currency pair according to business practices.
Default premium quotation	Premium quotation mode displayed by default in the Pricing screen, the Deal entry ticket, and in Simulation.
Default rollover quotation	Default quotation mode used within the rollover procedure.
Default spot quotation mode	Spot quotation mode EUR-USD <> EUR = x USD <> x USD per EUR

F

Face amount	Contract size. Can be expressed in the underlying currency or base currency according to market conventions.
Financing	Applies to past cash flows. It is also called cost of carry. It assumes that all past flows have been reinvested every day at the overnight financing rate.
Floating rate reference	Default archiving group for FRA transactions
Flow rounding rule	Determines if the amount is rounded (nearest, by excess, by default)
Foreign currency	See "Position currency "(Underlying)"
Format Spot/Strike/Forward/ Swap/Volatility	Display of spot, strike, forward and swap points is defined for both quotation modes.

Term	Definition
Forward shifter	Computes the Spot date from the Horizon date or the Delivery date from the Expiry date. More generally to compute the Value date. It relies on the Date to value date calendar.
Future cash proceeds	Are particular flows received in the future and not included in the market value. It could be the cash amount invested to enter into the transaction or any fixed but not yet settled flow e.g. the option premium which can be found in deal ticket.
Future prices	This flag determines which prices will be used and impact the futures contract prices. The values are: 1) Interpolated from OTC 2) Listed curve + OTC shift.
FX pricing settings & FX user pricing settings	Configuration flags mainly impact trade evaluation rules, cover models and specific market parameters assumptions. Can be modified locally for one user in the menu Configuration/Settings/FX user pricing settings.
FX SPOT	A spot trade is a contract between two parties to convert an amount in one currency into another currency (buying one, selling another) at a spot date i.e. very short term trade.

G

General settings	General settings are configured globally at the FO desk level. They can be customized locally by a given end user if he has the appropriate rights, given in the configurator session.
Generator	Is a template/model which defines a swap or a bond. Can be used "as is" in a trade or be locally modified. Enables fast insertion of trades. Contains everything that is needed: Calendar, Payment fixing and Calculation schedule, Stub period, Rate convention, In arrears or up front, An index, All... except figures. Can be changed "globally" or "locally".
Global change	Means modifying the generator template, which affects all new trades, including existing booked trades with the same generator.
Greeks	Delta: is the measure of the ratio of the change in price of the underlying asset to the change in the price of the option/ derivative Gamma : is the measure of the rate of change of the delta according to the price of the underlying asset Rho: is the measure of the sensitivity of an option to a change in the interest rate

Term	Definition
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Theta: is the measure of the rate of decline of an option due to the passing of time

Vega: is the measure of an option's sensitivity to the changes in the volatility of the underlying asset

I

If non business day	If the result date on the calendars in Calendar checking is not a business day, this setting defines the rule to be applied.
Interpolated index	A computed figure deduced from the interpolation of the previous and the next index (most common configuration). The system will compute the exact forward-forward rate on the archiving group rate curve.
Interpolator	Allows the user to compute capitalization factor, discount factor and discount rate between any two dates based on current curve.

L

Listed volatilities	This flag determines which volatilities will be used. The values are: 1) Interpolated from OTC 2) Listed curve + OTC shift
Local Change	Means modifying the generator at trade level. A local change only affects that particular trade. (See "Generators")
Long term schedule	Default frequency of coupon flows

M

Margining	Option to configure listed option with margining. Margining allows to pay the option premium in MtM.
Market	Organized Market on which the contract is traded.
Market Value	Market Value = Today's trade value. Market value is either the market quotation or the sum of all futures flows discounted to spot date.

Term	Definition
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O

Option cut off	Cut off time of the option (if any)
OTC	Over The Counter
OTC Contract	Underlying OTC currency pair

P

P&L currency (Base)	Currency in which the P&L for this currency pair will be expressed.
Payment calendar (FO Generators)	Used for coupon calculation period and coupon payment dates.
Payment currency (FO Generators)	Is the currency in which nominal and coupon payments are made.
Payment currency (FO Configuration)	Currency in which cash settlements are made.
Precision on amount	Determines the numbers of decimals of amounts expressed in this currency.
Premium decimals	Number of decimals displayed for the premium by display quotation mode.
Premium form factor	Factor to apply to the premium notation.
Premium quotation	Default premium quotation mode.
Position currency (Underlying)	Position currency for this currency pair. Also known as underlying, risky, or foreign currency.

R

Rate convention	Used for coupon flow calculation.
Rate entry mode	Constant swap points vs. constant interest rates when US rates move.
Reporting setup	Provides default calculation assumptions for reporting: printing reports, report batches, risk matrices, running a notepad.
Restore mode	Each parameter is shifted to new value then reset to its initial value before any other parameter is moved.

Term	Definition
Revenue	Revenue = Past returns + upcoming returns
Rights	Have to be granted in order for change of a generator.
Risky	See "Position currency (Underlying)"

S

Scenario	A set of overall settings and of ordered effects to be processed by the tool. The scenario is configured via the simulation module.
Schedules definition	Stores rules used for coupon calculation and payment dates.
Settlement	Available options are Delivery and Cash. This field indicates the settlement style by default of the spot forward on this contract
Settlement calendar	Set of settlement dates for the contract.
Shift definition	Sets the step (Number x Unit) to be added or subtracted (Forward/Backward) to the initial date.
Short & Long term rate conventions	Default interest rate calculation convention for short/long term loans.
Simple shifter / Relative shifter	A relative shifter first calculates the date according to the underlying shifter then applies the shift taking into account the rules defined in the current shifter definition.
Simulation settings	Simulation settings apply to the simulation screens. They are organized in a preliminary common configuration part, shared by all transaction types and additional setting screens specific to each activity (FX, Rates, FXMM, credit...)
Spot schedule	Defines the way to compute spot value date from a given date.
Spot to delivery calendar	Computes the Delivery date from the Spot date
Swap factor	Is a multiplying factor for display. Like the spot price, it follows the quotation mode of the currency pair.
Swap form factor	Factor to apply to the swap points notation.

Term

Definition

U

Underlying quotation

Quotation mode of the underlying currency pair.

Unit

Can be expressed as Business day, Day, Week, Month, Quarter, Semester or Year

V

Value date

Shifter used to compute Spot date from Horizon date and Expiry date from Delivery date.

Accenture: In-house FO Modules
August 2020
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